

Earthquakes and Communal Conflict: Evidence from Indonesia's Seismic Zones

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Abstract

Communal violence destroys assets, deters investment, and erodes the local cooperation on which public goods depend, and it is common in the places large earthquakes strike. Does a strong earthquake raise communal conflict or suppress it? The empirical record disagrees on the sign. To answer this question, we combine seven waves of Indonesia's village census (2005–2024) with USGS ShakeMap ground-shaking intensities for the 43,351 villages in the country's official seismic-hazard zones. We find that a village's first exposure to damaging shaking lowers the probability that it reports a mass brawl by 1.6 percentage points, against a pre-exposure mean of 6.7 percent. It is important to note, though, that such a decline is equally consistent with less fighting, with less reporting, and with better local order, and that no single outcome separates these readings. We therefore use several. The same village head, on the same form, over the same recall window, also

reports theft, robbery and assault, and these acts differ in what they demand of those who commit them. Communal violence falls while theft does not follow it down, and robbery and assault do not follow theft. We claim this divergence rather than theft's own rise, which is not separately identified under our preferred specification. Thus the evidence is consistent with a loss of the organisational capacity that collective violence requires and stealth theft does not.

Keywords: earthquakes, communal conflict, property crime, composition of disorder, Indonesia, difference-in-differences

JEL Codes: D74, K42, Q54, O12, O18, R12

1 Introduction

Large physical shocks are a recurring feature of life in hazard-prone developing countries, and communal violence is a development outcome in its own right: it destroys assets, deters investment, and erodes the local cooperation on which public goods provision depends (Barron, Kaiser, and Pradhan, 2009). The post-disaster period is also precisely when governments, security forces, and aid agencies decide how to allocate scarce resources between reconstruction and conflict prevention. Indonesia, the world’s most seismically active large economy and a country with a long record of localised communal violence (Barron, Kaiser, and Pradhan, 2009), is where both pressures meet most often.

Does a strong earthquake raise communal conflict or suppress it? The empirical record disagrees even on the sign. Brancati (2007) and Nel and Righarts (2008) find that disasters raise conflict risk, Slettebak (2012) finds, if anything, less, and newer localised evidence finds that disasters can lower conflict where they strike while raising it nearby through spatial displacement (Amarasinghe et al., 2026). The reviews divide the same way: Koubi (2019) finds no robust and general effect of climatic shocks on conflict onset, while the hierarchical meta-analysis of Burke, Hsiang, and Miguel (2015), pooling 55 studies, finds that climatic deviations systematically raise conflict risk.

Our own estimate takes a side. In Indonesia’s seismic hazard zones, a staggered difference-in-differences design finds that a village’s first exposure to damaging shaking lowers the probability it reports a mass brawl by 1.6 percentage points against a 6.7 percent pre-exposure mean, a decline of about a quarter. We defend that estimate: it is measured on the one item whose wording is identical in all seven waves, it survives province-by-wave fixed effects almost intact, and its pre-exposure record is flat. What the estimate does not do is say why, and that is where this paper makes its contribution. A decline in reported mass brawls is equally consistent with less fighting, with a village head too busy to file a report, and with a general improvement in local order under the eye of arriving responders. No

1 single outcome separates those readings, and the disaster-conflict literature has
2 been asking the question with one outcome at a time. We use several. What a
3 single shock does to *the composition* of disorder is a different question, and one
4 these data can answer.

5 The census that records communal conflict also records theft, robbery and
6 assault, filed by the same village head, on the same form, over the same twelve-
7 month recall window. Those four acts differ in what they require: communal
8 violence needs coordination and confrontation together, robbery an encounter
9 but no coordination, theft neither, and assault's encounters are largely among co-
10 resident acquaintances. Holding reporter, instrument and window fixed turns that
11 variation into an identification device, because the rival explanations make different
12 predictions across the four. Crucially, whatever moves all four acts alike within a
13 village, its fixed characteristics and any disturbance common to how disorder is
14 reported there, differences out of a contrast taken within the village. That contrast
15 is the paper's central result. What it leaves standing is an assumption, not a
16 guarantee: that untreated *relative* incidence would have moved in parallel, and we
17 test it directly. We combine seven waves of Indonesia's village census (PODES,
18 2005–2024) for the 43,351 villages in the official medium and high seismic-hazard
19 zones with USGS ShakeMap ground-shaking intensities. We date each village's
20 first exposure to Modified Mercalli Intensity (MMI) VI or above, where ordinary
21 masonry buildings begin to crack.

22 On a single common sample of 300,018 village-waves, first exposure lowers
23 reported communal conflict by 1.7 percentage points against a 2.6 percent base,
24 while reported theft does not fall with it. The five outcomes are stacked with
25 outcome-interacted fixed effects, so the standard errors respect the correlation
26 of a village's outcomes. Under the specification our selection rule prefers, which
27 absorbs province-by-wave effects, communal conflict falls by 1.7 points, which the
28 same model estimated as a Poisson puts at 33 percent of the base. Equality of
29 the theft and communal responses is rejected there on both scales ($p = 0.003$

1 linear, $p < 0.001$ proportional). Robbery and assault do not follow theft. Taken to
2 the earthquake, the unit at which treatment is assigned, the divergence between
3 theft and the definition-stable communal series is 7.6 points under treated-village
4 weighting and 6.3 points under an equal-event mean ($p = 0.023$), positive in 12 of
5 the 19 usable shocks.

6 We are deliberate about which half of that contrast carries weight. It is the
7 communal decline, not theft's level. Theft rises sharply under wave effects, by 4.3
8 percentage points on a 42 percent base, but its own coefficient is not separately
9 identified once province-by-wave effects are imposed, where the rise falls to 1.2
10 points and loses its precision. What the data establish is that the two series part
11 company, and that they part company because communal violence falls while
12 acquisitive crime does not. Nothing below requires theft to have risen.

13 That pattern discriminates. A uniform reporting shock moves all four acts
14 together; a general collapse of local order raises all four; a rise in the motive for
15 predation raises theft and robbery together, since both are acquisitive. Of the
16 accounts these four outcomes can tell apart, the pattern of theft up, coordination-
17 intensive violence down and encounter-based crime unmoved is most consistent
18 with a loss of the organisational capacity that collective violence requires and
19 stealth theft does not. What the outcomes cannot tell apart is which part of
20 that capacity gave way, the ability to coordinate or the confrontational contact
21 coordination operates on, and we use the word capacity for the two together.
22 A pre-determined moderator points the same way: the decline concentrates in
23 villages that had mobile-phone signal at the 2005 baseline, an interaction worth 2.1
24 percentage points that requires no parallel trend in the moderator and survives the
25 province-by-wave specification. We do not claim to have measured any channel's
26 share; we claim to have ruled out the alternatives these data can rule out.

27 The aggregate decline holds on the definition-stable measure and we defend
28 it, within one stated limit. The decline clears village-clustered inference and
29 earthquake-clustered inference when shocks are weighted by the villages they

1 treat ($p = 0.046$). It does not clear an equal-event mean across the nineteen
2 shocks. The estimand concerns the average exposed village rather than the average
3 earthquake. A first-time-conflict margin sits much lower. Among villages with no
4 conflict recorded in any earlier observed wave, exposure is associated with a 0.72
5 percentage point rise in first recorded conflict against a 2.2 percent counterfactual,
6 and the number survives a long battery of inference and specification checks. It
7 fails the test that matters. Run entirely inside the pre-exposure period, the same
8 design returns a treated-comparison gap of 1.5 percentage points where it should
9 return zero, and the gap does not widen when the earthquake arrives. These data
10 suggest the margin; this design cannot identify it. Section 5.3 gives the estimate,
11 the battery and the falsification together, and the falsification decides.

12 What the composition establishes, then, is not that earthquakes pacify or
13 inflame, but that one shock pushes different kinds of disorder in different directions,
14 and that the direction it pushes them discriminates among candidate mechanisms.

15 This paper contributes to the economics of conflict in three senses. First
16 and foremost, we show that the composition of disorder is identifying where the
17 level is not. Because the same official reports four acts on one form, a shock's
18 effect on their *relative* movement is measured within the village, so the differences
19 between shaken and unshaken places that defeat a single-outcome comparison
20 drop out. The disaster-conflict literature has often studied one outcome at a time.
21 Several outcomes here answer the question jointly. Their pattern fits a loss of
22 organisational capacity rather than a reporting shock, a collapse of order, or a
23 rise in the motive for predation. [Bazzi and Blattman \(2014\)](#) make the separation
24 of onset from continuation central to their study of economic shocks. We extend
25 it to physical shocks. A damaging earthquake changes both the incentives for
26 disorder and the technology of collective action, allowing the two margins to move
27 in opposite directions. It also supplies one reason for the disagreement above, since
28 an aggregate estimate then depends on the mixture of previously peaceful and
29 conflict-affected places a study observes. Whether this composition accounts for

1 the existing cross-national record remains unknown. Second, we provide evidence
 2 on mechanisms of a kind rarely available in this literature, which both [Koubi](#)
 3 [\(2019\)](#) and [Burke, Hsiang, and Miguel \(2015\)](#) call for: we observe not only *whether*
 4 the shock moves conflict but *which forms of disorder* move, in which directions,
 5 and for *whom*. Theft rises on the same census forms on which communal conflict
 6 falls. Robbery and assault do not follow theft up, and high-involvement collective
 7 action weakens. This pattern is consistent with degraded capacity for organised
 8 collective action. It is hard to reconcile with a uniform reporting shock or an
 9 opportunity-cost account in which income destruction raises all forms of violence
 10 ([Becker 1968](#); [Dube and Vargas 2013](#); within the organising framework of [Blattman](#)
 11 [and Miguel 2010](#)). We also confront the displacement concern directly: exploiting
 12 the geography of each shock, we find no detectable rise in conflict among nearby
 13 never-treated villages, which bounds, without eliminating, the concern that the
 14 suppression we estimate is merely the spatial relocation that [Amarasinghe et al.](#)
 15 [\(2026\)](#) document (Appendix D). Finally, we contribute the measurement design
 16 that makes such discrimination possible, and it travels.

17 The design deserves a statement of its own, because nothing in it is specific
 18 to Indonesia. PODES records four acts of disorder from the same village head,
 19 on the same form and over the same recall window. Their required inputs differ,
 20 yielding distinct predictions. A uniform reporting shock moves all four together; a
 21 general collapse of order raises all four. Stronger predatory motives raise theft and
 22 robbery together. Loss of organisational capacity raises stealth theft while lowering
 23 coordination-intensive violence. The last is the pattern we find, and theft and
 24 robbery differ at $p < 0.001$ on the common sample. No single-outcome estimate
 25 can make these distinctions, and reporting shocks operating at the level of the
 26 reporter difference out by construction. Any source recording several offence types
 27 from one reporter can implement the same test.

28 One methodological point follows, smaller than the rest but worth recording. In
 29 panels with unequally spaced waves, calendar-year exposure lags can misalign with

1 the enumeration month. On an identical sample, dating exposure by calendar year
2 attenuates the two-wave effect from 2.2 to 0.3 percentage points. It also produces
3 a significant coefficient one lag later that the aligned specification shows to be zero
4 (Appendix H, Table S31). We also provide the first village-level quasi-experimental
5 estimate of the earthquake–conflict relationship for Indonesia.

6 **What this design can and cannot establish.** Three limits govern the inter-
7 pretation. Theft’s own coefficient is not separately identified under the preferred
8 specification, so the contrast is carried by the communal decline; the paper claims
9 the divergence, not theft’s rise. The dynamic path is imprecise: the joint test over
10 the interior pre-exposure leads does not reject under the preferred specification
11 ($p = 0.31$), but the binned far-lead term remains positive and significant, and the
12 jump at exposure reaches only $p = 0.09$ there. And a first-time-conflict margin
13 appears in these data and then fails a placebo run entirely inside the pre-exposure
14 period; Section 5.3 reports it and nothing rests on it. Section 5.4.2 sets out what
15 remains: the pre-treatment record, the mechanism proxies, and the absence of an
16 independent violence series for these cohorts.

17 The remainder of the paper is organised as follows. Section 2 sets out the
18 conceptual framework and the predictions it generates. Section 3 describes the
19 village census, the crime module and the ShakeMap exposure measure. Section 4
20 presents the empirical framework and the identification assumptions. Section 5
21 reports the level effect, the composition of disorder, a first-time margin the design
22 cannot identify, and the limits of the evidence. Section 6 concludes.

23 **2 Conceptual Framework**

24 Strong earthquakes can plausibly raise or lower conflict, depending on which
25 mechanism dominates. The framework here centres on two competing channels.

1 **The inputs of collective violence.** Communal violence draws on four inputs:
2 grievance or economic stress; the opportunity for inter-group contact; the capacity
3 to mobilise and coordinate a group; and, working against them, the intensity of
4 monitoring by the state and community, whose capacity shapes whether grievances
5 escalate into organised violence (Besley and Persson, 2011). These inputs are
6 complementary: organised violence needs grievance, contact, and mobilisation
7 capacity present together and monitoring weak, so removing any one of the
8 capacity inputs suppresses it regardless of how strong the grievance is (Blattman
9 and Miguel, 2010; Esteban, Mayoral, and Ray, 2012; Ray and Esteban, 2017). A
10 strong earthquake damages the venues and roads on which inter-group contact
11 depends and absorbs survivors' time in recovery, so the assembly and coordination
12 that mobilisation requires become harder to supply. It may also deepen grievance
13 through asset destruction and income loss, though our data cannot show whether
14 it does, and the argument below does not need it to.

15 The organisational capacity a shock may degrade is not dedicated to violence.
16 The same standing arrangements that let a village assemble a crowd also let it staff
17 a night watch, so degrading them withdraws deterrence from property crime at the
18 moment they withdraw mobilisation from collective violence. In Indonesia these
19 are the same institutions: the village night watch, *siskamling* and its *poskamling*
20 posts (Barker, 1998). That observable guardianship deters property crime, and
21 does so locally, is established by Di Tella and Schargrodsky (2004). Tajima (2014)
22 supplies the instructive contrast: his shock, the collapse of authoritarian rule,
23 withdrew external monitoring while leaving village mobilisation intact, so disorder
24 rose, whereas an earthquake withdraws the mobilisation as well and the same
25 account predicts a decline. A strong earthquake also disperses households and
26 damages dwellings, lowering the chance that an offender encounters a resident;
27 both movements are the classical inputs of the routine activity approach (Cohen
28 and Felson, 1979), in which dispersion of activity away from households raises
29 property crime with motive held fixed. Temporary external presence may rise as

1 responders arrive, which can suppress organised violence ([Berman, Shapiro, and](#)
2 [Felter, 2011](#)), but it is directed at crowds and fighting rather than at stealth taking
3 from emptied houses, so it reinforces the first movement without offsetting the
4 second.

5 These consequences sort acts by the inputs each requires, and the sorting is
6 the identification idea behind the mechanism analysis in Section 5.2. Collective
7 violence needs both coordination and confrontation, so it should fall. Theft needs
8 neither: it is a stealth act, deterred by guardianship rather than prevented by any
9 coordination constraint, and an emptied dwelling is a better target rather than a
10 lost one, so both movements push it up. Lapsed guardianship pushes robbery up,
11 while scarcer encounters push it down, so theory signs only the comparison, that
12 robbery must rise by less than theft, and leaves its level to the data. Assault is
13 unsigned for the same reason with an added one: its encounters are largely among
14 co-resident acquaintances, which dispersal thins far less than it thins offender-
15 stranger encounters, so we count a null there as uninformative rather than as a
16 success. The four acts therefore separate without any assumption about whether
17 the motive for predation has changed, which is what makes their composition a
18 test of capacity and not of grievance.

19 One boundary of that test should be fixed here rather than conceded later. The
20 account has two inputs, the ability to coordinate and the confrontational contact
21 coordination operates on, and a strong shock degrades both at once. We use
22 *capacity* for the pair, because the four outcomes cannot tell them apart: a village
23 whose residents can no longer organise and a village whose residents simply meet
24 less often generate the same ranking across theft, robbery, assault and communal
25 violence. Nothing in what follows separates them, and the reader should take every
26 later use of the word in that composite sense.

27 **Two competing channels, and why a temporary shock persists.** The
28 canonical opportunity-cost channel ([Becker, 1968](#); [Miguel, Satyanath, and Sergenti,](#)
29 [2004](#)) implies a positive reduced-form effect from asset destruction. The net sign

1 of predation depends on wages relative to contestable rents (Dube and Vargas,
 2 2013). Our shock is physical rather than a growth shock. We find the opposite
 3 sign on the collective margin, which the composition evidence below seeks to
 4 explain. The competing channel runs the other way: contact disruption suppresses
 5 violence regardless of grievance, temporary external monitoring raises the cost of
 6 organising it (Berman, Shapiro, and Felter, 2011), and recovery reallocates time
 7 toward reconstruction. Disasters can also redirect participation toward recovery-
 8 oriented coordination (Bai and Li, 2021; Cassar, Healy, and von Kessler, 2017); in
 9 our setting the observed PODES proxies do not support a durable social-capital
 10 surge (Section 5.2.2). The shaking lasts seconds, but structural damage at MMI
 11 VI persists. Exposure therefore changes a village’s state rather than delivering
 12 only a pulse. This motivates an absorbing first-exposure model, an episodic
 13 design alongside it, and a direct test of whether the effect decays (Section 4).
 14 Supplementary Material Section S15 develops each channel and its prior evidence
 15 in full.

16 **Testable implications.** The framework yields five predictions, each tested
 17 below. First, the effect on communal conflict should be *state-dependent*, with onset
 18 and continuation moving in opposite directions. Sustaining organised inter-group
 19 violence requires coordination and draws on standing arrangements the shock
 20 degrades. A *first* episode requires less organisation and more of a trigger. Asset
 21 loss, contestable relief, and increased individual predation can supply those triggers.
 22 The two ingredients alone do not sign the margins, so we state the identifying
 23 assumption plainly. Continuation responds to capacity but not to triggers: where
 24 organisation already exists, triggers are not scarce. Onset is the reverse: where
 25 no organisation exists, the binding constraint is the incident that starts things.
 26 Prior evidence separates the two margins the same way: price shocks leave onset
 27 unmoved (Bazzi and Blattman, 2014), and Indonesian case evidence documents
 28 both halves (Barron and Sharpe, 2008), as Appendix C sets out. We therefore test
 29 the two margins separately (Section 5.3) rather than reporting only the average

1 they combine into.

2 Second, and most sharply, the four acts should separate by input rather than
3 by motive (Section 5.2.1). A rise in predatory motive alone predicts theft and
4 robbery moving together. The robbery margin fits our account and conflicts
5 with that prediction. It cannot separate our account from one combining greater
6 motive with physical displacement, which we test below. The comparative static
7 has been observed with the opposite sign on the input (García-Hombrados, 2020;
8 Cromwell et al., 1995); our contribution is not that earthquakes raise theft, which
9 is not general, but that the sign of the theft response tracks the sign of the
10 community-capacity response (Supplementary Material Section S15).

11 Third, the capacity inputs should be observably degraded. This is the prediction
12 our data test least well: collective participation is measured directly only from 2018,
13 formal policing does not move, and the community security post trends throughout
14 the panel, so each direct proxy is short, flat or confounded (Section 5.2.2). We
15 therefore treat the composition of crime, not the input measures, as the evidence
16 that carries this channel. Fourth, because communal violence is fought *between*
17 groups, the effect should be larger where pre-treatment polarisation is higher
18 (Appendix C). Fifth, stronger shaking should have a larger effect. We test this
19 with the same-earthquake band comparison rather than the threshold ladder
20 (Appendix S10). At the earthquake level, an unweighted event mean should be
21 attenuated relative to one weighted by exposed villages (Appendix C).

22 3 Data and Measurement

23 3.1 Conflict Outcome: PODES

24 The main outcome variable is a binary indicator of communal conflict from In-
25 donesia’s Village Potential Survey (PODES, *Potensi Desa*), a census of all villages
26 conducted by BPS (Statistics Indonesia) every three years. PODES asks the village
27 head (*kepala desa*) whether any intergroup conflict or mass brawl occurred in the

1 village in the past twelve months.

2 It is a census, not a sample: every village is asked in every wave, so coverage does
3 not vary with media attention or accessibility, the central weakness of event-based
4 conflict data in rural areas. What it misses is also clear: it is binary (no intensity
5 margin), retrospective over twelve months (no exact dates), and self-reported (a
6 village head may under-report events that reflect poorly on the village). Idiosyn-
7 cratic under-reporting that is uncorrelated with treatment attenuates the estimate
8 toward zero, making our estimates conservative; treatment-correlated reporting
9 is the serious threat, and we test it directly with a four-part reporting-artifact
10 analysis in Section 5.2 and an external event-data comparison in Appendix S14.

11 The analysis uses seven PODES waves enumerated between 2005 and 2024,
12 each fielded in May of its label year (2005, 2008, 2011, 2014, 2018, 2021, and
13 2024). We begin in 2005, the first wave in which the questionnaire asks specifically
14 about mass communal brawls. The analysis window of 2005–2024 covers six major
15 seismic events of national significance: the 2004–05 Sumatra-Andaman sequence,
16 the 2006 Yogyakarta earthquake, the 2009 Padang–Pariaman earthquake, and
17 several subsequent events in Lombok (2018), Sulawesi (2018), and West Java
18 (2022).

19 Because the conflict question’s wording changes across waves, we construct a
20 crosswalk that harmonises the definitions; exact wording and coding for each wave
21 are documented in the replication package.

22 **Which measure is primary, and why it matters.** The questionnaire asks
23 two things in sequence. A gate item asks whether any mass brawl occurred in the
24 village in the past twelve months, yes or no. Villages answering yes are then asked
25 to classify the episodes into named types: between groups of residents, between
26 villages, between residents and the security forces, between ethnic groups, and so
27 on. The two give different measures, and the difference is not innocuous.

28 Our primary outcome is the **gate**: whether any mass brawl occurred. Its
29 wording is identical in all seven waves from 2005 to 2024, so it is comparable across

1 the panel by construction. The alternative, which we report alongside it, is the
2 union of the named communal subtypes. The composite coincides with the gate in
3 2005, when the parent item is the bare gate. As subtypes are added, it becomes
4 narrower: the wedge is 0 percent in 2005, 11.7 in 2008, 17.1 in 2014, 23.7 in 2018,
5 and 33.7 in 2024. The composite thus carries a definitional trend, from a 0 percent
6 gap with the gate to a gap of one third. Wave effects absorb its overall level but
7 cannot absorb a component that moves with treatment.

8 The choice matters for identification (Section 5.1, Table 2). Under the preferred
9 province-by-wave effects, the gate retains 94 percent of its wave-effects estimate, the
10 composite 68 percent, and the *warga* subtype 33. The gate is, in other words, the
11 measure least dependent on comparisons made across provinces, and it is also the
12 one whose wording does not move. We therefore lead with it, report the composite
13 and the *warga* subtype beside it throughout, and note where they disagree.

14 **Village boundary harmonisation and analysis sample.** Indonesia under-
15 went extensive administrative subdivision (*pemekaran daerah*) in recent decades,
16 the number of villages growing from roughly 68,000 to 84,000, which poses a
17 fundamental panel-construction challenge. We address it with a chain-linkage
18 crosswalk that maps every village in each wave back to the boundary it occupied in
19 2005, its “2005 parent” village. We call the resulting panel the *2005-boundary panel*
20 throughout. Because villages created by later pemekaran are mapped back to their
21 2005 parent polygon, administrative splits cannot mechanically enter or exit the
22 sample; Appendix B summarises the construction and Appendix S13 documents
23 the procedure, match rates by wave, and two crosswalk-robustness re-estimations.
24 Controls are baseline village characteristics from PODES 2003 (enumerated August
25 2002). We enter them in levels for the Callaway–Sant’Anna nuisance model and
26 interact them with wave dummies in two-way fixed-effects specifications. After
27 dropping missing controls or outcomes, the absorbing two-way fixed-effects sample
28 has $N = 292,712$ village-waves. The benchmark Callaway–Sant’Anna estimator
29 uses $N = 225,372$ and identifies villages first shaken in waves 3–8 (2008–2024).

1 3.2 Crime Outcomes: The Rest of the Security Module

2 The composition argument requires outcomes beyond communal conflict, and
3 PODES supplies them on the same page of the same form. Immediately after the
4 mass-brawl item, the security module asks the village head whether each of ten
5 categories of criminal act occurred in the village over the same twelve-month recall
6 window, answered yes or no. We use three: theft (*pencurian*), robbery, which the
7 questionnaire defines as theft with violence (*pencurian dengan kekerasan*, labelled
8 *perampokan* in the earlier waves), and assault (*penganiayaan*). These are item
9 codes 01, 02 and 04 of that list, and the codes are stable across all seven waves.
10 We do not use the remaining seven categories, which include fraud, arson, sexual
11 offences, drugs, gambling, homicide and trafficking, because none of them is signed
12 by the framework of Section 2.

13 Three features of these items matter for what follows. First, the crime items
14 share a reporter, instrument, and recall window with communal conflict. A common
15 shift in village reporting thus moves all four and differences out of the within-village
16 contrast. Second, the column conventions in the released microdata differ across
17 waves even though the questionnaire item numbers do not; Appendix B describes
18 the mapping and Appendix S13 verifies it item by item. Third, the module is
19 fielded in every wave, so the crime outcomes and the conflict outcome are observed
20 on the same seven-wave grid, and event time is contiguous.

21 Theft is common, at roughly two villages in five, while robbery and assault
22 are rare, at three and five percent, and communal conflict rarer still, at between
23 two and three percent (Appendix Table 10). The gap in base rates is the reason
24 Section 5.2.1 reports the composition contrast on a proportional scale as well as a
25 linear one; Appendix B reports incidence by wave.

26 The composition analysis runs on the subsample of village-waves for which all
27 five outcomes are non-missing, $N = 300,018$ village-waves covering 43,351 villages.
28 This is smaller than the panel used for the level analysis because the security
29 module has item-level non-response, and it is the sample on which every number

1 in Section 5.2.1 is computed.

2 3.3 Seismic Exposure: USGS ShakeMap MMI

3 The treatment variable is ground-shaking intensity, measured on the Modified
4 Mercalli Intensity (MMI) scale, a twelve-point scale (I to XII) of how strongly
5 an earthquake is felt and how much damage it does at a given location. We take
6 MMI from USGS ShakeMap, which interpolates instrument and felt-report data
7 into a fine grid of shaking values for each earthquake (Worden et al., 2018). For
8 each event, we compute the average MMI across the grid cells that fall inside a
9 village’s boundary; across events within a calendar year, we keep the maximum, so
10 the measure reflects the most severe shaking a village experienced that year.

11 The USGS ComCat catalogue spans 1990–2024 (72,745 events), beginning fifteen
12 years before the first analysis wave. This prevents pre-panel shaking from placing
13 a village in a spuriously late cohort or the never-treated group. Supplementary
14 Material Section S7 tests the catalogue start year.

15 Treatment is defined two ways: (1) **continuous**: the village-year MMI value;
16 and (2) **binary**: $D_{mmi6} = \mathbf{1}\{\text{MMI} \geq \text{VI}\}$, corresponding to “Strong” shaking per
17 the EMS-98 scale at which widespread damage to ordinary masonry structures
18 begins (Wald et al., 1999). Exposure is mapped to the first PODES enumeration
19 following the earthquake, that is the first wave whose May enumeration date falls
20 after the event. Because the PODES conflict item covers the preceding twelve
21 months, this mapping can leave the first treated wave only partially exposed when
22 the earthquake falls inside that recall window; Section 5.4 audits the event dates and
23 reports a clean-post restriction that drops those cases. Robustness to the intensity
24 cutoff is assessed with alternative MMI thresholds (V and VII) in Section 5.4;
25 robustness to the intensity *measure*, using an MMI derived independently from
26 ShakeMap Peak Ground Acceleration, is reported in Supplementary Material
27 Section S17.

28 For the staggered difference-in-differences estimators, which compare villages

1 by *when* they are first treated, we identify for each village the first calendar year
2 in which its shaking reaches MMI VI or above, and map that to the corresponding
3 PODES survey wave. Treatment is therefore first *observed* strong-shaking exposure
4 during the ShakeMap catalogue period, not necessarily a village’s first strong
5 earthquake in history. Village fixed effects absorb fixed differences associated with
6 historical seismicity, though not any part of a past shock’s effect that itself evolves
7 over the panel. Villages that never experience such shaking over 2005–2024 form
8 the never-treated comparison group.

9 The estimation sample is restricted to villages in the official medium and high
10 seismic-hazard zones, which fixes the comparison set to places that could have
11 been shaken; Appendix C gives the construction.

12 **Why a threshold and not a dose.** Modified Mercalli Intensity is an ordinal
13 scale, one step of which corresponds to a factor of about 1.9 in peak ground
14 acceleration, so we do not enter it linearly: treatment is the threshold $\mathbf{1}\{\text{MMI} \geq \text{VI}\}$
15 and intensity contrasts compare bands within the same earthquake. Averaging
16 raster-cell intensity within a village nonetheless treats it as cardinal. Appendix B
17 explains this choice and bounds its consequences. The estimate retains its sign using
18 the pixel maximum, median, or a representative point. Exposure reconstructed
19 from cardinal Peak Ground Acceleration yields a decline of 1.3 percentage points,
20 significant at 1 percent.

21 **Where the ShakeMap products come from, and whether the early ones**
22 **inform.** One feature of this catalogue has to be stated plainly because it bears
23 on every early cohort. USGS ShakeMap coverage for Indonesian events breaks at
24 2013. Products for events before that date were not produced at the time: they
25 are reconstructions, generated in a single later batch from the ShakeMap Atlas,
26 built from no station data and from modelled ground-motion relations rather than
27 recorded intensities. From 2013 onward the products are instrumental. Our 2008
28 and 2011 cohorts, and most of the 2014 cohort, therefore rest on modelled shaking,

1 while the 2018, 2021 and 2024 cohorts rest on measured shaking.

2 The village census contains an independent check. From 2008 onward, PODES
3 asks whether an earthquake occurred in the village during its recall window.
4 ShakeMap exposure predicts that report by 0.302 in the reconstructed era and
5 0.398 in the instrumental era, both significant at 1 percent. The base rate is 7 to
6 13 percent. The correspondence is weaker in the reconstructed era by roughly a
7 quarter, which is the size and direction greater measurement error would produce
8 (Appendix B, Appendix Table 11).

9 The validation supports a narrower conclusion. Our treatment is a threshold,
10 $\mathbf{1}\{\text{MMI} \geq \text{VI}\}$, so error in the underlying intensity does not enter as classical
11 error in a continuous regressor; it misclassifies the treatment itself, and it does so
12 most often for villages whose true intensity sits near the cutoff. Under classical
13 misclassification this would tend to attenuate the treatment contrast, but non-
14 classical error can generate more complicated bias, and error that is correlated with
15 magnitude or with distance from the epicentre is exactly the kind this catalogue
16 might carry. The validation therefore establishes that the reconstructed products
17 contain substantial information about which villages were shaken, which is the
18 substantive question the provenance objection raises. It does not establish the
19 direction of any remaining bias.

20 An ACLED-based cross-check is described in Appendix C.

21 **3.4 Descriptive Statistics**

22 Table 1 presents summary statistics for the primary analysis sample. The pooled
23 conflict probability is 2.6 percent; roughly 1.2 percent of village-waves experience
24 $\text{MMI} \geq \text{VI}$ in the prior wave. The low baseline matters for reading magnitudes
25 later: communal conflict is a rare event, so a treatment effect of a few percentage
26 points is large in relative terms. The pooled mean, however, understates the
27 relevant benchmark: ever-treated villages reported conflict at 6.2 percent across
28 their pre-treatment waves, roughly twice the pooled rate, and Section 5 interprets

all estimates against this treated-cohort baseline rather than the pooled mean.

Table 1: Descriptive Statistics: Medium and High Seismic-Hazard Zones

Variable	Mean	Std. Dev.	Min	Max	<i>N</i>
<i>Panel A: Outcomes</i>					
Any communal conflict (composite)	0.026	0.158	0	1.000	303,457
Communal conflict, definition-stable (<i>warga</i>)	0.016	0.126	0	1.000	303,457
<i>Panel B: Treatment</i>					
Ever exposed $\text{MMI} \geq \text{VI}$ (absorbing)	0.273	0.445	0	1.000	303,457
Strong shaking, $\text{MMI} \geq \text{VI}$ (window t-1)	0.012	0.109	0	1.000	303,457
MMI intensity (mean-then-max, window t-1)	2.298	1.903	0	8.330	303,457
<i>Panel C: Baseline Control Variables (PODES 2003, enumerated August 2002)</i>					
Any household electricity	0.945	0.225	0	1.000	294,357
Paved road access	0.562	0.494	0	1.000	294,357
Internet available	0.032	0.175	0	1.000	294,357
Police post present	0.102	0.301	0	1.000	294,357
Health centre present	0.414	0.490	0	1.000	294,357
Number of primary schools	2.480	2.365	0	32.0	294,357
Distance to district capital (km)	44.1	54.6	0	985.0	292,712

Sample: KRB Medium/High seismic-hazard zones, PODES waves 2005–2024, native 2005-boundary baseline (43,351 villages $\times$ 7 waves). Panel C reports the baseline controls used throughout: PODES 2003 village characteristics, entered in levels for the Callaway–Sant’Anna nuisance model and interacted with wave dummies in the two-way fixed-effects specifications. Control *N* falls below the outcome *N* because baseline characteristics are missing for a small number of villages.

4 Empirical Framework

4.1 Choosing Among Specifications

The counterfactual this design needs is a village in the same province observed in the same survey wave, and that is where the economics points rather than a statistical convenience. Indonesia’s seismic zones lie disproportionately outside Java, in post-conflict provinces on their own declining conflict trajectories. Village fixed effects absorb the level of such a trajectory but not its slope, so a comparison that pools across provinces credits the earthquake with a decline the province was already experiencing. The preferred specification therefore absorbs village and province $\times$ wave fixed effects and excludes the always-treated cohort.

1 Concretely, for village v observed in PODES wave t we estimate

$$2 \quad y_{vt} = \beta D_{vt} + \gamma' \mathbf{X}_{vt} + \mu_v + \lambda_{p(v)t} + \varepsilon_{vt}, \quad g_v \neq 2005, \quad (1)$$

3 where y_{vt} is the binary communal-conflict indicator and $D_{vt} = \mathbf{1}\{t \geq g_v\}$ is an
 4 absorbing indicator equal to one from the wave of first exposure at $\text{MMI} \geq \text{VI}$,
 5 g_v , onward. The term μ_v denotes village fixed effects and $\lambda_{p(v)t}$ province $\times$ wave
 6 fixed effects, where $p(v)$ is the province containing village v ; it is this second
 7 term, rather than a common wave effect, that lets each province follow its own
 8 conflict trajectory. $\mathbf{X}_{vt}$ collects the seven PODES 2003 baseline characteristics
 9 listed in Appendix D, each interacted with the six post-baseline wave dummies,
 10 so villages differing in initial infrastructure are permitted to diverge over time.
 11 Standard errors are clustered at the village (Abadie et al., 2023). The restriction
 12 $g_v \neq 2005$ drops the always-treated cohort, which contributes no pre-exposure
 13 wave, so every identified cohort is first exposed between 2008 and 2024. Replacing
 14 $\lambda_{p(v)t}$ with a common wave effect λ_t recovers the conventional two-way benchmark
 15 of Equation (4); the difference between the two is precisely the specification choice
 16 at issue here.

17 To test the pre-period and trace the path of the effect, we replace the single
 18 absorbing indicator with a set of lead and lag dummies in event time,

$$19 \quad y_{vt} = \sum_{k \in \mathcal{K}} \beta_k \mathbf{1}\{t - g_v = k\} + \gamma' \mathbf{X}_{vt} + \mu_v + \lambda_{p(v)t} + \varepsilon_{vt}, \quad (2)$$

20 with $\mathcal{K} = \{\leq -5, -4, -3, -2, 0, 1, 2, 3, 4, \geq 5\}$. The wave immediately before
 21 first exposure, $k = -1$, is the omitted category, and the endpoint cells are binned so
 22 that this reference contains only pre-exposure observations rather than a mixture
 23 that includes post-treatment waves. The joint test of the pre-exposure coefficients
 24 $\beta_{-4}, \beta_{-3}, \beta_{-2}$ in Equation (2) is the pre-trend test reported below.

25 Raw cohort paths are consistent with both features of that structure, and we
 26 read them as motivation for the fixed-effect design rather than as evidence on

1 parallel trends (Appendix Figure S2).

2 Staggered designs admit many defensible specifications, and choosing among
3 them after seeing the estimates invites exactly the criticism such designs attract.
4 Our preferred specification is therefore selected on the economically relevant
5 counterfactual set out above, not on how the diagnostics come out. The pre-
6 trend test then serves as validation rather than as the selection criterion, and
7 it validates: the joint test of the pre-treatment leads does not reject for either
8 outcome ($p = 0.68$ composite, $p = 0.66$ *warga*), whereas under wave fixed effects
9 the Callaway–Sant’Anna joint Wald test over pre-treatment group-time cells rejects
10 for both ($p < 0.001$ and $p = 0.014$). These are different statistics, for a reason
11 given in Appendix H. A reader who prefers a mechanical rule reaches the same
12 design, since this is also the simplest specification that satisfies the test. Moreover,
13 the choice is conservative on magnitude: of the two, it yields the *smaller* estimate.

14 We report the estimators below on the wave-fixed-effects panel as well, both
15 because they are the conventional benchmarks and because the contrast between
16 the two is itself informative.

17 The Callaway–Sant’Anna estimator, its weighting, and the implementation
18 choices this panel forces are described in Appendix C.

19 Villages enter unweighted, so the target is the effect on the average exposed
20 *village*, not on the average exposed *person*; a population-weighted version, which
21 would answer a different and equally reasonable question, cannot be reported
22 because BPS stopped disseminating village population counts after the 2011 round
23 (Appendix C).

24 4.2 Identification Assumptions

25 **Physical exogeneity.** The location and intensity of earthquakes are determined
26 by fault geometry, seismic-wave propagation, and local geophysical conditions,
27 which are plausibly orthogonal to short-run village-level social and political con-
28 ditions, conditional on the hazard-zone restriction, village fixed effects, and pre-

1 treatment trajectories (Kirchberger, 2017; Gignoux and Menendez, 2016). Realised
2 shaking nevertheless varies nonrandomly within an exogenous earthquake: villages
3 near faults face higher expected exposure (Borusyak and Hull, 2023). Restricting
4 comparisons to the same broad hazard band addresses expected exposure, while
5 village fixed effects absorb fixed differences in fault proximity.

6 **Parallel trends.** Under the CS(2021) estimator with covariates, the relevant
7 assumption is *conditional* parallel trends: after accounting for observable pre-
8 treatment differences in growth and infrastructure trajectories, the potential un-
9 treated outcomes of eventually-treated cohorts evolve in parallel with those of
10 not-yet-treated villages (Sant’Anna and Zhao, 2020). We assess pre-trends in two
11 ways and they give a mixed verdict. Pre-treatment coefficients are cumulative
12 deviations from the omitted wave immediately before first exposure. The average is
13 small and insignificant for either outcome (0.8 percentage points, $p = 0.33$, on the
14 composite; 0.4 points, $p = 0.56$, on *warga*). A joint Wald test of all pre-treatment
15 group-time coefficients, which does not net offsetting signs against each other, is
16 less reassuring: it *rejects* for the composite ($\chi^2(15) = 42.9$, $p < 0.001$) and, less
17 strongly, for *warga* ($\chi^2(15) = 29.4$, $p = 0.014$).

18 Two features bound that rejection. It is specific to wave fixed effects, and
19 the residual drift runs against the finding rather than toward it, so it biases a
20 subsequent decline toward zero rather than manufacturing one. Appendix H sets
21 out both, with the caveat that a transitory elevation that would have reverted on
22 its own is not excluded.

23 We therefore treat conditional parallel trends as an assumption to probe rather
24 than assert, and do not read either pre-period as clean. Appendix M documents
25 the covariate trajectories being conditioned on (Appendix Table 25) and shows
26 that the residual growth imbalance is conservative for our finding, and we report
27 HonestDiD sensitivity bounds (Rambachan and Roth, 2023), under which the
28 composite sign survives only moderate relative-magnitude bounds: on both the
29 Sun–Abraham event study and the CS(2021) benchmark the identified set admits

1 zero by $\bar{M} = 0.5$ (Section 5.4.1). The always-treated 2005 cohort is excluded;
 2 identified cohorts enter between 2008 and 2024. In 2005 the composite uses the
 3 broader mass-brawl gate. We therefore check it against both the definition-stable
 4 *warga* series and the identical-wording gate. Both move in the same direction
 5 (Table S30). Figure 1 shows the event study on that preferred specification, and it
 6 is the exhibit on which a reader should form their own view of the pre-period. The
 7 five leads are individually indistinguishable from zero, the largest reaching $t = 1.4$,
 8 and they do not reject jointly ($\chi^2(5) = 4.67$, $p = 0.46$). The decline appears in the
 9 wave of exposure at 1.8 percentage points and holds at about two points for three
 10 waves before fading to zero by the fifth. We none the less do not rest the argument
 11 on this figure, for the reason just given: the same series does not pass under wave
 12 fixed effects, and a pre-test that one specification passes and another fails is weak
 13 evidence either way.

14 Figure 1 and the other event studies plot pointwise intervals. Simultaneous
 15 bands are about 35 percent wider (sup- t multiplier 2.65, versus pointwise 1.96).
 16 Under these bands all pre-exposure coefficients remain indistinguishable from zero.
 17 The effect three waves later becomes marginal, while the exposure-wave and first
 18 post-exposure effects remain significant (Appendix C).

19 Because the pre-period is not clean, we rest the composition result on the
 20 composition contrast, whose own identifying assumption we state in Section 5.2.1
 21 rather than treat as absent.

22 **No anticipation.** Seismic events at the frequency of PODES surveys (every 3
 23 years) are unforecastable. Villages cannot systematically adjust conflict behaviour
 24 in anticipation of a future earthquake.

25 **Measurement and reporting validity.** The outcome is reported by the village
 26 head on the PODES form, and the same official leads the local disaster response,
 27 so one might worry that a decline in *reported* communal conflict reflects reduced
 28 reporting capacity rather than reduced conflict. We treat this as a threat to

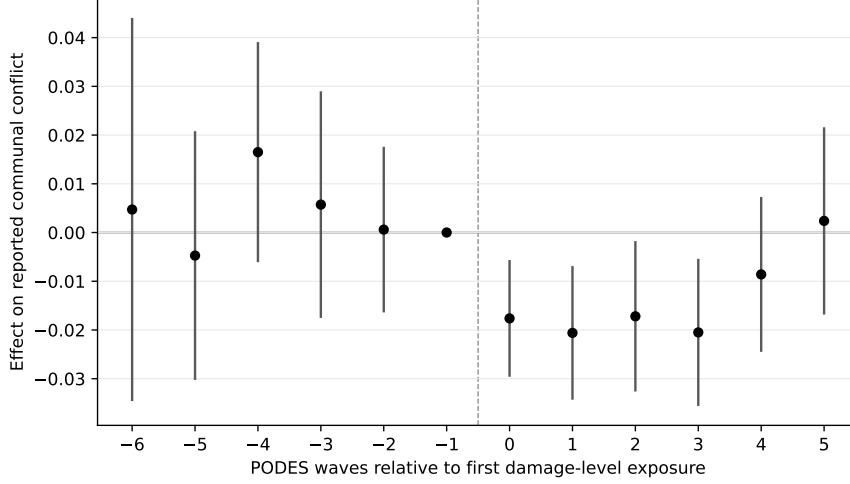


Figure 1: Event Study on the Preferred Specification

Note: Sun–Abraham (2021) interaction-weighted event study of first exposure at $\text{MMI} \geq \text{VI}$ on the composite communal-conflict indicator, with village and province-by-wave fixed effects and 95 percent *pointwise* confidence intervals clustered at the village. Bands valid for the entire path simultaneously are wider by the *sup-t* factor reported in Appendix C. Event time is in PODES waves relative to first exposure; the endpoint cells are binned so the omitted category is the wave before exposure alone rather than a mixture that includes post-treatment cells. The always-treated 2005 cohort is excluded, so every identified cohort is first exposed between 2008 and 2024. Appendix E reports the Callaway–Sant’Anna counterpart on the wave-fixed-effects panel, where the joint pre-trend test behaves differently, and Section 5.4.1 reports the sensitivity bounds that we, rather than this figure, treat as the binding constraint.

1 identification, not merely to interpretation. The central defense is behavioural:
2 theft is recorded by the same respondent on the same form, and it *rises* sharply while
3 communal conflict falls, a composition no uniform reporting shock can produce.
4 Section 5.2 develops this argument in full, adds three further observational checks,
5 and reports placebo items from the same form (village-head age and sex). Under
6 the preferred specification the sex placebo is a precise null and head age moves in
7 the direction opposite to the one a reporting story predicts.

8 **Resolving a pre-period signal.** An earlier calendar-year coding of first exposure
9 produced a marginally significant pre-period coefficient, traced to wave-boundary
10 timing that misplaced onset by one wave for about 21 percent of ever-treated
11 villages; re-timing exposure to the first wave enumerated after the event removes
12 it (Appendix H). The joint Wald rejection under wave fixed effects remains and
13 we carry it as an open caveat; the HonestDiD bounds (Section 5.4.1), not the
14 pre-trend tests, are the binding constraint.

1 **Pre-treatment balance.** In the 2005 KRB Medium/High sample, eventually-
2 treated and never-treated villages are close on nine of ten normalised differences,
3 each below the 0.25 threshold (Imbens and Wooldridge, 2009). The health-centre
4 count is the exception (0.423). Interacting it with wave moves every estimate by
5 less than one percent (Appendix Table 18). The imbalance is visible but it is not
6 doing the work. Remaining level differences are absorbed by village fixed effects,
7 and the pre-trend diagnostics of Appendix H address differential trends.

8 5 Results and Discussion

9 This section reports two things and keeps them distinct. Section 5.1 gives the effect
10 on the outcome the design intends, whether any mass brawl was reported, and the
11 evidence that it is identified: a flat near pre-exposure path under the preferred
12 specification, near-invariance to province-by-wave trends, and earthquake-level
13 inference under one of two weightings. Section 5.2.1 then asks what a decline
14 in reported mass brawls means, which the level alone cannot say, and answers
15 it with the composition of disorder across four acts recorded on the same form.
16 Section 5.3 sets out a first-time margin these data will not identify, and reports it
17 in full because the falsification it fails is instructive about what this design can
18 and cannot do. Section 5.4 takes up the threats.

19 5.1 The Effect on Reported Mass Brawls

20 Under the selected village and province-by-wave effects, first exposure at MMI
21 $\geq$ VI lowers reported mass brawls by 1.55 percentage points ($p = 0.0002$). The
22 pre-exposure mean among eventually exposed villages is 6.7 percent. That is a
23 decline of 23 percent relative to what these villages reported before they were
24 shaken.

25 Three features of Table 2 do the work. The first is the comparison across
26 measures. Under village and wave fixed effects the subtype composite gives the

Table 2: The Level Effect on Three Measures of Communal Conflict

The definition-stable gate is the measure least dependent on comparisons made across provinces, and the only one whose near pre-exposure path is flat.

	Mass-brawl gate (definition-stable)	Subtype composite (<code>conflict_any</code>)	Warga subtype
<i>Panel A. Effect of first exposure at $MMI \geq VI$</i>			
Village + wave FE	-0.0165 (0.0035)	-0.0199 (0.0033)	-0.0075 (0.0026)
Village + province $\times$ wave FE	-0.0155 (0.0042)	-0.0136 (0.0040)	-0.0025 (0.0032)
p	0.0002	0.0008	0.4418
Pre-exposure mean	0.0670	0.0624	0.0372
Effect as % of that mean	-23.2	-21.7	-6.7
% of wave-FE estimate retained	94.3	68.1	33.4
<i>Panel B. Gate in event time, endpoints binned, reference t_{-1}</i>			
$t_{\leq -5}$ (binned)	-0.0238 (0.0087)	-0.0070 (0.0098)	
t_{-4}	-0.0118 (0.0088)	-0.0033 (0.0095)	
t_{-3}	-0.0037 (0.0088)	-0.0086 (0.0099)	
t_{-2}	0.0120 (0.0066)	0.0048 (0.0079)	
t_0	-0.0215 (0.0054)	-0.0183 (0.0063)	
t_{+1}	-0.0195 (0.0057)	-0.0242 (0.0069)	
joint p , interior leads	0.067	0.594	
joint p , incl. binned	0.005	0.682	
Observations	303,457	303,457	303,457

The mass-brawl gate is the survey's own yes/no item on whether any mass brawl occurred in the village in the past twelve months, asked in identical terms in every wave from 2005 to 2024. The subtype composite is the union of the named communal subtypes, which coincides with the gate in 2005 and is progressively narrower thereafter, so it carries a definitional trend that wave effects absorb in level but not in interaction with treatment. Panel A reports the absorbing first-exposure effect under the two fixed-effect structures the paper compares; the row labelled % of wave-FE estimate retained is the ratio of the province-by-wave coefficient to the wave-FE coefficient, and is the quantity of interest for how much of each estimate is between-province rather than within. Panel B reports the gate in event time on the same sample with the always-treated cohort dropped, endpoints binned rather than discarded so that the reference period is t_{-1} alone. Standard errors clustered at the village.

larger decline, 2.0 percentage points against the gate's 1.7, which is what a definition that narrows over the panel mechanically delivers. Imposing province-by-wave effects reverses the ranking: the gate moves only to 1.6 points, retaining 94 percent of its estimate, while the composite falls to 1.4 points and retains 68 percent, and the *warga* subtype falls to 0.3 points and retains 33 percent, losing significance entirely. The concern that motivated the province-by-wave specification, that seismic zones sit disproportionately in provinces already on declining conflict trajectories, turns out to bite hard on the narrower measures and barely at all on the gate.

The second is the pre-exposure record. Panel B traces the outcome in event time with the endpoints binned rather than discarded, so the reference period is the wave before exposure alone. Under the preferred specification all four pre-exposure coefficients are individually insignificant, the joint test over the interior leads does not reject ($p = 0.59$), and the joint test including the binned far endpoint does not reject either ($p = 0.68$). The outcome then falls by 1.8 points in the wave of exposure and 2.4 points in the wave after. Under wave effects alone the same leads are jointly marginal ($p = 0.07$) and reject once the binned endpoint is included ($p = 0.005$), which is a further reason to prefer the province-by-wave counterfactual rather than a reason to doubt it.

The pre-exposure record remains inconclusive for two reasons. Non-rejection provides little reassurance here. At 2.802 times its standard error, the largest lead could detect only a pre-trend of 0.028 against an effect of 0.018. A violation would have to be half again as large as the effect to be detected (Roth, 2022). And the constraint this paper names as binding is not the pre-trend test but the relative-magnitudes bound. The gate is no more robust to this bound than the subtype composite. The identified set for its first post-exposure effect excludes zero at $\bar{M} = 0.25$ ($[-0.034, -0.003]$) but admits zero at $\bar{M} = 0.50$ ($[-0.039, +0.001]$). The composite also breaks down there; the aggregated target admits zero already at $\bar{M} = 0.25$. The gate buys definitional stability and near-immunity to province

1 trends. It does not weaken the parallel-trends requirement.

2 The third is the unit of inference, and here the result is partial. Treatment is
3 assigned by earthquakes, not by villages, so village clustering overstates precision.
4 We estimate effects across the 19 usable shocks and resample whole earthquakes.
5 The treated-village-weighted mean is a decline of 1.7 percentage points, with a
6 block-bootstrap 95 percent interval of $[-0.036, -0.000]$ and $p = 0.046$. It is negative
7 in 12 of 19 events, with a leave-one-out range of $[-0.023, -0.011]$. An equal-event
8 mean, which gives a shock that treated twenty villages the same weight as one that
9 treated two thousand, is a decline of 0.4 points and does not approach significance
10 ($p = 0.78$). The effect concerns the average exposed *village*; it is not detectable as
11 an average across *earthquakes* (Table 3).

Table 3: Earthquake-Level Inference on the Mass-Brawl Gate

The effect holds when earthquakes are weighted by the villages they treat, and is not detectable as an unweighted average across shocks.

	Mass-brawl gate
Treated-village-weighted mean	-0.0165
earthquake block-bootstrap SE	(0.0091)
95% CI	$[-0.0359, -0.0002]$
two-sided p	0.046
Equal-event mean	-0.0036
SE	(0.0125)
two-sided p	0.777
Leave-one-earthquake-out range	$[-0.0227, -0.0114]$
Earthquakes with a negative effect	12 of 19
Usable earthquakes	19

For each usable earthquake, defined as one whose first-treated cohort contains at least twenty in-sample villages and whose onset wave falls in 2008–2024, we estimate the absorbing treated-cohort effect on the definition-stable mass-brawl gate against never-treated controls, with village and wave fixed effects. The treated-village-weighted mean weights each earthquake by the number of villages it first treats; the equal-event mean weights every earthquake alike. The block bootstrap resamples whole earthquakes with replacement (9,999 draws, seed 42) and recomputes the weights in each draw, which is the correct unit of assignment and gives a materially larger standard error than village clustering.

12 Appendix E reports the estimator comparison, the full event study and the
13 specification-selection arithmetic. What the level does not tell us is why. A decline
14 in reported mass brawls is consistent with less fighting, with less reporting, and
15 with a general improvement in local order, and no single outcome can separate

1 them. That is what the next section is for.

2 **5.2 The Composition of Disorder: What One Shock Does** 3 **to Four Acts**

4 Why would one shock raise reported theft while communal fighting falls? This
5 section presents the paper’s central result, the composition of disorder: PODES
6 records not only communal violence but also village-level crime, and competing
7 theories of the earthquake–conflict link make sharply different predictions across
8 these outcomes. The evidence establishes which *margins* of disorder move together,
9 which is enough to discriminate among theories of the sign, but it does not *identify*
10 the underlying economic channel, because PODES measures neither welfare nor
11 labour-market outcomes. We therefore read these results as reduced-form patterns
12 consistent with a capacity account rather than as proof of it, and we flag direct
13 welfare measures, for example from the National Socio-Economic Survey (Susenas),
14 as what would convert that consistency into identification.

15 **5.2.1 The Composition Result: Communal Conflict Falls Where Crime** 16 **Does Not**

17 We estimate five outcomes on a single *common sample*: 43,351 villages over seven
18 waves, with the same absorbing treatment and no controls. Stacking outcomes
19 with outcome-interacted fixed effects accounts for the correlation of a village’s
20 outcomes in the standard errors (Table 4).

21 Unlike the level specifications, which exclude the cohort first exposed in 2005,
22 the composition sample retains it, because it is the sample the crime module
23 defines; dropping that cohort moves the estimates in the direction that favours our
24 reading rather than against it (Appendix J).

25 Like the communal-conflict outcome, the crime outcomes are constructed from
26 wave-specific PODES crime-module items. Section 3.2 gives the item for each
27 wave and Appendix S13 the verification; a change in an item’s wording is common

1 to every village in that wave, so it is absorbed by the outcome-specific wave
2 effects rather than left in the contrast. What the design cannot absorb is a
3 treatment-correlated shift specific to one item, which we take up next.

Table 4: Composition of Disorder on a Common Sample

Same villages, waves, and specification for all five outcomes; a formal test rejects equal treatment effects across outcomes, including against the definition-stable warga outcome.

	(1) Communal	(2) Theft	(3) Robbery	(4) Assault	(5) Warga
Absorbing D_{treat}	-0.0218*** (0.0034)	0.0429*** (0.0080)	-0.0141*** (0.0035)	-0.0008 (0.0045)	-0.0085*** (0.0027)
FDR q (Benjamini-Hochberg)	0.000	0.000	0.000	0.864	0.002
Baseline mean	0.026	0.418	0.031	0.055	0.016
Observations	300,018	300,018	300,018	300,018	300,018

Each column is a separate absorbing regression (village and wave fixed effects, no controls) of the indicated outcome on window-aligned exposure D_{treat} , on the common sample of all seven crime-module waves (2005–2024), restricted to village-waves where all five outcomes are non-missing. Communal is the collective violent composite; Theft is individual economic crime; Robbery is property crime with violence; Assault is individual violent crime; Warga is the civilian-versus-civilian communal subtype whose survey definition is stable across all waves. The Westfall–Young family-wise correction is not reported: it requires a separate bootstrap run and is available on request. FDR q is the Benjamini–Hochberg false-discovery-rate value computed on the five per-outcome p-values. Formal cross-outcome composition tests, village-clustered, from the outcome-interacted stacked model on this same sample: theft vs. communal $F = 57.70$ $p = 0.000$; theft vs. warga $F = 38.78$ $p = 0.000$; theft vs. assault $F = 26.47$ $p = 0.000$; theft vs. robbery $F = 48.83$ $p = 0.000$, the comparison the framework signs. Kabupaten-clustered robustness: theft vs. communal $F = 7.07$ $p = 0.008$; theft vs. warga $F = 5.04$ $p = 0.025$; theft vs. robbery $F = 10.89$ $p = 0.001$. Standard errors clustered at the village level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4 The contrast requires parallel trends in the theft-minus-communal difference and
5 a reporting response common to the two items, and we test the first directly rather
6 than assert it. Under the preferred specification the joint test over the interior
7 pre-exposure leads does not reject ($p = 0.31$; $p = 0.18$ against the definition-stable
8 series), and the two leads nearest exposure are precise nulls. Under wave effects
9 the same test rejects ($p = 0.018$). One term does not settle down in either: the
10 binned coefficient on periods five waves and more before exposure is 4.5 percentage
11 points and individually significant, so a joint test that includes it sits at $p = 0.06$.
12 Appendix Table 20 reports both tests, including the binned distant endpoint.
13 Appendix J states the assumption, the three threats that survive the differencing,
14 and what bounds each of them.

On identical data, communal conflict falls (-0.0218^{***}) while theft does not follow it down and instead rises, by 4.3 percentage points: the two move in opposite directions (Figure 2, Panel A), and a test of their equality is decisively rejected ($p < 0.001$), on a gap of 6.5 percentage points. What we lean on is this composition contrast, and within it the communal decline, which is the half that survives every specification we report.

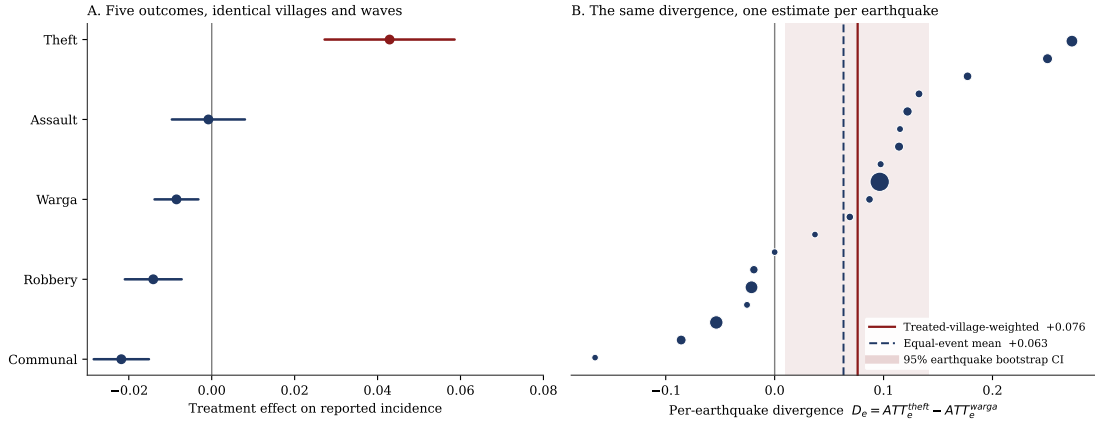


Figure 2: The Composition of Disorder

Communal violence falls where acquisitive crime does not, and the divergence holds when the earthquake, rather than the village, is the unit of inference.

Note: Panel A plots the absorbing treatment effect on each of five outcomes estimated on the single common sample described in this section, with village and wave fixed effects, no covariates, and 95 percent village-clustered confidence intervals, so a coefficient of -0.022 is a fall of 2.2 percentage points. Theft is drawn in maroon. Panel B plots the per-earthquake divergence $D_e = ATT_e^{theft} - ATT_e^{warga}$ for each of the 19 usable earthquakes, sorted by magnitude, with marker area proportional to the earthquake’s number of unique first-treated villages. The solid maroon line is the treated-village-weighted mean, the dashed navy line the equal-event mean, and the shaded band the 95 percent earthquake block-bootstrap interval for the treated-village-weighted estimate (9,999 draws, seed 42). Individual earthquakes are not labelled here because the common-sample restriction changes the treated counts; the named per-event estimates for communal conflict appear in Figure 3 and the full event list in Supplementary Material Section S7.

Because the composite changes definition in 2008, we also anchor the contrast on *warga*, the civilian-versus-civilian subtype with near-stable wording. It falls by 0.9 percentage points ($p = 0.002$). Equality with theft is rejected ($p < 0.001$), including under kabupaten clustering. The result thus holds beyond the composite and its 2008 definition change (Appendix J).

The same holds when the communal outcome is the mass-brawl gate that Section 5.1 makes primary. On the identical common sample, the gate falls by 1.8 percentage points under wave effects and by 1.9 points under province-by-wave

1 effects, both significant at 1 percent. Theft is 4.3 and 1.2 points, respectively.
2 Equality is rejected at $F = 50.8$ under wave effects and $F = 10.0$ ($p = 0.002$) under
3 the preferred specification, a sharper rejection than the composite delivers. The
4 contrast is therefore not an artefact of which communal measure is used: it holds
5 on the gate, on the subtype composite, and on the *warga* subtype.

6 The outcomes have very different base rates, so a linear equality test cannot
7 rule out a common proportional disturbance. If every act moved by the same
8 percentage, theft on a 42 percent base would move by far more percentage points
9 than communal conflict on a 2.6 percent base. We therefore re-estimate the same
10 stacked specification as a fixed-effects Poisson model, in which coefficients are
11 proportional responses. The divergence survives the change of scale. Communal
12 conflict falls by 31 percent while theft rises by 14 percent. Equality is rejected
13 ($p < 0.001$), including against the definition-stable series ($p = 0.003$). All three
14 of the equality contrasts we report survive kabupaten clustering, and correcting
15 the five per-outcome p -values for multiple testing leaves the communal, *warga* and
16 robbery declines and theft's rise significant at 5 percent, with assault the single
17 null ($q = 0.864$; Table 4). Appendix J reports the full proportional estimates, the
18 robbery and assault coefficients, and the clustered and corrected variants.

19 Estimated outcome by outcome under the CS(2021) benchmark, the same
20 gradient appears (Appendix Table 23): theft rises by 7.6 percentage points, robbery
21 does not rise, assault is null and communal conflict falls. Theft's own pre-trend fails
22 (4.2 percentage points, $p = 0.04$). Theft can also rise from weaker guardianship,
23 abandoned dwellings, aid inflows, or changed reporting. The benchmark thus
24 supports the *contrast* with communal conflict rather than a claim about theft's
25 level (Appendix J).

26 Three inferences follow. First, the decline is not part of a general improvement
27 in local order, since property crime moves sharply the other way. Second, the
28 asymmetry is what a single degraded input predicts. Theft is a stealth act that a
29 lapsed night watch stops deterring; mass fighting is a coordinated act that the same

lapsed capacity stops enabling; and police-post presence does not rise. The two therefore part company without any change in the motive for predation. Studies of income shocks find the same asymmetry (Mehlum, Miguel, and Torvik, 2006; Blakeslee and Fishman, 2018; Bignon, Caroli, and Galbiati, 2017; Hardiman, 2025), though we cannot separate that reading from a strengthening economic motive, because household welfare is unobserved here. Third, with individual violence flat while theft rises, the decline is concentrated in *organised, collective* violence, which requires assembly and arenas of inter-group contact (Ray and Esteban, 2017; Esteban, Mayoral, and Ray, 2012; Yanagizawa-Drott, 2014). Assault, the closest individual analogue of fighting, does not move, but the framework leaves it unsigned, so that null is uninformative. Robbery is the margin that discriminates. A raised-motive story predicts that it rises alongside theft. It does not. The framework signs the theft–robbery comparison but leaves robbery’s level to the data. Their effects differ at $F = 48.8$ ($p < 0.001$) under village clustering and $F = 10.9$ ($p = 0.001$) under kabupaten clustering. The latter is the strongest pairwise contrast in Table 4. They also differ on the proportional scale ($p = 0.005$), relevant given base rates of 0.031 for robbery and 0.418 for theft. The pattern therefore does not require a decline in the motive for conflict; the reading it selects is disruption of the social technology of mass fighting. We do not read robbery’s level as a causal estimate, any more than theft’s, because its own pre-trend rejects (Appendix J).

The contrast under the preferred specification. Section 4.1 prefers province-by-wave fixed effects for the level outcome, and consistency requires applying that standard to the contrast rather than only to the level. We do so. We replace outcome $\times$ wave effects with outcome $\times$ province $\times$ wave effects in the stacked model. The communal decline attenuates from 2.2 to 1.7 percentage points and remains significant at 1 percent. Theft’s rise falls from 4.3 to 1.2 points and is no longer distinguishable from zero. Table 5 reports all five outcomes and every equality test under both specifications and under the proportional model.

1 The equality tests survive. On the linear scale, theft and communal conflict
2 differ at $F = 8.61$ ($p = 0.003$). They also differ proportionally ($p < 0.001$):
3 communal conflict falls by 33 percent of its base rate while theft moves 5.7 percent
4 in the opposite direction. Against the definition-stable *warga* series the linear
5 test sits just outside 5 percent ($p = 0.061$) and the proportional test outside it
6 ($p = 0.104$), which is the weakest of the three comparisons and we do not present it
7 otherwise. The framework signs the theft–robbery comparison. It rejects equality
8 on the linear scale ($F = 4.39$, $p = 0.036$), but not on the proportional scale
9 ($p = 0.143$).

10 The preferred specification narrows the wave-effects result. What province-
11 by-wave effects remove is theft’s own rise, not the communal decline: the decline
12 attenuates by about a fifth and stays significant, while theft’s coefficient loses three
13 quarters of its size and all of its precision. The divergence therefore continues to
14 be rejected, but it is carried by communal violence falling rather than by theft
15 climbing. A substantial part of theft’s movement is between-province rather than
16 within, which is what one would expect of a shock whose footprint is regional,
17 but which we cannot separate here from province-specific trends in reporting or in
18 acquisitive crime. Nothing in the mechanism argument below requires theft’s level,
19 only that it fails to fall with communal violence, and that much holds under every
20 specification we report.

21 **The divergence at the earthquake level.** Because treatment is assigned
22 by earthquakes, we also take the divergence to the shock level, the honest unit
23 given the concentration of the communal effect in a minority of earthquakes
24 documented in Appendix C. For each of the 19 earthquakes we estimate the
25 treated-cohort effect on theft and on *warga* and form the per-event divergence
26 D_e , the difference between the two treatment effects for earthquake e (Table 22).
27 The treated-village-weighted mean divergence is 7.6 percentage points. Unlike
28 the communal decline, it is significant under earthquake-level inference (Figure 2,
29 Panel B). A bootstrap resampling whole shocks returns a 95 percent interval of

Table 5: The Composition Contrast Under the Preferred Specification
One common sample, all five outcomes stacked; the theft-versus-robbery comparison the framework signs is the one that survives.

	Linear		Proportional
	(1)	(2)	(3)
	Wave	Province $\times$ wave	Province $\times$ wave
<i>Panel A. Effect of first exposure, outcome by outcome</i>			
Communal conflict	-0.0218*** (0.0034)	-0.0169*** (0.0041)	-0.3345*** (0.1047)
Theft	0.0429*** (0.0080)	0.0123 (0.0093)	0.0568** (0.0231)
Robbery	-0.0141*** (0.0035)	-0.0075** (0.0037)	-0.1363 (0.1330)
Assault	-0.0008 (0.0045)	-0.0044 (0.0053)	-0.0148 (0.0764)
Warga conflict	-0.0085*** (0.0027)	-0.0057* (0.0033)	-0.1629 (0.1348)
<i>Panel B. Equality of treatment effects across outcomes</i>			
Theft = communal, p	0.000	0.003	0.000
Theft = warga, p	0.000	0.061	0.104
Theft = robbery, p	0.000	0.036	0.143
Observations	1,500,090	1,500,090	457,943

All five outcomes are stacked and interacted with first exposure so that each column is one regression, with outcome-by-village effects throughout. Columns (1) and (2) run on one common sample of 43,351 villages across 28 provinces and five waves. Column (3) does NOT: the Poisson estimator drops any outcome-village group whose outcome is zero in every wave, and because the base rates differ by more than an order of magnitude, theft retains almost every village while the rarer communal series retain only those with at least one recorded episode. The proportional equality tests in column (3) therefore compare outcomes measured on different village sets, which is why we read them alongside the linear tests rather than in place of them. Columns (1) and (2) differ only in whether the time effects are outcome-by-wave or outcome-by-province-by-wave; column (3) repeats column (2) as a Poisson model, so its coefficients are proportional rather than in percentage points. Panel B tests equality of the treatment effect between theft and each of the other outcomes within the same regression. The theft-versus-robbery comparison is the one the framework signs, since both are property offences recorded on the same form. Standard errors are clustered at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

1 $[+0.010, +0.142]$ ($p = 0.025$). The leave-one-out range is $[+0.057, +0.095]$. The
 2 divergence is positive in 12 of the 19 events, and the equal-event mean is positive
 3 and significant in its own right ($0.063, p = 0.023$), so the result does not depend on
 4 weighting toward the larger earthquakes. The compositional shift, in other words,
 5 is the part of the result that holds up at the level of the shock itself under either
 6 weighting, even where the communal decline alone is only marginal and, on the
 7 equal-event estimand, absent. The *warga* response is negative and separated from
 8 zero at the shock level. On the full *warga* sample its treated-village-weighted effect
 9 is a decline of 1.1 points, with an earthquake block-bootstrap 95 percent interval of
 10 $[-0.0260, -0.0007]$. Its failure to rise is therefore informative rather than a mere
 11 failure to reject (Appendix J).

12 **The same divergence under the preferred counterfactual.** Correct infer-
 13 ence and a credible counterfactual are different requirements, and we impose them
 14 jointly rather than leave the gap open. We re-estimate divergence event by event
 15 using province-by-wave rather than wave effects. We also compare above-threshold
 16 villages with those that felt the same shock below the damage threshold. Diver-
 17 gence remains positive but insignificant in both designs: the weighted mean falls
 18 from 7.6 to 1.6 percentage points, and the within-earthquake estimate is 2.2 points,
 19 neither distinguishable from zero. The same within-earthquake design returns a
 20 pre-period placebo of 3.1 points ($p = 0.13$), which is larger than the estimate it is
 21 meant to validate. We therefore do not count that placebo as a pass, and we apply
 22 the same standard here that Section 5.3 applies to the first-time margin: a placebo
 23 of that size leaves the within-earthquake contrast uninformative in either direction
 24 rather than confirmed. The attenuation limits the composition result: both the
 25 point estimates and their precision fall. What survives at the shock level is the
 26 between-earthquake divergence under wave effects, reported above; what does
 27 not is the within-footprint intensity contrast. Appendix J works the comparison
 28 through specification by specification, including the province-by-wave panel tests,
 29 the over-absorption arithmetic, and the cohort composition behind the estimates.

1 This is outcome discrimination, not mediation: the crime outcomes are alter-
2 native responses to the same shock, and their signs select among theories instead
3 of decomposing the treatment effect. We read it as a shift in the *composition*
4 of village-level social disorder, not as individual-level substitution, since PODES
5 records incidence at the village level and cannot track perpetrators. Accordingly
6 we estimate no channel-specific mediation shares, and we do not condition on theft
7 in the conflict regression, since theft is itself a post-treatment outcome that would
8 act as a bad control.

9 **Could this be a reporting artifact?** All four outcomes are reported by the
10 same village head on the same PODES form, so one must ask whether post-
11 earthquake administrative disruption suppresses *reporting* of communal conflict
12 rather than conflict itself. Four features of the data argue against it. The items
13 sit in the same security module with the same respondent and format yet move
14 in opposite directions, where a disruption to reporting capacity would depress
15 them together. Under-reporting depresses items, it does not inflate them, and
16 theft rises on the same form in the same wave. Salience runs against the artifact,
17 since a mass brawl is more memorable than diffuse petty theft, so degraded recall
18 would understate the theft increase. And the conflict decline is negative at all six
19 post-treatment horizons and significant as late as t_{+3} , well beyond any plausible
20 horizon of administrative disruption. The ACLED cross-check adds an independent
21 media-based lens whose incident-level reading does not show the broad masked
22 rise in communal fighting the artifact story requires (Appendix S14). A reporting
23 story consistent with all four facts would have to be as specific as the behavioural
24 one it replaces.

25 Two placebo items appear on the same form from the same respondent: the
26 village head's age and a female-head indicator. Under the causal estimator, the
27 latter is a precise null. Head age moves significantly, but opposite to the direction
28 a reporting explanation predicts (Appendix G).

29 The channels through which a shock can move each outcome, and why the

1 framework does not sign them without the composition contrast, are set out in
2 Appendix C.

3 **5.2.2 Which Capacity Inputs Move: Communication, Access, and** 4 **Social Capital**

5 This subsection asks which of the framework’s inputs the shock moves, and we
6 state the evidentiary standard once rather than at each item. Where an input
7 can be used as a *pre-determined moderator*, the treatment is interacted with its
8 pre-exposure value and the estimate carries the same identifying assumption as
9 the main effect. Where an input is instead treated as an outcome in its own right,
10 that is a second causal claim requiring its own parallel trends, and none of the
11 input series PODES offers passes that test. We therefore lead with the moderator
12 evidence and report the outcome regressions as descriptive.

13 **Mobilisation capacity: the moderator test.** Strong mobile-phone signal
14 proxies contact and mobilisation capacity (Yudhistira et al., 2021). Mobile coverage
15 is associated with protest in Africa, as documented by Manacorda and Tesei (2020);
16 social media and protest in Russia are documented by Enikolopov, Makarin, and
17 Petrova (2020). In those settings it lowers coordination costs rather than simply
18 spreading information. If that is the operative channel, the decline should be
19 concentrated in villages that had signal to lose. It is. Table 6 interacts first
20 exposure with signal presence at the 2005 baseline, a value fixed before any
21 identified cohort is treated and held by 76.4 percent of villages. The interaction is
22 negative and significant on both outcomes and under both fixed-effects regimes.
23 In the preferred composite column the implied split is stark: among villages with
24 no baseline signal the effect is -0.0014 (standard error 0.0065), indistinguishable
25 from zero, against a decline of 2.2 percentage points among villages that had it.

26 Two features make this the strongest mechanism evidence in the paper. The
27 moderator is pre-determined, so the test does not require parallel trends in the
28 input itself, which is the standard route in this literature (Bai 2023; Amarasinghe

1 et al. 2026). And it survives the province-by-wave specification, which most of the
2 paper’s other margins do not.

3 It is not a decomposition. Panel B races signal against the other 2005 infras-
4 tructure inputs the design can measure. The signal interaction weakens when other
5 inputs enter. It reaches 10 percent significance on the composite (1.7 percentage
6 points, $p = 0.088$) but does not clear conventional levels on *warga* (1.3 points,
7 $p = 0.11$). The paved-road interaction independently predicts *warga* (1.6 points,
8 $p = 0.003$). More than one input is doing work, and signal is not cleanly separable
9 from road access. Neither a police post nor distance to the district capital mod-
10 erates the effect. Supplementary Material Section S2 reports the moderators the
11 design cannot cleanly separate.

12 **The same inputs as outcomes, and why they do not corroborate.** Treating
13 the same inputs as outcomes in their own right is a second causal claim, requiring
14 its own parallel trends, and we report the result rather than only the verdict.
15 Table 7 estimates each input on all seven waves under the preferred specification,
16 alongside the endpoint-binned test of its own pre-exposure leads. One input passes
17 that test, *gotong royong* participation, and it is a ceiling variable held by 97 percent
18 of villages that returns a precise null. Every input with a detectable effect fails it.

19 We could stop there, as earlier drafts did, and call these regressions descriptive.
20 That would understate the problem. The inputs that move most move in the
21 direction opposite to the account we favour. Places of worship and mosques *rise*
22 after exposure, by 3.7 and 3.1 percent, so the venues on which inter-group contact
23 depends are not being destroyed. The *linmas* count *rises*. Migrant workers *fall* by
24 a tenth, so more residents are present rather than fewer, which cuts against the
25 routine-activity reading in which emptied dwellings raise theft. And the sharpest
26 direct test of the guardianship version of the account fails outright, with theft
27 rising by more where no security post existed to lose.

28 One further input is worth its own sentence, because its null is precise rather
29 than uninformative. A referee may reasonably ask whether the decline reflects a

Table 6: Mobilisation Capacity: The Pre-Determined Signal Moderator

The decline is concentrated in villages that had mobile signal to lose, and the moderator is fixed before any identified cohort is treated.

	Any conflict		Warga conflict	
	(1)	(2)	(3)	(4)
<i>Panel A. First exposure interacted with 2005 mobile signal</i>				
First exposure	0.0030	-0.0014	0.0078	0.0058
(villages without 2005 signal)	(0.0062)	(0.0065)	(0.0049)	(0.0052)
First exposure $\times$ 2005 signal	-0.0307***	-0.0205***	-0.0203***	-0.0135**
	(0.0072)	(0.0074)	(0.0057)	(0.0058)
Implied effect, villages with signal	-0.0277***	-0.0219***	-0.0126***	-0.0077**
	(0.0038)	(0.0046)	(0.0030)	(0.0037)
Village fixed effects	Yes	Yes	Yes	Yes
Wave fixed effects	Yes	—	Yes	—
Province $\times$ wave fixed effects	—	Yes	—	Yes
Observations	232,778	232,778	232,778	232,778
<i>Panel B. Horse race against other 2005 infrastructure, province $\times$ wave fixed effects</i>				
First exposure $\times$ 2005 signal	-0.0205***	-0.0165*	-0.0135**	-0.0126
	(0.0074)	(0.0096)	(0.0058)	(0.0079)
$\times$ paved road	—	-0.0112	—	-0.0157***
		(0.0071)		(0.0052)
$\times$ police post	—	-0.0044	—	-0.0028
		(0.0103)		(0.0085)
$\times$ distance to district capital	—	0.0027	—	-0.0023
		(0.0038)		(0.0032)
Observations	232,778	210,294	232,778	210,294

The moderator is an indicator for mobile-phone signal recorded in the 2005 PODES wave, fixed before any identified cohort is treated and held by 76.4 percent of villages. Because it is pre-determined, the test does not require a parallel trend in the moderator itself. All columns absorb village fixed effects and cluster standard errors at the village. Panel A reports the effect of first exposure among villages without 2005 signal, its interaction with signal, and the implied effect among villages with signal. Panel B re-estimates the province-by-wave columns adding interactions of first exposure with three other 2005 inputs: an indicator for a paved road, an indicator for a village police post, and distance to the district capital standardised to mean zero and unit standard deviation. The signal indicator itself is absorbed by the village fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

1 surge of state capacity after the shock rather than a loss of the village's own. The
2 census records whether a village has a disaster early-warning system, in four waves
3 from 2014, and first exposure does not produce one: +0.009 (standard error 0.011)
4 against a base of 11.6 percent, with a pre-trend that passes ($p = 0.45$). Formal
5 disaster preparedness does not arrive in the wake of the earthquake, so that version
6 of the state-capacity story is not what is driving the result.

7 None of the remaining inputs can be read causally, because each fails its own
8 pre-trend, and some plainly reflect post-disaster reconstruction rather than a
9 response of social organisation. The direct input measures do not corroborate the
10 capacity reading. The composition evidence rests on the pattern across the four
11 acts and on the pre-determined moderator, and the input series are uninformative
12 or adverse. A reader who requires direct measurement of the channel should treat
13 this paper as not providing it.

Table 7: Capacity Inputs Treated as Outcomes

Only one input passes its own pre-exposure test, and the inputs that move most move against the account we favour.

	Waves obs.	Pre-exp. mean	Wave FE	Prov. $\times$ wave (preferred)	% of mean	Pre-trend p
Civil-defence members	7	16.702	0.1347 (0.1950)	0.4133 (0.2200)	2.5	0.000
Security post present	6	0.672	-0.0122 (0.0085)	-0.0099 (0.0099)	-1.5	0.000
Places of worship	7	7.146	0.3046 (0.0499)	0.2639 (0.0618)	3.7	0.000
Mosques	7	6.168	0.2297 (0.0418)	0.1919 (0.0493)	3.1	0.001
Migrant workers	7	74.164	-8.6067 (1.5645)	-7.6899 (1.3776)	-10.4	0.006
Agriculture main income	6	0.847	-0.0058 (0.0043)	-0.0157 (0.0048)	-1.9	0.000
Any <i>gotong royong</i>	6	0.969	-0.0021 (0.0032)	-0.0022 (0.0036)	-0.2	0.399
Disaster early-warning system	4	0.116	-0.0008 (0.0102)	0.0086 (0.0112)	7.4	0.448

Each row is a separate absorbing regression of the input on first exposure at $\text{MMI} \geq \text{VI}$, with village fixed effects and either wave or province-by-wave effects. % of mean expresses the preferred-specification coefficient against the pre-exposure mean among villages that are eventually exposed. The final column is the joint test of the pre-exposure leads from an endpoint-binned event study on the preferred specification, with the always-treated cohort dropped; treating an input as an outcome is a second causal claim, and a small p there means the series was already diverging before exposure and the coefficient should not be read causally. Standard errors clustered at the village.

Supplementary Material Section [S19](#) reports the moderator horse-race and the guardianship falsification in full. A PODES module on *gotong royong* corroborates the capacity reading on the three waves that carry it, with the power limits that implies, and heterogeneity by pre-treatment polarisation points the same way (Appendix [C](#)).

The contrast in event time. The falsification the first-time margin fails has to be put to the contrast this paper rests on, and it does better, but not cleanly. Under the preferred effects, the three interior pre-exposure coefficients of the theft-minus-communal difference are 2.7, -0.9 and -0.1 percentage points; their joint test does not reject ($p = 0.31$). The difference then rises to 2.5 percentage points in the exposure wave and 4.0 points afterward. The binned distant endpoint remains 4.5 points (Section [5.2.1](#)). That term suggests a pre-exposure level difference more than a trend. Appendix [J](#) reports the full event-time path under both specifications (Appendix Table [20](#)) and records how this table came to differ from earlier drafts.

5.3 A First-Time Margin This Design Cannot Identify

A matched counterfactual holds the shock itself fixed: for each earthquake, villages shaken at $\text{MMI} \geq \text{VI}$ are compared with villages in the same footprint that felt sub-damaging shaking (mean MMI in $[4, 6)$), so the design identifies the *incremental* effect of crossing the damage threshold, and we take same-earthquake-and-same-province as its primary counterfactual for the reason set out in Appendix [K](#). Splitting the matched sample by pre-period conflict is the informative step, and it cuts against the average. Among the roughly nine in ten treated villages with no conflict recorded in *any* earlier observed PODES wave, exposure is associated with a 0.8 percentage point rise in first recorded conflict ($p = 0.001$). The counterfactual onset rate is 2.2 percent among matched controls, making this an increase of about a third. With an earthquake bootstrap the estimate is 0.8 percentage points ($p = 0.017$), positive in 16 of 23 events. Requiring treated and comparison

1 villages to share a province gives 0.9 points ($p = 0.008$ with earthquake clustering).
2 The aggregate local decline instead loads on the minority with a prior conflict
3 history, where it reaches 6.2 points and is significant at 1 percent. Appendix K
4 gives the matched design and the composite estimate on the full matched sample
5 with its own pre-period, the choice between province restriction and province-
6 by-wave absorption, the classification rule behind “no prior conflict”, and the
7 differential-reversion caveat. The full battery for this margin, the longer look-back,
8 the local-geography calipers and their power calculation, the independent soil
9 measurements, and the version under the counterfactual the composition contrast
10 is held to, is in Supplementary Material Section S11. None of it rescues the margin,
11 and the reason is the test below.

12 **A falsification test the onset design does not pass.** Every placebo reported
13 so far tests the level or tests chance. None tests confounding of the onset comparison
14 itself, and that is the test this design most needs, because on the onset arm the
15 pre-exposure outcome is zero by construction. We run the same design inside
16 the pre-exposure period, moving exposure back one wave. A village is previously
17 peaceful if it reports no conflict through two waves before its earthquake. The
18 outcome is first recorded conflict in the last wave *before* the earthquake. Under
19 exchangeability this coefficient should be zero.

20 It is not. Villages about to be shaken are 1.5 percentage points more likely to
21 report a first episode in the pre-exposure wave ($p = 0.004$). The gap grows under
22 the local restrictions, to 1.6 points within a province and 1.9 points within 30 km
23 (Table 8). These are two to three times the effect we estimate at exposure. This
24 pre-exposure gap is the sharpest limit on the onset result.

25 We therefore report the margin and do not carry it. Three things point the
26 same way: the gap predates the earthquake, it does not widen when the earthquake
27 arrives, and it is confined to villages on firm ground (Supplementary Material
28 Section S11).

Table 8: Falsification: the Onset Design Run Before the Earthquake
Villages about to be shaken were already more likely to report a first episode, by more than the effect estimated at exposure.

Comparison set	Estimate	(SE)	p	p , quake
Same earthquake	0.0149***	(0.0051)	0.004	0.010
and same province	0.0164***	(0.0061)	0.007	0.030
and within 30 km	0.0186**	(0.0078)	0.017	0.084
<i>Panel B. Treated-minus-comparison gap, wave by wave</i>				
One wave before the earthquake (t_{-1})	0.0145	(0.0051)	0.005	
Wave of exposure (t_0)	0.0038	(0.0041)	0.347	
t_{+1}	0.0073	(0.0046)	0.116	
t_{+2}	0.0149	(0.0061)	0.015	
Change from t_{-1} to t_0	-0.0107	(0.0065)	0.101	

The same-earthquake matched design run entirely inside the pre-exposure period: exposure is moved back one wave, a village counts as previously peaceful if it reports no conflict through two waves before its earthquake, and the outcome is first recorded conflict in the last wave *before* the earthquake arrives. A positive coefficient would mean villages about to be shaken were already starting conflict sooner than their comparison villages, which is the confounding the onset design cannot rule out from its own pre-period. Panel B places that gap wave by wave on one sample and one regression, so the pre-exposure and post-exposure gaps are directly comparable: the last row asks whether the earthquake widens the gap it inherits. Stack-by-village and stack-by-wave effects throughout; the last column clusters on the earthquake.
 $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

1 5.4 Threats to Identification and Robustness

2 This section asks whether the level result survives the main threats to the design,
3 and is organised around those threats rather than around specifications. The
4 estimate stays negative and close to the benchmark across most margins, but not
5 under every weighting and unit of assignment, and we collect the exceptions in
6 the scope-and-limitations subsection below. Supplementary Material Section [S7](#)
7 reports every check in full; here we state the question, the answer, and the bearing
8 on the main claim. How many distinct shocks identify the effect, and what changes
9 when the earthquake rather than the village is the unit of inference, is set out in
10 Appendix [C](#).

11 5.4.1 Identification Threats

12 **Treatment-outcome timing.** PODES conflict at wave t covers the twelve
13 months before May enumeration, so an earthquake inside that window gives a
14 partial-exposure $t = 0$ outcome. An exact-date audit shows 74 percent of the

1 19 stacked earthquakes are clean-post, and every major earthquake carrying the
2 effect is clean-post; restricting to clean-post cohorts leaves the estimate essentially
3 unchanged (2.1 against 2.0 percentage points on the full sample) and *warga* at 0.7
4 points (Supplementary Material Section S7).

5 **Parallel trends.** Section 4 states the assumption and the evidence in full; the
6 residual drift under wave fixed effects is positive, so it biases a subsequent decline
7 toward zero rather than manufacturing one, and the sensitivity analysis below, not
8 the pre-trend tests, is the binding constraint.

9 **Is the effect a region-type artifact?** Village fixed effects absorb region levels;
10 the sharper test is region-specific trends. Province $\times$ wave effects attenuate the
11 no-controls full-panel absorbing effect on the subtype composite by about a third,
12 from 2.0 to 1.4 percentage points. It remains significant at 1 percent under
13 village clustering and at 5 percent under kabupaten clustering ($p = 0.035$). The
14 *warga* margin does not survive ($p = 0.44$). On the definition-stable gate the same
15 comparison attenuates by only six percent (Table 2), which is the reason Section 5.1
16 rests on the gate. Appendix D reports what happens when the same trends are
17 conditioned on rather than absorbed.

18 **Exposure measurement.** Restricting to village-waves with a measured rather
19 than imputed first lag leaves the composite at 1.5 percentage points ($p = 0.01$).
20 Bounding missing ShakeMap events suggests that at most 0.22 percent of never-
21 exposed in-sample villages could be misclassified, attenuating the estimate by four
22 percent (Supplementary Material Section S7).

23 **Functional form.** The estimate keeps its sign across four raster-to-village ag-
24 gregation rules and a five-percent area trim, and its significance except under
25 the pixel-maximum rule. Instrumental Peak Ground Acceleration reproduces it,
26 at 1.3 percentage points under Callaway–Sant’Anna and significant at 1 percent.
27 A conditional logit also yields the same sign and significance, relevant because

1 parallel trends in levels and after nonlinear transformation cannot generally both
2 hold (Roth and Sant’Anna, 2023) (Supplementary Material Section S7).

3 **Cohort dependence, placebos and randomisation inference.** No single
4 cohort drives the result: dropping each in turn leaves every estimate negative
5 and significant, ranging from 1.3 to 2.5 percentage points, though because two
6 of the largest events fall in the same cohort this is a leave-one-cohort-out and
7 not a leave-one-earthquake-out check. A forward-lag placebo is a null (0.3 points,
8 $p = 0.60$). Spatial randomisation relocates each earthquake footprint to a random
9 never-treated village. Only one of 500 placebo draws is at least as negative as the
10 actual estimate of 2.0 points (one-sided $p = 0.004$). Yet placebo dispersion is 2.1
11 times the clustered standard error, indicating that village clustering understates
12 uncertainty when earthquakes assign treatment. Applied to the compositional
13 divergence rather than to the level the same exercise is weaker, at a one-sided
14 $p = 0.044$, so both clear 5 percent and the divergence by the narrower margin
15 (Supplementary Material Section S7; Supplementary Material Table S15).

16 **HonestDiD sensitivity.** This is the binding constraint, not the pre-trend
17 tests. On the preferred specification, the composite identified set excludes zero at
18 $\bar{M} = 0.25$ ($[-0.034, -0.005]$) and admits it at $\bar{M} = 0.50$ ($[-0.039, -0.000]$). The
19 finer breakdown grid, 0.40 to 0.45, instead comes from the Sun–Abraham event
20 study under wave effects, not the preferred specification. The definition-stable
21 *warga* target breaks down at $\bar{M}$ between 0.05 and 0.10. That bound comes from
22 the Sun–Abraham event study, not the preferred level specification. Since $\bar{M}$ is not
23 comparable across estimators, it does not bound the level estimate (Appendix L).

24 Regional trends and the weight on forbidden comparisons, and the clustering
25 and spatial-structure choices, are treated in Appendix C.

1 5.4.2 Scope and Limitations

2 Several results are genuinely sensitive and we set them out together. HonestDiD
3 is the binding constraint (Section 5.4.1): the composite’s breakdown lies between
4 $\bar{M} = 0.25$ and 0.50 on the preferred specification, and the definition-stable *warga*
5 target breaks down an order of magnitude earlier. The specification choice itself
6 moves the magnitude only from -0.0213 to -0.0165 . Inference also depends
7 on the outcome and estimator. A province-level wild-cluster bootstrap gives
8 the composite $p = 0.030$ in the two-way benchmark, but *warga* does not clear
9 ($p = 0.215$). Clustering the CS(2021) influence function on 28 provinces yields
10 $p = 0.043$ for the pooled ATT. Thus the composite still clears 5 percent under
11 these conservative checks, while the definition-stable subtype does not. The level
12 also holds when earthquakes are weighted by the villages they treat ($p = 0.01$) but
13 not under an equal-event mean ($p = 0.34$).

14 The onset margin has four boundaries, and the first is the most serious: run
15 inside the pre-exposure period, the design returns 1.5 percentage points where it
16 should return zero, and more under the local restrictions (Table 8). The villages
17 driving it are excluded from the headline arm by construction, but exchangeability
18 on onset risk is the assumption the design needs and it does not hold where it can
19 be checked. Appendix K documents three further limits. The risk-set wild-cluster
20 bootstrap gives $p = 0.18$, versus $p = 0.017$ under earthquake-level aggregation.
21 Local comparisons by distance caliper or epicentral-distance bin remain positive
22 but imprecise; none can detect an effect of the estimated size (Tables S34 and S35).
23 Shear-wave velocity and the site-response proxy preserve the sign. Adding province-
24 by-wave effects to the matched design leaves 0.4 points ($p = 0.10$ with village
25 clustering; $p = 0.27$ with earthquake clustering). The onset level therefore does
26 not pass the rule applied to the level outcome in Section 4.1. Across all three
27 the contrast between margins is the firmer object than either level, clearing the
28 wild cluster bootstrap at $p = 0.026$ under the full observed history. The onset
29 sign, negative average, and composition pattern persist. The evidence supports

1 a decline in reported communal conflict after strong shaking within hazard-zone
2 villages. It does not establish a universal or mechanism-decomposed law: no
3 reversal appears, but only the first post-exposure horizon is precisely estimated.
4 Population displacement is real but far too small to carry the result; the evidence
5 is in Appendix C.

6 **Internal and external validity.** Identification rests on parallel trends condi-
7 tional on baseline covariates rather than on experimental assignment, and the sign
8 of the staggered level estimate survives relative-magnitudes violations of a quarter
9 of the largest observed pre-period deviation but not of a half. Three weaknesses
10 recur across Table 15. Conditional pre-trends pass only with province-by-wave
11 effects. Treatment comes from 19 to 24 earthquakes, so village clustering overstates
12 precision. The definition-stable *warga* outcome moves in the same direction but
13 reaches only 10 percent on the specification that passes the pre-trend test. We
14 therefore do not rest the paper on the average level effect. The paper’s claim
15 is the composition of disorder: the relative movement of four acts recorded by
16 one reporter on one form, identified by a contrast taken within the village. That
17 contrast is better behaved in the pre-exposure period than the first-time margin is,
18 with a flat near pre-period under the preferred specification, but it is not clean
19 either: a distant binned endpoint remains positive and significant, and we report
20 it rather than drop it.

21 The estimand is local and intent-to-treat. The sample is restricted to villages
22 inside Indonesia’s officially mapped medium and high seismic-hazard zones, and
23 within it the treated villages are more conflict-prone and faster-growing, reporting
24 communal conflict at 6.2 percent before exposure against a pooled 2.6 percent. The
25 estimate concerns first observed damage-level exposure among villages actually
26 struck in a seismically active setting. It does not imply the same effect for an
27 arbitrary Indonesian village or for settings with different conflict institutions.

1 6 Conclusion

2 A strong earthquake does not simply raise or lower communal conflict in an
3 Indonesian village. It moves several kinds of disorder at once, and the pattern of
4 that movement is what these data identify. On one common sample of village-waves,
5 first exposure at $\text{MMI} \geq \text{VI}$ lowers reported communal conflict by 1.7 percentage
6 points against a 2.6 percent base, about a third of it, while reported theft does
7 not fall with it and instead rises. Robbery and assault do not follow theft. Under
8 the preferred specification, theft and communal responses differ on the linear
9 ($p = 0.003$) and proportional ($p < 0.001$) scales. Against definition-stable *warga*,
10 equality rejects under wave effects but sits just outside 5 percent under preferred
11 effects.

12 That is a contrast measured within the village, on one form, by one reporter.
13 Whatever moves all four acts alike, fixed village characteristics and any common
14 reporting disturbance, drops out. A differential trend in theft *relative* to communal
15 conflict does not drop out; the design assumes its absence and cannot guarantee it.
16 Under the preferred specification, the near pre-exposure period is flat. The two
17 closest leads are precise nulls, and the interior leads do not jointly reject. At distant
18 horizons a level difference remains, but one cohort supplies most of that cell. The
19 first-time margin in the same data fails a sharper version of that test, returning
20 before exposure the gap it should return only after. The contrast also discriminates
21 among explanations. A uniform reporting shock would move all four acts together.
22 A general collapse of order would raise all four. A rise in the motive for predation
23 would raise theft and robbery together. A loss of the organisational capacity that
24 collective violence needs, and stealth theft does not, would produce exactly the
25 pattern we observe. A pre-determined moderator agrees: the decline concentrates
26 in villages that had mobile-phone signal at the 2005 baseline, an interaction worth
27 2.1 percentage points that needs no parallel trend in the moderator and survives
28 the specification most other margins do not.

29 The aggregate decline is a result and we defend it, with its limit stated. On

1 the definition-stable measure it survives province-by-wave effects almost intact, its
 2 pre-exposure path shows no statistically detectable differential trend, and it clears
 3 earthquake-level inference when shocks are weighted by the villages they treat. It
 4 does not clear an unweighted average across shocks, so what we estimate is the
 5 effect on the average exposed village. The composition result carries two boundaries
 6 of its own. Theft's own rise is not separately identified once province-by-wave
 7 effects are imposed, so the divergence is carried by the communal decline rather
 8 than by theft climbing; nothing in the mechanism argument requires theft's level,
 9 only that it fails to fall. And at the shock level the divergence survives between
 10 earthquakes but not within a single earthquake's footprint, where the design is too
 11 coarse to detect an effect of the size it produces. A first-time-conflict margin that
 12 these data suggest fails its own falsification: run inside the pre-exposure period,
 13 the design returns the gap it should not, and the earthquake does not widen what
 14 it inherits. The margin and the test that undoes it appear together in Section 5.3.

15 Every sign we estimate is consistent with two consequences of one shock. On the
 16 reading we favour, a strong earthquake degrades the village's capacity for organised
 17 collective action, and that capacity was doing two jobs at once: it supplied the
 18 mobilisation communal violence requires, and the routine guardianship, the night-
 19 watch rotations, that deters property crime. Remove it and coordination-intensive
 20 violence falls while stealth theft rises, whereas acts that require an encounter do not:
 21 robbery does not follow theft up, falling by 1.4 percentage points on the common
 22 sample, and assault does not move. That asymmetry is the part a rising-motive
 23 story alone does not deliver, since with encounters and guardianship unchanged
 24 a stronger motive would raise theft and robbery together. A second, narrower
 25 piece of direct evidence points the same way, on the narrow footing the data allow:
 26 high-involvement public-purpose *gotong royong* falls by 3.1 to 3.2 points while
 27 emergency mutual aid does not. We do not observe household income, so the theft
 28 rise is consistent with lapsed guardianship, with economic stress, or with both, and
 29 the sharpest test our data permit of the guardianship reading fails (Section 5.2.2).

1 What we can say is that no change in motive is needed to generate the pattern we
2 find.

3 Two limits of scope remain. The identified cohorts enter treatment between
4 2008 and 2024, so villages already exposed before the panel opens are not estimated.
5 And the crime comparison discriminates among mechanisms without decomposing
6 the effect into income, guardianship, aid and mobilisation shares; direct household
7 evidence on welfare and labour-market margins would turn the motive side of
8 the argument from consistency evidence into a measured channel. The estimates
9 come from rural and peri-urban villages in Indonesia’s medium and high seismic-
10 hazard zones, a setting with dense village-level social organisation and a state
11 that mobilises visibly after disasters. The production-function logic travels as
12 a *prediction*; the net sign does not necessarily travel. Displacement into camps
13 can concentrate rival groups, and weak or predatory state presence can favor
14 violence. Under either condition the framework predicts more violence, as in the
15 legitimacy-and-coordination channel through which earthquakes *raised* conflict in
16 imperial China (Bai, 2023).

17 Two implications follow, and the design’s limits keep both modest. The first
18 concerns the post-disaster window, when governments, security forces, and aid
19 agencies divide scarce resources between reconstruction and conflict prevention.
20 A quiet communal-conflict count in that window is not reassurance. The decline
21 in the count concentrates where conflict was already established, and it fits a loss
22 of the capacity to organise better than an easing of grievances. Latent tensions
23 could therefore resurface as external monitoring and reconstruction wind down, a
24 dynamic our design does not test. The first-time margin yields no implication. Its
25 design returns a positive coefficient, but the gap it reports is already present before
26 the earthquake arrives, so it cannot tell a planner where a first episode becomes
27 more likely. The second implication is for research. An aggregate disaster-conflict
28 estimate is an average over places with and without a conflict history, and its sign
29 can move with that mixture. Separating onset from continuation may well be the

1 right question for this literature. Our evidence shows how demanding that question
2 is: the design that appears to answer it fails a falsification that the composition
3 contrast, run on the same villages and the same waves, survives.

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1

Online Appendix

2

Earthquakes and Communal Conflict: Evidence from Indonesia's Seismic Zones

3

Not for print; available online and in the replication package.

A Staggered DiD: Group-Level ATTs

Cohort heterogeneity

Group-level heterogeneity. No single cohort drives the effect: five of six estimable cohort ATTs are negative, and the one positive cohort, G6, is predominantly Aceh villages beginning from a near-zero floor less than a decade past the Helsinki peace agreement. Dropping G6, or Aceh altogether, leaves the estimate unchanged or larger (-0.0203^{***} ; Appendix Tables 9 and S7).

Table 9: Callaway–Sant’Anna Group-Level ATT(g)

Takeaway: five of the six cohorts have negative ATTs; the only positive cohort is G6, the Aceh Pidie Jaya cohort, whose pre-treatment conflict rate sits at a post-conflict floor and can only revert upward.

Cohort	First treated wave	ATT	SE	z	p
<i>Panel A: Composite (any communal conflict)</i>					
G3	Wave 3 (2008)	−0.0151	0.0082	−1.85	0.065
G4	Wave 4 (2011)	−0.0121	0.0070	−1.73	0.084
G5	Wave 5 (2014)	−0.1281	0.0704	−1.82	0.069
G6	Wave 6 (2018)	+0.0323	0.0121	+2.66	0.008
G7	Wave 7 (2021)	−0.0527	0.0118	−4.47	0.000
G8	Wave 8 (2024)	−0.0244	0.0174	−1.40	0.160
Simple ATT	(all cohorts)	−0.0177	0.0047	−3.81	0.000
<i>Panel B: Warga (definition-consistent)</i>					
G3	Wave 3 (2008)	−0.0067	0.0070	−0.96	0.336
G4	Wave 4 (2011)	−0.0056	0.0054	−1.05	0.293
G5	Wave 5 (2014)	−0.0846	0.0569	−1.49	0.137
G6	Wave 6 (2018)	+0.0252	0.0097	+2.60	0.009
G7	Wave 7 (2021)	−0.0202	0.0088	−2.30	0.022
G8	Wave 8 (2024)	−0.0182	0.0166	−1.10	0.271
Simple ATT	(all cohorts)	−0.0075	0.0037	−2.04	0.042

Group-time ATTs from Callaway–Sant’Anna with baseline PODES 2003 covariates, doubly-robust weighting and not-yet-treated controls. G6, the Aceh Pidie Jaya cohort, is the only positive-ATT group; Supplementary Material Section S7 shows that dropping it leaves the pooled estimate essentially unchanged. G5 contains 37 treated villages and its estimate is correspondingly imprecise.

B Data and Exposure Detail

This appendix carries the measurement detail that Section 3 states in brief: the incidence of the four acts by wave, the column conventions of the crime module,

- 1 why exposure is a threshold rather than a dose, and the validation of the pre-2013
- 2 ShakeMap reconstructions against the village's own earthquake report.

3 Incidence of the four acts by wave

Table 10: Incidence of the Four Acts of Disorder, by Wave

All four are recorded by the same official, on the same form, over the same recall window; theft is an order of magnitude more common than the rest.

Wave	Communal conflict	<i>Warga</i> subtype	Theft	Robbery	Assault
2005	0.024	0.018	0.425	0.035	0.054
2008	0.027	0.016	0.474	0.037	0.074
2011	0.029	0.017	0.402	0.032	0.062
2014	0.032	0.018	0.465	0.041	0.059
2018	0.032	0.021	0.505	0.038	0.073
2021	0.017	0.011	0.306	0.022	0.033
2024	0.019	0.012	0.353	0.012	0.034
All waves	0.026	0.016	0.418	0.031	0.055

Share of village-waves reporting each act in the preceding twelve months, on the composition common sample: village-waves inside the medium and high seismic-hazard zones for which all five outcomes are non-missing. Communal conflict is the composite mass-brawl measure; *warga* is the civilian-versus-civilian subtype whose wording is near-stable across waves; theft, robbery and assault are items 01, 02 and 04 of the same security module. $N = 300,018$ village-waves, 43351 villages.

- 4 Table 10 reports incidence by wave. Theft is common, at roughly two villages
- 5 in five, while robbery and assault are rare, at three and five percent, and communal
- 6 conflict rarer still, at between two and three percent. The gap in base rates is the
- 7 reason Section 5.2.1 reports the composition contrast on a proportional scale as
- 8 well as a linear one: a shock that moved every act by the same proportion would
- 9 move theft by far more percentage points than communal conflict, so a linear test
- 10 of equal effects is not by itself evidence against a common proportional disturbance.
- 11 All four series decline after 2018, which the wave effects absorb.

12 Village boundary harmonisation

- 13 The 2005-boundary panel maps every village in each wave back to its 2005 parent
- 14 village. It is built from BPS Master File Desa records and supplemented by
- 15 GPS spatial joins and manual verification where name matching fails. Because

1 villages created by later pemekaran are mapped back to their 2005 parent polygon,
2 administrative splits cannot mechanically enter or exit the sample. Appendix [S13](#)
3 documents the procedure, match rates by wave, and two crosswalk-robustness
4 re-estimations.

5 **Column conventions in the crime module**

6 The questionnaire item numbers of the crime module are stable across the seven
7 waves, but the column conventions in the released microdata are not. Two waves
8 interleave the occurrence column with a follow-up asking whether the act became
9 more or less frequent than the year before, so the position of an item in the data
10 file is not its position on the form. Appendix [S13](#) gives the exact variable name
11 used for each outcome in each wave, together with the cross-tabulation against the
12 follow-up item that verifies each mapping.

13 **Why a threshold and not a dose**

14 Modified Mercalli Intensity is an ordinal scale, and that governs what can be done
15 with it. Its categories are defined by what people feel and what buildings do, not
16 by a measured quantity, and the instrumental relations that map ground motion
17 onto it are logarithmic: [Wald et al. \(1999\)](#) estimate $\text{MMI} = 3.66 \log_{10}(\text{PGA}) - 1.66$
18 over $\text{V} \leq \text{MMI} \leq \text{VIII}$, so one step on the scale corresponds to a factor of about
19 1.9 in peak ground acceleration rather than to a fixed increment. The scale does
20 not even rest on one physical quantity throughout its range: below about MMI
21 VII it tracks acceleration and felt reports, above it velocity and damage, which is
22 why [Wald et al. \(1999\)](#) recommend different relations on either side.

23 A coefficient on MMI entered linearly would therefore assert that a move from
24 IV to V means the same thing as a move from VIII to IX, which is false in ground
25 motion and unsupported in damage. We do not make that assertion. Treatment is
26 a threshold, $\mathbf{1}\{\text{MMI} \geq \text{VI}\}$, which is an operation an ordinal scale admits, and the
27 intensity contrasts we report compare bands within the same earthquake rather

1 than fitting a slope in MMI.

2 One step in our construction is nonetheless cardinal, and we flag it rather than
3 let it pass. Averaging intensity across the raster cells inside a village treats MMI as
4 though differences were comparable, which by the logarithmic relation above they
5 are not: the average of MMI V and MMI VII understates the intensity implied by
6 the average acceleration. Three things bound the consequence. The bias grows
7 with within-village dispersion, which is small for the village polygons here; the
8 estimate keeps its sign under the pixel maximum, the median and a representative
9 point as well as the mean; and an exposure rebuilt from Peak Ground Acceleration,
10 which is cardinal, reproduces the result at -0.0129 (standard error 0.0039).

11 **Validating the reconstructed ShakeMap products**

12 Section 3 notes that USGS ShakeMap products for Indonesian events before 2013
13 are later reconstructions built from modelled ground-motion relations rather than
14 recorded intensities. The usual response is to concede the point and restrict the
15 sample. We can do better, because the village census contains an independent
16 check that we had not previously used. From 2008 onward PODES asks each
17 village, in its disaster block, whether an earthquake occurred there over the recall
18 window. That is the village head's own report, collected by a different agency for
19 a different purpose, and it is available for the same village-waves as the outcome.

20 Table 11 asks whether the ShakeMap treatment predicts it. It does, strongly
21 and in both eras. Villages whose ShakeMap intensity reaches $\text{MMI} \geq \text{VI}$ within the
22 window are 34 percentage points more likely to report an earthquake themselves
23 (standard error 0.7), against a base of 7 to 13 percent. Splitting at the provenance
24 break, the relationship is $+0.302$ (standard error 0.011) in the reconstructed era
25 and $+0.398$ (0.011) in the instrumental era. The difference is statistically clear,
26 and its size and direction are what greater measurement error would produce: the
27 correspondence is weaker in the reconstructed era by roughly a quarter.

Table 11: Does the ShakeMap Treatment Predict the Village’s Own Report?
An independent check on the pre-2013 reconstructed products, from the village census disaster block.

	Reconstructed 2008–2014	Instrumental 2018–2024	All waves
<i>Panel A. Raw agreement</i>			
P(village reports a quake ShakeMap $\text{MMI} \geq \text{VI}$)	0.422	0.517	
P(village reports a quake no ShakeMap $\text{MMI} \geq \text{VI}$)	0.069	0.126	
difference	0.352	0.391	
<i>Panel B. Village and wave fixed effects</i>			
ShakeMap $\text{MMI} \geq \text{VI}$ in the window	0.3019 (0.0107)	0.3979 (0.0114)	0.3411 (0.0072)
Observations	128,658	130,046	258,904
p , equality of the two eras		0.000	

The outcome is the village’s own report, in the PODES disaster block, that an earthquake occurred over the recall window; it is collected independently of USGS. The regressor is episodic ShakeMap exposure at $\text{MMI} \geq \text{VI}$ within the same window rather than the absorbing indicator, because the survey item asks about the window and not about history. USGS ShakeMap products for events before 2013 in this catalogue are model reconstructions built years afterwards from no station data; from 2013 they are instrumental. Columns one and two split the panel at that break. A relationship of similar size on both sides is evidence that the reconstructed products carry real information about which villages were shaken; a much weaker one on the left would quantify how much the early treatment is attenuated. Standard errors clustered at the village.

1 C Mechanism Narrative, Event Study, and De- 2 sign Detail

3 This appendix holds the material the main text summarises in one sentence
4 each: the channel narrative, the *gotong royong* corroboration, the polarisation
5 heterogeneity, the event study, the count of identifying shocks, regional trends, and
6 the clustering and spatial-structure choices.

7 Population displacement

8 If shaking drives residents out, the pool for collective violence shrinks and the
9 decline is mechanical rather than behavioural. We can test this only in part.
10 PODES records population and household counts through 2011 and not from 2014
11 onward, so the 2008 and 2011 cohorts can be checked while the later cohorts, which
12 supply most of the identifying variation, cannot. On the 130,052 village-waves

1 with counts, the point estimates do point that way, but neither is distinguishable
2 from zero: log population falls by -0.0072 (standard error 0.0060 , $p = 0.23$) and
3 log households by -0.0089 (0.0063 , $p = 0.16$), about 0.7 and 0.9 percent. That
4 residential location is the margin a disaster moves durably, while income recovers
5 faster than expected, is what [Deryugina, Kawano, and Levitt \(2018\)](#) document for
6 Hurricane Katrina, and it is why we read headcounts rather than income as the
7 first stage available to us.

8 Displacement is therefore not established on the one sample where we can look
9 for it, and even read at its point estimate it is far too small to carry the result.
10 On the same restricted sample the communal-conflict effect is -0.0147 against a
11 5.1 percent pre-exposure rate, a fall of roughly 29 percent, so for a 0.7 percent
12 loss of population to produce it mechanically the elasticity of reported conflict
13 with respect to village population would have to be about forty. The PODES
14 counts are a village-head report rather than an enumeration, but matching the
15 2010 Population Census to the 2011 wave for 35,628 in-sample villages gives a
16 correlation of 0.99 in levels and logs and a log-log slope of 0.970 (standard error
17 0.001), so correcting for the implied compression would move the first stage from
18 -0.72 to about -0.74 percent. The composition bounds the mechanical version
19 directly: depopulation predicts that *every* category of reported disorder falls, yet
20 theft rises sharply over the same villages and waves while communal conflict falls.
21 That rules out depopulation as the sole account. It does not rule out selective
22 displacement of particular groups, which neither PODES nor our design observes,
23 and we flag that as an open threat rather than one we have closed (Supplementary
24 Material Section S7).

25 Channels Consistent with the Evidence

26 What follows is mechanism-consistent evidence, not a mediation decomposition:
27 we do not claim to identify any channel's share. The organising logic is that a
28 strong earthquake *may* raise the economic *motive* for disorder, through asset and

1 income losses PODES cannot observe, while degrading the *capacity* to organise it
2 collectively. The two outcomes resolve that tension differently. Individual theft,
3 which needs only motive, rises. Communal violence, which also needs contact,
4 coordination and the ability to evade monitoring, falls, because on this account
5 the shock erodes those inputs faster than it may raise the motive.

6 **The capacity-degrading channels.** Three forces plausibly degrade the capacity
7 for organised violence. *Contact disruption*: communal violence emerges from
8 contested spaces and the inter-group contact central to relations in Indonesia ([Bazzi](#)
9 [et al., 2019](#); [Lowe, 2021](#); [Rao, 2019](#)), which strong shaking can render unusable.
10 We flag in advance that the one venue type we can count, places of worship, does
11 not decline, so within this channel the evidence rests on communication rather
12 than venue destruction. *Post-disaster monitoring*: a temporary surge of external
13 actors raises the cost of organising violence, which the counterinsurgency evidence
14 suggests can suppress it ([Berman, Shapiro, and Felter, 2011](#)), though the net effect
15 of aid and presence is context-dependent and can be conflict-increasing where aid
16 becomes a contestable prize ([Crost, Felter, and Johnston, 2014](#); [Nunn and Qian,](#)
17 [2014](#)). This is the most tentative channel: PODES cannot measure the surge, and
18 the one durable state-presence proxy it records, police-post presence, does not rise
19 after exposure. *Recovery-period opportunity costs*: reconstruction can reallocate
20 labour into construction and raise wages rather than collapse them ([Kirchberger,](#)
21 [2017](#)), lowering the return from conflict, opposite to the income-destruction channel
22 ([Dube and Vargas, 2013](#)).

23 The rising theft signals active individual-predation incentives. Whether that
24 reflects a lower return to honest work or a higher return to appropriation, the
25 two income-side forces of [Dube and Vargas \(2013\)](#), neither requires inter-group
26 coordination and both push individual predation upward. What the composition
27 establishes is that these incentives, however generated, do *not* convert into collective
28 violence. The -1.5 point communal estimate is therefore best read as consistent
29 with a setting in which degraded collective capacity dominates any countervailing

1 rise in the motive for disorder: if the shock also raises the motive it is a net
2 effect in which capacity dominates, and if it does not it is the capacity effect
3 alone. Our data do not separate these, and the composition is the same under
4 both. Susenas microdata would turn the motive from consistency evidence into
5 a measured channel; because the opportunity-cost force predicts *more* conflict, a
6 finding that welfare fell would sharpen this reading rather than weaken it.

7 **Onset and continuation in prior evidence.** The first prediction of Section 2
8 separates onset from continuation, and the prior evidence separates them the same
9 way. [Bazzi and Blattman \(2014\)](#) make the separation of these margins an explicit
10 methodological lesson, and their price shocks leave onset unmoved, which is what
11 a shock that transfers income without destroying the physical settings of daily
12 life should do; a damaging earthquake differs precisely in supplying the incidents
13 an onset requires. [Barron and Sharpe \(2008\)](#) document both halves in Indonesia,
14 describing established conflict as cyclical and retaliatory and tracing the path into
15 first-time violence through isolated incidents that escalate into group clashes.

16 **Corroborating Evidence on Collective Capacity**

17 The channels above are inferred from what the shock does to inputs. PODES also
18 measures collective capacity directly. From 2018 onward the survey asks whether
19 villagers customarily engage in *gotong royong*, mutual-help collective action, and
20 how widely they participate, on an identical three-point scale in 2018, 2021 and
21 2024. Two items are asked separately: collective action for public purposes, which
22 the questionnaire illustrates with communal labour, night-watch rotations and
23 village festivals, and collective action to help residents struck by misfortune such as
24 death, illness or accident. The first is a direct measure of the capacity to organise
25 the village; the second is a measure of disaster response.

26 Table 12 estimates the effect of first strong exposure on both. High-involvement
27 public-purpose *gotong royong* falls by 3.1 to 3.2 percentage points against a mean

1 of 86 percent, significant under both sets of fixed effects. The misfortune-response
2 item does not fall; its extensive margin rises slightly. The two move in opposite
3 directions, which is what the capacity reading predicts and what a general collapse
4 in village functioning would not: routine collective organisation weakens while
5 emergency mutual aid, which needs no standing coordination, does not.

6 We put weight on this cautiously. The item exists in only three waves, so
7 only the 2021 and 2024 cohorts are identified within the window, 942 villages in
8 all. A Callaway–Sant’Anna estimate on the same item corroborates the two-way
9 specifications, at -0.0326 (standard error 0.0142 , $p = 0.021$), and the single pre-
10 treatment lead that three waves permit does not reject ($+0.022$, standard error
11 0.040 , $p = 0.58$), although a test built on one lead has little power. The calibration
12 row of the table makes the power problem explicit: on this restricted sample the
13 conflict effect itself is no longer significant under the preferred fixed effects. Two
14 further limits bound the reading. The estimand is not the paper’s main one: it
15 averages the 2021 and 2024 cohorts only, so it covers different earthquakes over
16 a different period and corroborates the direction of the effect without being a
17 component of its magnitude. And the 2021 wave was enumerated in May 2021,
18 during the pandemic, which depresses collective activity in level; that wave is also
19 the first post-exposure wave for the 2021 cohort. Wave fixed effects absorb that
20 shock where it is common across villages, and it would bias the estimate only if
21 restriction intensity covaried with earthquake exposure. We therefore read the
22 result as corroborating the capacity interpretation, not as establishing it.

23 **Suggestive Heterogeneity**

24 Interacting exposure with pre-treatment religious polarisation deepens the decline
25 by 1.6 percentage points per standard deviation, as a contact-and-mobilisation
26 account predicts, but the gradient does not survive the proportional-scale test that
27 governs the rest of the paper, so we do not read it as evidence for the channel
28 (Supplementary Material Section S6).

Table 12: Collective Capacity After Strong Shaking

Outcome	Wave FE	Province $\times$ wave FE	Pre-exposure mean	N
Public-purpose <i>gotong royong</i> , high involvement	−0.0311** (0.0138)	−0.0324** (0.0164)	0.863	78,225
any involvement	−0.0027 (0.0018)	0.0030 (0.0026)	0.994	78,225
Misfortune-response <i>gotong roy-</i> <i>ong</i> , high involvement	−0.0111 (0.0099)	−0.0033 (0.0127)	0.911	78,225
any involvement	0.0023 (0.0020)	0.0059** (0.0025)	0.996	78,225
<i>Power calibration on the same restricted sample</i>				
Any communal conflict	−0.0395*** (0.0097)	−0.0148 (0.0112)	0.024	93,603

Gotong royong is measured on an identical three-point scale in PODES 2018, 2021 and 2024 (1 = present, most residents involved; 2 = present, few involved; 3 = no such custom); the item is absent or differently coded in earlier waves, so the window is three waves. Only cohorts first exposed in 2021 and 2024 are identified within it (942 villages, 1.8 percent of village-waves treated); villages exposed at or before 2018 are always-treated here and dropped. The calibration row shows that the conflict effect itself loses significance on this restricted sample under the preferred fixed effects, so these estimates are low-powered by construction. A Callaway–Sant’Anna estimate on the same item gives −0.0326 (standard error 0.0142, $p = 0.021$), and the single pre-treatment lead the three waves permit does not reject (+0.0222, standard error 0.0404, $p = 0.58$), though a test built on one lead has little power. The 2021 wave was enumerated during the COVID-19 period, which depresses collective activity in level; wave fixed effects absorb that shock where it is common across villages. Standard errors are clustered at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

1 Event Study

Figure 1 presents the event study on the preferred specification. The diagnostics that bind are those of the CS(2021) event study on the conventional wave-fixed-effects panel (Appendix Figure S7): there the average lead is small and insignificant for both outcomes ($p = 0.33$ composite, $p = 0.56$ *warga*) but a joint Wald test over all pre-treatment group-time cells rejects for both ($\chi^2(15) = 42.9$ and $\chi^2(15) = 29.4$), so we do not read either pre-period as a clean pass. A violation the pre-test would miss half the time exceeds the pooled *warga* ATT (Roth, 2022), which is one reason we do not read a passed pre-test as evidence, and the HonestDiD bounds of Section 5.4.1, not the pre-trend tests, are the binding constraint (Appendix H). We carry the definition-stable *warga* measure throughout

1 because its wording is far closer to comparable across the seven waves and its
2 pre-trend is the milder of the two; Supplementary Material Section S9 quantifies
3 its residual 2005 incomparability at 3.8 percent of *warga* villages against 22.9
4 percent for the composite. We do not read it as a second estimate of the level
5 effect, for which it is underpowered. The longer-run estimates show no reversal,
6 but the later lags are imprecise nulls rather than evidence of persistence, so we
7 read the absorbing average as attenuated by them rather than as showing that the
8 post-earthquake state carries the effect on its own.

9 **Simultaneous inference on the event-study path.** One inference point
10 belongs with any event study and we report it rather than leave it implicit (Baker
11 et al., 2025). The intervals plotted in Figure 1, and those in every event study we
12 show, are pointwise. A statement about the whole path of coefficients, which is
13 what a pre-trend reading is, compares several estimates at once and therefore needs
14 a critical value that accounts for the multiplicity. On the Callaway–Sant’Anna
15 counterpart we compute that value by simulating from the estimated covariance
16 matrix of the event-study vector: the 95 percent sup- t multiplier is 2.65 against the
17 pointwise 1.96, so bands valid for the entire path are about 35 percent wider. The
18 pre-period conclusion survives the stricter standard, since all three pre-exposure
19 coefficients remain indistinguishable from zero under it, which is the direction that
20 matters for reading pre-trends. One post-exposure coefficient does not: the estimate
21 three waves after exposure is -0.0167 with a pointwise interval of $[-0.029, -0.004]$
22 but a simultaneous interval of $[-0.034, +0.000]$, so it is significant read alone and
23 marginal read as part of the path. The effect in the wave of exposure and one wave
24 later remain significant under both.

25 Under wave fixed effects the joint pre-test rejects in every estimator family
26 we run; under province-by-wave effects it passes in every family that can absorb
27 them. The verdict turns on the fixed effects, not on the estimator, which is what
28 Section 4.1 predicts (Appendix H).

29 Conflict in neighbouring villages does not rise as exposed villages’ conflict falls,

1 so the decline is not displacement (Appendix D).

2 **How Many Shocks? Stacked Event Design and Event-Level** 3 **Inference**

4 The design has 2,995 ever-treated villages, but they are not 2,995 independent
5 assignments: villages in the same footprint receive the same shock. Assigning
6 each village to the earliest earthquake exposing it to $\text{MMI} \geq \text{VI}$, **19 distinct**
7 **earthquakes** with at least twenty in-sample first-treated villages each account for
8 93 percent of the staggered-treated villages (Appendix S12.1).

9 Treating the earthquake rather than the village as the unit of assignment sepa-
10 rates two estimands, and that distinction is the substantive point of this subsection.
11 The *treated-village-weighted* average weights each earthquake’s effect by the villages
12 it first exposed and answers what happened to the average exposed village; it is
13 -0.0268 (standard error 0.0105, earthquake bootstrap interval $[-0.047, -0.007]$).
14 The *equal-event* mean counts each earthquake once and answers what the average
15 earthquake does to its footprint; it is -0.0165 (standard error 0.0150), not distin-
16 guishable from zero, and its coincidence with the village-level ATT is numerical
17 rather than substantive.

18 We claim the first, because the question posed in the introduction is what
19 strong shaking does to the villages it strikes; on the second the paper establishes
20 no decline, and we regard that as the honest ceiling on what the staggered result
21 can claim. The pooled evidence is carried by the six large-footprint earthquakes,
22 but that should not be read as a footprint gradient: estimated directly the slope of
23 the per-event effect on log footprint is -0.0004 (standard error 0.0016, $p = 0.81$),
24 estimated across the 46 events that clear the lower inclusion threshold rather
25 than the 19 the figure displays, and weighted by the precision of each event’s
26 own estimate. Unweighted the slope is -0.0097 (0.0074, $p = 0.20$), still not
27 distinguishable from zero. Larger events carry the pooled estimate through their
28 weight; the six largest-footprint events do average -0.038 against -0.004 for the

rest (Supplementary Material Section S7), a step the linear slope does not detect and which we read as post hoc. The composition result does not depend on the choice: the theft-versus-*warga* divergence is positive and significant under *both* estimands (Section 5.2). Supplementary Material Section S7 collects every estimand, weighting and inference procedure, Figure 3 shows all 19 event-specific estimates, and Supplementary Material Section S7 gives the stacked construction and the clustering, disjoint-control and aftershock-independence checks.

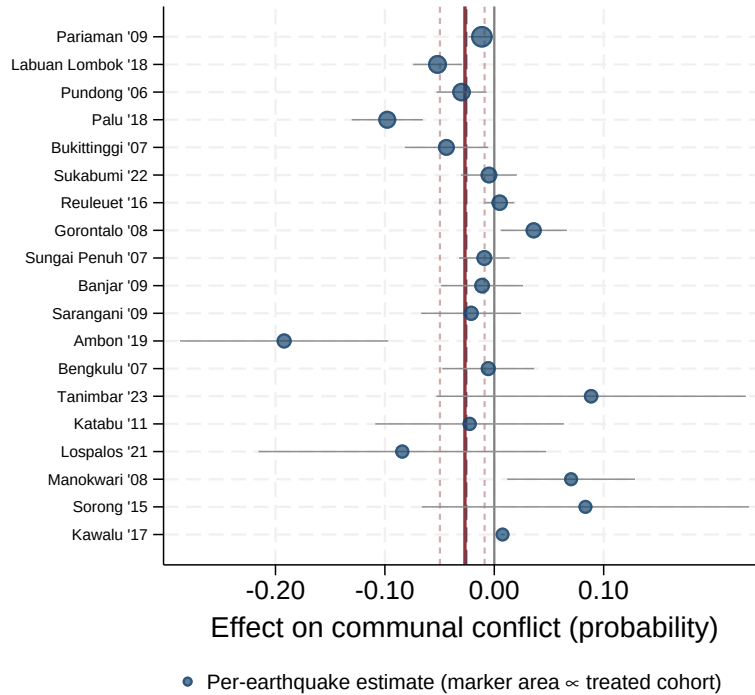


Figure 3: Event-Specific Difference-in-Differences Estimates

Note: Each row is one earthquake’s own difference-in-differences estimate of the effect on communal conflict, with a 95% confidence interval. Rows are ordered by treated-cohort size and labelled by the earthquake’s nearest place (USGS ShakeMap catalogue) and year; marker area is proportional to the treated cohort, counted as unique first-treated villages. The solid grey line is zero. The dashed navy line is the pooled stacked estimate (-0.0257), variance-weighted toward the large, sharply-identified shocks. The solid maroon line is the treated-village-weighted estimate (-0.0268), with its earthquake-level bootstrap 95% confidence interval shown as dashed maroon lines. Thirteen of the 19 estimates are negative, and the effect is concentrated in, though not exclusive to, the major earthquakes.

8 Regional Trends and Forbidden Comparisons

Two objections apply to the absorbing design at once: the regional-trajectory problem of Section 4.1, and the already-treated comparison problem of de Chaisemartin

1 and D’Haultfoeulle (2020), on which a Goodman–Bacon decomposition puts 3.8
2 percent of the weight, though those comparisons return -0.0376 against -0.0186
3 for the clean ones and so pull in the same direction.

4 The same critique can be put in terms of the weights the estimator places on
5 the individual group-period effects rather than on comparison types, and that is
6 the form in which de Chaisemartin and D’Haultfoeulle (2020) state it, so we report
7 it too (Table 13). Of the 82,832 underlying effects, 32,347 receive positive weight
8 summing to 1.1775 and 50,485 receive negative weight summing to -0.1775 , so
9 negative weights are 13.1 percent of the gross weight mass. More effects carry a
10 negative weight than a positive one, but the positive weights carry almost all of
11 the magnitude. This bounds the concern rather than dismissing it: an estimate
12 whose negative weights are this small cannot be produced by sign reversal alone,
13 though the decomposition does not by itself establish convexity, which is why the
14 heterogeneity-robust estimators and not this benchmark carry the paper.

Table 13: Weights the Two-Way Estimator Places on the Underlying Effects
Negative weights are a small share of the gross mass, so the estimate is not an artifact of sign reversal.

	Number of ATTs	Sum of weights
Positive weights	32,347	1.1775
Negative weights	50,485	-0.1775
Total	82,832	1.0000
Negative share of gross weight	13.10 percent	

Decomposition of the two-way fixed-effects coefficient into the weights it places on the underlying group-period average treatment effects, following de Chaisemartin and D’Haultfoeulle (2020). The weights are a property of the treatment design and the panel structure alone, not of the outcome, so a single decomposition characterises every outcome the paper reports; we verified that the composite returns identical weights. Negative weights arise because already-treated units serve as controls for later-treated ones. A decomposition in which the negative weights are a small share of the total, as here, means the estimate cannot be driven by sign reversal alone, though it does not by itself establish that the weighted sum recovers a convex average of the underlying effects.

15 Province×wave effects absorb the first; dropping the always-treated cohort
16 removes the units with no pre-period and barely moves the estimate (-0.0199 to
17 -0.0193), and the estimators of Section 4, none of which uses an already-treated
18 unit as a control, answer the residual concern. The joint pre-trend clears in all four

1 columns (Appendix Table S47), which it does not under wave fixed effects, and
2 the kabupaten $\times$ wave column is reported for completeness but is uninformative,
3 absorbing 62 percent of the remaining treatment variance (Appendix Table S48).

4 The effect is concentrated rather than general: inter-village conflict falls by 1.7
5 percentage points, 45 percent of its pre-exposure mean, while inter-group conflict
6 within the village does not move detectably (-0.0035 , standard error 0.0032 ,
7 against a pre-exposure mean of 3.7 percent). That interval still admits a decline of
8 about a quarter of the base, so we read it as an absence of evidence rather than
9 as a precise zero. That margin also decays, strongest within the first year and
10 indistinguishable from zero beyond roughly six years, with earthquake fixed effects
11 identifying each lag within the same event and the decay test rejecting equality
12 between recent and distant bins (Appendix Table S49); neither the composite nor
13 the mass-brawl gate shows a comparable profile, and identification here is exposed
14 villages against themselves over time rather than exposed against unexposed.

15 Design, Clustering and Spatial Structure

16 The estimate survives the exposure, clustering, and spatial choices: alternative MMI
17 thresholds ($\geq V$ and $\geq VII$, though the ladder traces no dose-response), kabupaten
18 clustering, Conley spatial-HAC errors, Oster (2019) coefficient stability under
19 which adding controls moves the estimate *away* from zero, and spillover checks
20 leave it intact, though a 25 km donut keeps the sign and loses precision (-0.0057 ,
21 $p = 0.24$), while the conservative 28-cluster province bootstrap leaves the composite
22 at 5 percent and *warga* short of it. Repeated exposure is rare and where it occurs
23 we detect no additional effect (0.004 , $p = 0.75$); statistical insignificance is not
24 equivalence, so we read this as consistent with, not establishing, a first-shock state
25 change (Supplementary Material Sections S7 and S9).

1 Conventional Staggered-DiD Benchmark: Callaway and 2 Sant’Anna (2021)

3 Our conventional staggered-DiD benchmark is the group-time average treatment
4 effect estimator of Callaway and Sant’Anna (2021), hereafter CS(2021):

$$5 \quad ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g], \quad (3)$$

6 where $G = g$ denotes the cohort of villages whose first treatment wave is g , and
7 $Y_t(0)$ is the potential outcome the same villages would have had at wave t absent
8 any earthquake. Because it never uses already-treated villages as a comparison
9 group, it avoids the contamination that biases conventional two-way fixed effects
10 when treatment timing is staggered (Callaway and Sant’Anna, 2021; Roth et al.,
11 2023). We aggregate $ATT(g, t)$ to a single average effect across post-treatment
12 periods using doubly-robust inverse-probability weighting (Sant’Anna and Zhao,
13 2020), taking villages not yet treated at each point in time as controls. Standard
14 errors are analytical, based on the Callaway–Sant’Anna influence function. A wild
15 cluster bootstrap is not available for this estimator, so its few-cluster check is
16 province-clustered inference (Section 5.4); the fixed-effects estimates are checked
17 with a wild cluster bootstrap (9,999 Webb-weight replications, seed 42).

18 **Why absorbing treatment?** An earthquake is an episodic physical event, so
19 treating a village as exposed forever after its first strong shock requires justification:
20 the framework of Section 2 predicts hysteresis, the precisely estimated response is
21 concentrated in the first two waves while later estimates are imprecise, a profile that
22 governs how the pooled estimate should be read rather than whether the absorbing
23 definition is admissible, and the episodic design of Section D lets treatment switch
24 off; Appendix S10 gives the full justification.

25 **Implementation.** Waves are indexed ordinally because the survey is unevenly
26 spaced, and estimation uses the default pairwise-balancing rule of `csdid` on the

1 unbalanced panel, which retains cohorts a fully balanced restriction would drop.
2 All CS(2021) estimates use `csdid` version 1.72; Appendix [S10](#) records the imple-
3 mentation details and the replication package the version of every user-written
4 command.

5 **Cohort composition.** Villages first exposed before or during 2005 are always-
6 treated in the 2005-boundary panel and are excluded from CS(2021) by the not-yet-
7 treated control requirement. Cohorts G3–G8 (first treated 2008–2024) identify the
8 estimator, with group-level ATTs for all six cohorts reported in Appendix Table 9.
9 The exclusion is not a technicality at the margin: of the 13,092 in-sample villages
10 ever exposed to damage-level shaking, 10,097, or three quarters, are always-treated,
11 so the identified sample is the quarter first shaken after the panel opens and it
12 contains neither the 2004 Sumatra–Andaman nor the 2005 Nias rupture. Because
13 state capacity, reconstruction regimes, and conflict history differ across the period,
14 treatment effects need not be homogeneous across cohorts; the estimand applies to
15 cohorts entering treatment between 2008 and 2024, and we do not extrapolate it
16 to the excluded early cohorts.

17 **Weighting and the estimand.** One weighting choice shapes the estimand and
18 we state it plainly, because it decides whose average the paper reports ([Baker et al.,](#)
19 [2025](#)). Villages enter unweighted, so every group-time cell counts each exposed
20 village equally and the target is the effect on the average exposed *village*, not on
21 the average exposed *person*. Indonesian villages differ greatly in population, so the
22 two are not the same quantity, and a population-weighted version would answer
23 a different and equally reasonable question. We cannot report it. BPS stopped
24 disseminating village population counts after the 2011 round, so population is
25 structurally missing from 2014 onward and therefore unavailable for the 2018,
26 2021 and 2024 cohorts, which supply most of the identifying variation. Where
27 population does enter, in the matched within-earthquake designs, we use the 2005
28 value fixed before any identified cohort is treated rather than a contemporaneous

1 one, for the same reason the covariates are fixed at baseline.

2 The two-way fixed-effects benchmark and its Goodman–Bacon decomposition
3 are set out in Appendix D; no claim here rests on it.

4 The episodic estimand of [de Chaisemartin and D’Haultfoeuille \(2026\)](#), which
5 lets treatment switch off, is described in Appendix D.

6 **Sample Restriction: Seismic Hazard Zones**

7 The analysis is restricted to villages in Indonesia’s Kawasan Rawan Bencana (KRB)
8 Medium or High seismic-hazard zones (Appendix Figure S3), mapped by the
9 Center for Volcanology and Geological Hazard Mitigation (PVMBG/ESDM) and
10 consistent with Indonesia’s official national seismic-hazard assessment ([Irsyam et al.,](#)
11 [2020](#)). Limiting to KRB zones ensures that all included villages have non-trivial
12 earthquake exposure, so the not-yet-treated comparison villages are drawn from
13 the same seismic risk set rather than from areas that face no earthquake hazard at
14 all. The restricted sample covers Sumatra, Java, Bali, Nusa Tenggara, Sulawesi,
15 Maluku, and Papua. Of the 63,115 villages in the 2005-boundary panel, 43,351
16 fall in a Medium or High zone and 19,362 do not; a further 402, six tenths of one
17 percent, have no KRB record at all, because the polygon does not intersect the
18 published map, and we drop them rather than assign them to either side.

19 A natural concern is that the hazard classification is endogenous to the earth-
20 quakes we study. It is not: the KRB map is a hazard assessment based on fault
21 and subduction-zone geometry rather than realized damage ([Irsyam et al., 2020](#)),
22 the restriction enters the design only through village fixed effects, which absorb
23 any time-invariant hazard level, and dropping it entirely and re-estimating on the
24 full national sample leaves the effect intact (Appendix S10). Appendix Figures S4
25 and S5 map the treated villages by cohort and place the cohorts on the survey
26 calendar.

1 **Alternative Conflict Measure: ACLED**

2 For robustness, we use ACLED (Armed Conflict Location and Event Data Project)
3 event data geo-matched to village boundaries, available for 2015–2024. ACLED
4 covers riots, violence against civilians, and battles involving civilian groups. Because
5 temporal overlap with PODES is limited to three waves (2018, 2021, 2024), ACLED
6 is used only as an external comparison, not as a primary outcome or a validation
7 of PODES.

8 **D Benchmark Estimators, Province Trends, and** 9 **Displacement**

10 This appendix holds the estimator comparisons and two checks that the main text
11 summarises in one sentence each: the two-way benchmark and its decomposition,
12 the episodic estimand, the Callaway–Sant’Anna benchmark in full, the two routes
13 for province trends, and the displacement check.

14 **Benchmark Estimator: TWFE**

15 As a benchmark, we estimate a two-way fixed-effects linear probability model
16 (TWFE-LPM):

$$17 \quad y_{vt} = \alpha D_{treat,vt} + \gamma' \mathbf{X}_{vt} + \mu_v + \lambda_t + \varepsilon_{vt}, \quad (4)$$

18 where $D_{treat,vt} = \mathbf{1}\{t \geq g_v\}$ is a binary absorbing treatment indicator equal to one
19 from the first treatment wave onward, $\mathbf{X}_{vt}$ is a vector of seven village characteristics
20 measured at the PODES 2003 baseline (household electrification, paved road access,
21 internet availability, police post, health-centre presence, number of primary schools,
22 and distance to the district capital), entered as levels in the doubly-robust estimators
23 and interacted with wave dummies in the two-way specifications, μ_v are village
24 fixed effects, and λ_t are wave fixed effects. Standard errors are clustered at the
25 village level following [Abadie et al. \(2023\)](#).

1 We also estimate a distributed-lag TWFE with exposure lags aligned to May-
2 to-May windows, so each lag falls fully before the conflict reference window (full
3 specification in Appendix S10). Alongside CS(2021) we report the interaction-
4 weighted event-study estimator of Sun and Abraham (2021).

5 The distributed-lag results, and the timing artifact that unequally spaced
6 survey panels create when exposure windows are misaligned, are documented in
7 Appendix S10.

8 Two terms recur below and we fix them here. The preferred *counterfactual*
9 *specification* is the province-by-wave two-way fixed-effects design of Section 4.1.
10 Our primary *heterogeneity-robust staggered estimator* is the interaction-weighted
11 event study of Sun and Abraham (2021). The reason is the design, not the
12 diagnostics: it is the heterogeneity-robust estimator that can absorb interacted
13 fixed effects, so it implements the preferred counterfactual directly rather than
14 approximating it. CS(2021) can hold province trends fixed only by carrying
15 province indicators inside the nuisance models it fits cell by cell, which is a different
16 operation and, as Section D shows, one this panel does not support well. We report
17 CS(2021) throughout as the conventional benchmark and for its explicit treatment
18 of staggered timing, and we report what the two routes give side by side rather
19 than selecting between them on the basis of which pre-trend test passes. Every
20 estimate below is labelled by the design it uses.

21 **Complementary Estimand: Episodic Exposure (dCdH)**

22 Because earthquakes are episodic physical events, we complement the absorbing
23 design with an estimand in which treatment can switch off. We estimate the de
24 Chaisemartin and D’Haultfoeuille (2026) estimator with a non-absorbing treatment
25 indicator: $D_{interval,vt} = \mathbf{1}\{\exists \text{ month in interval}[t-1, t) : \text{MMI} \geq 6\}$, which switches
26 between zero and one across waves. The dCdH Effect₁ captures the ATT one wave
27 after switching to $D_{interval} = 1$, that is, the effect of fresh exposure in the current
28 inter-wave window regardless of the village’s exposure history. Together, the two

1 designs answer two distinct questions: what is the effect of having entered the
2 post-earthquake state (absorbing), and what is the effect of being freshly shaken
3 (episodic)?

4 **The Conventional Benchmark: CS(2021)**

5 Table 14 reports the benchmark for two outcomes. Panel A gives the CS(2021)
6 aggregate ATT: -0.0177^{***} for the composite “any communal conflict” indicator
7 ($SE = 0.0047$, $N = 225,372$) and a smaller -0.0075^{**} ($SE = 0.0037$, $p = 0.042$)
8 for the definition-stable *warga* indicator. Panel B gives the Sun–Abraham (2021)
9 interaction-weighted estimator, with significant effects in the wave of exposure
10 ($t_0 = -0.0223^{***}$ composite; -0.0096^{**} *warga*). Neither outcome has a clean joint
11 pre-trend under wave fixed effects; Appendix H sets out the diagnostics across
12 estimator families and Section 5.4.1 bounds what the residual violation can do to
13 the estimate.

14 We report the two columns together rather than as a lead estimate plus a
15 robustness check, because each trades one property for another. The composite is
16 the more powerful but its coding rule changes at 2008, in a direction that makes
17 the 2005 wave broader rather than narrower. *Warga* is definition-stable across all
18 seven waves but rarer, and on the level effect that imprecision binds: it agrees in
19 sign with the composite and is not significant under the preferred specification
20 (-0.0031 , standard error 0.0031). We therefore do not present it as a second level
21 estimate. What it delivers is an outcome whose survey definition is fixed, which is
22 what the composition contrast of Section 5.2.1 requires and where it is decisive.

23 **Economic magnitude.** First strong exposure lowers the communal-conflict
24 probability by 1.8 percentage points, with a 95 percent interval of $[-2.69, -0.85]$
25 points. The benchmark is not the pooled sample mean of 2.6 percent but the
26 treated villages’ own counterfactual: ever-treated villages reported conflict at 6.2
27 percent across their pre-treatment waves, roughly twice the pooled rate, because

1 earthquakes strike the more conflict-prone, faster-growing parts of the hazard zone.
2 The estimand is accordingly local to hazard-zone villages that are actually struck.
3 The identifying variation comes from cohorts G3–G8 (first treated 2008–2024),
4 principally the Padang 2009 earthquake and later events. Post-treatment dynamic
5 ATTs are negative at all six estimable horizons and individually significant at t_{+0} ,
6 t_{+1} and t_{+3} ; the rest are negative but imprecise, so the longer-run path shows no
7 reversal without establishing persistence.

8 **Group-level heterogeneity.** No single cohort drives the effect. The one positive
9 cohort is post-Helsinki Aceh, and dropping it, or Aceh entirely, leaves the estimate
10 unchanged or larger (Appendix A).

11 **The conventional benchmarks.** The two-way fixed-effects benchmark gives
12 -0.0213 with controls and -0.0193 without; the latter is not a negative-weighting
13 artifact, since the already-treated comparisons carry only 3.8 percent of the
14 Goodman–Bacon weight and return -0.0376 against -0.0186 for the clean ones,
15 pulling in the same direction rather than against it, and the episodic estimator of de
16 Chaisemartin and D’Haultfoeuille (2026) gives -0.015^{***} with matching dynamics.
17 Across heterogeneity-robust and conventional estimators, and across absorbing and
18 episodic treatment definitions, the estimate ranges from -0.014 to -0.022 and
19 every confidence interval lies below zero (Appendix Table S46); the decomposi-
20 tions, distributed lags, the timing artifact for unevenly spaced panels, and the
21 placebo panel are in Appendix S10. CS(2021) is the estimator we carry through
22 the robustness battery because it has the largest set of published diagnostics.

23 Table 15 makes the point on a common sample and control set: six designs,
24 including the same-earthquake stack that holds the shock itself fixed, range from
25 -0.0165 to -0.0227 , between 26 and 36 percent of the pre-exposure mean. The
26 stacked row targets the *incremental* effect of crossing the damage threshold, smaller
27 than the total effect by construction, and we do not call it the strongest evidence:
28 its own pre-lead is marginal ($+0.0136$, $p = 0.06$) and its earthquake-clustered

Table 14: Conventional Staggered-DiD Benchmarks: Effect of First $\text{MMI} \geq \text{VI}$ Exposure on Communal Conflict

2005-boundary panel, 2005–2024, KRB Medium/High

	Warga conflict (consistent def.)	Any communal conflict (composite)
<i>Panel A: Callaway–Sant’Anna (2021), not-yet-treated control</i>		
Overall ATT (simple)	–0.0075** (0.0037)	–0.0177*** (0.0047)
Pre-trend, average lead (p)	–0.0038 (0.56)	–0.0077 (0.33)
Pre-trend, joint Wald (p)	$\chi^2(15) = 29.5$ (0.014)	$\chi^2(15) = 42.9$ (0.000)
<i>Panel B: Sun–Abraham (2021) IW, never-treated control</i>		
Effect at t_0	–0.0096** (0.0041)	–0.0223*** (0.0051)
Effect at t_{+1}	–0.0067 (0.0043)	–0.0194*** (0.0054)
Pre-lead t_{-2} (placebo)	+0.0077	+0.0138**
<i>Panel C: Earthquake-level inference (stacked-event design, 19 events)</i>		
Treated-village-weighted ATT	–0.0114	–0.0268
event block-bootstrap SE [p]	(0.0065) [0.034]	(0.0105) [0.006]
Equal-event mean, secondary [p]	–0.0080 [0.49]	–0.0155 [0.32]
Earthquake events (clusters)	19	19
Baseline mean (pooled)	0.016	0.026
Pre-treatment mean, ever-treated	0.037	0.062
Observations	225,372	225,372

Notes: Panel A reports analytical influence-function SEs and Panel B village-clustered SEs in parentheses. The average pre-trend does not reject, but a joint Wald test of all pre-treatment group-time $\text{ATT}(g, t)$ coefficients rejects for both outcomes under wave fixed effects (the aggregated event-study leads oscillate in sign rather than trending toward the post-treatment decline); the test is passed once province-by-wave effects are absorbed (the robustness discussion in the main text). Panel C treats the earthquake, not the village, as the unit of assignment: the treated-village-weighted ATT is stacked across the 19 earthquakes and its SE comes from an event-level block bootstrap that resamples whole earthquakes (9,999 draws); the naive village-clustered SE for this estimand understates uncertainty because treatment is driven by 19 shocks. The equal-event mean weights each earthquake equally (13 of 19 negative). Conley spatial-HAC SEs at 50–150 km cutoffs and kabupaten-level clustering leave the composite reduced-form estimate significant at the 1 percent level (the robustness discussion in the main text); under a deliberately conservative province-level wild-cluster bootstrap (few clusters), the pooled staggered ATT is marginal under province clustering ($p = 0.064$), which we report rather than suppress. Pre-treatment mean = mean outcome among ever-treated villages before first exposure. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

1 standard error is far larger. The intensity ladder is not monotone under CS(2021)
2 and changing the threshold also changes which cohorts and earthquakes enter,
3 so we rest the dose argument on the same-earthquake band comparison instead
4 (Appendix S10).

Table 15: Six Designs, One Sign: Estimates Across Identification Strategies

Design	Estimate (SE)	% of pre- exposure mean	Pre-trend joint p	N
Absorbing TWFE, province $\times$ wave FE	-0.0165*** (0.0041)	-26.4	0.678	225,372
Finite duration (two waves)	-0.0173*** (0.0037)	-27.8	0.678	225,372
Event study: wave of exposure	-0.0194*** (0.0063)	-31.0	0.678	225,372
Event study: one wave after	-0.0227*** (0.0069)	-36.4	0.678	225,372
Callaway-Sant’Anna (2021)	-0.0177*** (0.0047)	-28.4	—	225,372
Stacked within-earthquake	-0.0174*** (0.0040)	-27.9	0.063	147,905

Rows 1–5 are estimated on the same sample (KRB Medium/High, always-treated cohort excluded) with baseline PODES 2003 controls; standard errors clustered at the village level. Pre-trend joint p tests leads at -4 , -3 and -2 against zero. The pre-exposure mean is 0.0624. Callaway-Sant’Anna uses wave fixed effects, since the estimator does not accommodate province $\times$ wave absorption; its pre-trend is reported separately via **estat pretrend**. The stacked design compares villages exposed at $\text{MMI} \geq \text{VI}$ to villages that felt the *same* earthquake at $\text{MMI} \in [4, 6)$, with event $\times$ village and event $\times$ wave fixed effects over 24 earthquakes; clustering by earthquake widens its standard error to 0.0098. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Province Trends: Absorbed or Conditioned On

6 Section 4.1 argues that the counterfactual should hold province-specific trajectories
7 fixed; that argument does not say how. Province trends can be absorbed as
8 saturated fixed effects or conditioned on inside the estimator’s nuisance models,
9 and the two routes disagree: absorbing them leaves the result intact, conditioning on
10 them does not. Either the level decline is partly carried by provincial trajectories, or
11 conditioning on thirty-two province and island indicators per group-time cell breaks
12 the estimator’s overlap; our data do not separate the two, and Supplementary
13 Material Section S7 reports both routes in full. What follows limits the level
14 estimate, not the paper: the composition contrast differences out disturbances

1 common to the outcomes recorded on the same form by the same respondent, and
2 the earthquake-level divergence is estimated event by event rather than from this
3 panel at all. This is the second reason, alongside the pre-trend record, why the
4 level stands as a motivating average and why the composition evidence rests on
5 the divergence rather than on either level.

6 **Does the Decline Displace? Conflict in Neighbouring Vil-** 7 **lages**

8 Disasters can *lower* conflict where they strike while *raising* it nearby ([Amarasinghe](#)
9 [et al., 2026](#)). Assigning each never-treated village to a mutually exclusive proximity
10 ring by distance to the nearest treated village, displacement predicts positive
11 ring coefficients; we find the opposite, with all three rings negative under village
12 and wave fixed effects and no ring ever significantly positive under either fixed-
13 effect structure (Supplementary Material Section S7, Supplementary Material
14 Section S7). The local decline shows no detectable offset onto neighbours; we read
15 this as bounding, not eliminating, the displacement concern, since PODES cannot
16 track individuals who relocate across village boundaries.

17 **E The Level Estimate in Full**

18 This appendix carries the detail of the average effect that Section 5.1 states in
19 brief.

20 Table 17 reports the specification selected by the rule of Section 4.1: village
21 and province $\times$ wave fixed effects, always-treated cohort excluded. First exposure
22 to damage-level shaking lowers the probability that a village reports communal
23 conflict by 1.7 percentage points against a pre-exposure mean of 6.2 percent, a
24 reduction of roughly a quarter. The estimate is 1.5 points without covariates and
25 1.7 with them, so it does not rest on the control set.

26 It does not rest on *how* the covariates enter either, which is a separate question

and one the design can answer directly. Covariates can be brought into a DiD estimate in three ways, and each is biased under a different failure: an outcome-regression adjustment fails if the model for untreated outcome trends is misspecified, inverse probability weighting fails if the propensity-score model is, and the doubly robust combination is consistent when either one is right (Baker et al., 2025). Table 16 reports all three on the same specification. They agree to the fourth decimal, at -0.0176 , -0.0177 and -0.0177 for the composite and -0.0074 , -0.0075 and -0.0075 for *warga*, so no working model is carrying the result.

Table 16: Covariates Entered Three Ways

Takeaway: regression adjustment, inverse probability weighting and the doubly robust combination give the same estimate, so it does not depend on which working model is assumed correct.

	Regression adjustment	Inverse probability weighting	Doubly robust
Any communal conflict	-0.0176^{***} (0.0046)	-0.0177^{***} (0.0047)	-0.0177^{***} (0.0047)
<i>Warga</i> conflict	-0.0074^{**} (0.0037)	-0.0075^{**} (0.0037)	-0.0075^{**} (0.0037)
Observations	225,372		

Notes: Callaway–Sant’Anna aggregate ATT on the preferred panel, not-yet-treated controls, `long2` base period, baseline covariates fixed at the 2005 wave. The three columns differ only in how those covariates enter. Regression adjustment models the untreated outcome trend given covariates and is biased if that working model is wrong; inverse probability weighting models the propensity score instead and is biased if *that* model is wrong; the doubly robust estimator combines the two and is consistent if either is correct. Reporting all three follows Baker et al. (2025): agreement across the columns means the estimate does not rest on any one working model. Standard errors are analytical, based on the Callaway–Sant’Anna influence function and clustered at the village. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The event-study rows show where the effect sits in time. It is present in the wave of exposure (-0.0194) and the wave after (-0.0227), and the three subsequent lags are jointly indistinguishable from zero ($p = 0.38$). The wave of exposure is the first PODES enumeration after the shock, which the survey calendar can place anywhere from a few months to three years later, so t_0 is a first observed post-exposure effect and not a short-run one; we use that language throughout rather than calling it immediate. Because the absorbing indicator averages over later waves in which villages have already recovered, the pooled estimate is best

1 read as smaller than the wave-of-exposure response.

2 Both outcomes pass the joint pre-trend test in this specification ($p = 0.68$
3 composite, $p = 0.66$ *warga*), as the economic reasoning of Section 4.1 anticipates.
4 The definition-stable *warga* outcome moves in the same direction throughout but
5 does not attain conventional significance here. Its pre-exposure mean is 3.6 percent
6 against 6.1 for the composite, so it carries roughly half the power; we read it as
7 corroborating in sign rather than as independent confirmation.

8 Because treated villages enter the window from elevated conflict levels, the
9 obvious competing account is mean reversion, and it is worth stating the floor
10 it leaves. Letting villages with different fixed 2005-baseline conflict follow their
11 own wave trends, which is precisely the design that allows conflict-prone villages
12 to revert along their own trajectory, still leaves the effect at -0.0131 (standard
13 error 0.0028), and at -0.0140 (0.0035) when province-by-wave effects are added as
14 well. Reversion can absorb part of the decline but not the whole of it. What it
15 does bound is where the decline lives: the effect loads on the fewer than one in ten
16 villages with a prior conflict episode, so the average decline is concentrated where
17 conflict had already occurred; it does not describe prevention of first-time conflict.

18 The Callaway-Sant’Anna benchmark reproduces the sign and broad magnitude
19 of the preferred estimate on both outcomes; Appendix D reports it in full, together
20 with the definition-stable *warga* column and what the two outcome codings trade
21 against each other.

22 Absorbing province trends as fixed effects and conditioning on them inside
23 the estimator’s nuisance models give different answers; Appendix D reports both
24 routes and what separates them.

25 The event study places the response in the wave of exposure and the wave after,
26 with later horizons imprecise, and its pre-treatment coefficients are the object the
27 pre-trend tests interrogate (Appendix C).

Table 17: Preferred Specification: Province $\times$ Wave Fixed Effects, Always-Treated Cohort Excluded

	Any communal conflict	<i>Warga</i> (definition-stable)
Absorbing exposure, no covariates	−0.0154*** (0.0040)	−0.0035 (0.0032)
Absorbing exposure, baseline covariates	−0.0165*** (0.0041)	−0.0042 (0.0033)
Event study: wave of exposure	−0.0194*** (0.0063)	−0.0070 (0.0050)
Event study: one wave after	−0.0227*** (0.0069)	−0.0064 (0.0054)
Pre-trend joint p (leads −4, −3, −2)	0.678	0.660
Pre-exposure mean	0.0624	0.0372
Effect as % of pre-exposure mean	−26.4	−11.3
Observations, covariate rows	225,372	225,372
Observations, no-covariate row	232,778	232,778

All columns include village and province $\times$ wave fixed effects, with the always-treated cohort excluded, so no already-treated village serves as a control. Standard errors clustered at the village level. Covariates are PODES 2003 baseline characteristics interacted with wave dummies. The two event-study rows come from a single regression that also includes leads at −4, −3 and −2 and lags at +2, +3 and +4; the latter are jointly indistinguishable from zero (composite $p = 0.38$), so the effect is confined to the two waves following exposure. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

1 F Pre-Treatment Balance

2 How comparable were eventually-treated and never-treated villages before any
3 treatment occurred? Table 18 compares the two groups on ten village characteristics
4 measured in levels at the 2005 wave, which pre-dates first exposure for every
5 identified cohort. The purpose is not to demonstrate perfect random assignment;
6 it is to show that, within the KRB Medium/High hazard-zone sample, treated and
7 control villages were closely comparable before the shocks arrived.

8 Nine of the ten normalised differences (Imbens and Wooldridge, 2009) fall
9 below the conventional 0.25 threshold. The exception is the number of health
10 centres, at 0.423, which we flag rather than average away; the next largest are
11 the primary-school count (0.246) and the police post (0.176). Eventually treated
12 villages are larger than never-treated ones (4,308 against 3,656 residents, normalised
13 difference 0.148) and better provisioned on several margins, which is what one

Table 18: Pre-Treatment Balance at the 2005 Wave: Eventually-Treated vs. Never-Treated Villages, KRB Medium/High

	Eventually treated	Never treated	Norm. diff.
Communal conflict (binary)	0.040	0.026	0.081
Population	4308.494	3655.616	0.148
Number of households	1071.929	920.837	0.139
Paved road access	0.669	0.604	0.136
Household electricity	0.947	0.960	-0.061
Distance to district capital (km)	26.350	29.344	-0.126
Number of primary schools	3.374	2.672	0.246
Police post present	0.175	0.114	0.176
Number of health centers	0.757	0.456	0.423
Internet available	0.040	0.035	0.025

Village characteristics measured in levels at the 2005 PODES wave, which pre-dates first exposure for every identified treatment cohort. Eventually treated: villages with window-aligned first $\text{MMI} \geq \text{VI}$ exposure in 2008–2024 ($n = 2995$); never treated: villages in the same KRB Medium/High hazard zones with no $\text{MMI} \geq \text{VI}$ exposure through 2024 ($n = 30259$). Norm. diff. is the normalized difference of Imbens and Wooldridge (2009), $(m_T - m_C) / \sqrt{(s_T^2 + s_C^2)/2}$, a scale-free measure whose conventional threshold for concern is 0.25. This is a descriptive balance check, not a hypothesis test, so no standard errors or significance stars apply.

would expect if seismically active districts differ systematically from the rest of the country. Baseline conflict differs by 1.4 percentage points (0.040 against 0.026, normalised difference 0.081).

Because the health-centre difference sits above the conventional threshold, we ask directly whether it is load-bearing. Adding the baseline health-centre count interacted with wave moves the estimate by less than one percent on every outcome, and trimming villages in the top decile of baseline health-centre provision leaves every estimate significant, with a largest movement of 10 percent on inter-village conflict. The imbalance is visible but it is not doing the work: remaining level differences are absorbed by village fixed effects, and the pre-trend tests of Section 5.4 address differential trends.

G Reporting-Artifact Placebo Outcomes

An empirical falsification test. The four arguments of Section 5.2.1 are observational; we also test the artifact story directly. If post-earthquake conditions distorted what village heads report, the distortion should surface in other items filed

1 by the same respondent on the same form. We construct two placebo outcomes
2 with no plausible earthquake channel, the village head’s age and an indicator
3 for a female head, and estimate them under both fixed-effect schemes alongside
4 communal conflict on the identical subsample. The two specifications disagree,
5 and the disagreement is itself informative.

6 Under wave fixed effects the placebo fails. Head age moves under both estima-
7 tors, by -0.54 years under CS(2021) and -0.72 under two-way fixed effects, each
8 significant at 1 percent. The female-head indicator moves under two-way fixed
9 effects (-0.0094 , significant at 5 percent) though not under CS(2021) (-0.0060 ,
10 $p = 0.26$). Both items reject their joint pre-trend test ($p = 0.004$ for age, $p < 0.001$
11 for the female indicator), so under wave effects these series are not behaving like
12 placebos at all. Under the province-by-wave effects that Section 4.1 selects, both
13 pass their pre-trend test ($p = 0.76$ and $p = 0.59$) and the female-head placebo
14 becomes a precise null (-0.0001 , $p = 0.98$). Head age does not become a null: it
15 reverses sign to $+0.45$ years, about five months, and is significant at 5 percent.
16 We report that rather than describe the placebo as passing outright. A reporting-
17 capacity story predicts that exposure degrades what the head records, and the
18 sign reversal runs the other way; what the exercise establishes is that no common
19 distortion moves the apparatus items and the conflict item together, not that the
20 apparatus items are motionless. Communal conflict on the identical subsample
21 falls by 2.7 percentage points under wave effects and 2.3 under province-by-wave,
22 significant at 1 percent in both.

23 The naive TWFE benchmark does move on these placebos, but only because
24 both trend strongly over the panel, which is exactly where staggered-timing
25 weighting bias appears and the heterogeneity-robust estimator is required. Table 19
26 tabulates the full results.

27 This is the same pathology the conflict outcome shows, and it has the same
28 source. Both apparatus variables trend strongly across the panel, head age from
29 45.0 to 48.3 years and the female share from 0.027 to 0.072, and those trends differ

Table 19: Placebo Outcomes from the Same PODES Form: Village-Head Age and Sex
2005-boundary panel, window-aligned first exposure; waves 2005–2024 excluding 2011

	Placebo outcomes		Same-sample
	Head age (years)	Female head	Communal conflict
<i>Panel A: wave fixed effects</i>			
CS(2021) ATT	-0.5392*** (0.2023)	-0.0060 (0.0054)	
pre-trend χ^2 p	0.004	<0.001	
TWFE	-0.7171*** (0.1726)	-0.0094** (0.0042)	-0.0270*** (0.0040)
<i>Panel B: province $\times$ wave fixed effects</i>			
TWFE	0.4452** (0.2178)	-0.0001 (0.0049)	-0.0225*** (0.0050)
pre-trend joint p	0.755	0.594	0.121
Observations	159,449	159,449	159,449

Placebos are village-head age and a female-head indicator from the raw PODES apparatus module (six waves; 2011 excluded, coding unverifiable), aggregated to constant 2005-parent boundaries. The third column re-estimates communal conflict on the identical subsample, so effect and placebo are read off the same villages and waves under the same specification. Pre-trend tests bin the endpoints ($\text{rel} \leq -5$ and $\text{rel} \geq +5$) so that the reference category is $\text{rel} = -1$ alone rather than a mixture that includes post-treatment cells. Standard errors are clustered at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

by region. A staggered design that absorbs only wave effects attributes part of such a trend to treatment whenever cohorts enter at different points on it; absorbing province-by-wave effects removes it. That the placebos fail exactly where the conflict pre-trend fails, and pass exactly where it passes, is independent support for the specification rule rather than a separate problem.

H Pre-Trend Diagnostics Across Estimators

This appendix collects the pre-trend evidence that Sections 4 and C summarise: what bounds the rejection under wave fixed effects, the power of the pre-test itself, and how the verdict moves across estimator families.

What bounds the wave-effects rejection

Two things bound what that rejection can imply. It is specific to wave fixed effects, though the comparison cannot use the identical statistic: `csdid` cannot absorb

1 interacted fixed effects, so the province-by-wave evidence comes from the joint
 2 test of the event-study leads, which does not reject ($p = 0.68$ composite, $p = 0.66$
 3 *warga*), and the saturated Sun–Abraham event study, which can absorb them,
 4 agrees ($\chi^2(5) = 4.69$, $p = 0.46$). And the drift the data do exhibit runs *against*
 5 the finding rather than toward it: the unconditional group-time average pre-trend
 6 is positive ($+0.0107$, $p = 0.01$), as is that of the specification least exposed to
 7 bad-control concerns ($+0.0151$, $p < 0.01$), and a positive pre-existing drift biases
 8 a subsequent decline toward zero instead of manufacturing it. That argument
 9 holds against a drift that *continues* into the post-treatment period, not against a
 10 transitory elevation that *reverts* by the time of exposure; conditioning on a passed
 11 pre-test is itself a selection step, and it displaces the wave-of-exposure estimate by
 12 about 0.004, again toward zero.

13 **Pre-test power**

14 We also ask whether the test had the power to detect a violation in the first place
 15 (Roth, 2022), and two points follow. The calculation refers to the pre-test the
 16 **pretrends** package implements, which flags a violation when any individual pre-
 17 treatment coefficient is significant, and not to the joint Wald test reported above.
 18 First, a pre-trend large enough to survive the test would displace the estimate
 19 *toward* zero, not away from it, so it would understate rather than manufacture
 20 the negative composite effect, the same direction as the growth imbalance in
 21 Appendix M. Second, and less comfortably, the *warga* effect is small relative to
 22 the violation the test would miss half the time: a linear pre-trend of $+0.0046$ per
 23 wave on the Callaway–Sant’Anna estimates accumulates over two waves to more
 24 than the pooled *warga* ATT of -0.0075 . We therefore do not read “*warga* passes
 25 its pre-trend test” as evidence in its own right, and rest the definition-stability
 26 argument on the composition contrast of Section 5.2.1, which rests on a different
 27 assumption rather than on none, and which we state there.

1 The pre-trend verdict across estimator families

2 The pre-trend verdict turns on the fixed-effect structure rather than on the estima-
3 tor, and we state the pattern in one place because it is the evidence behind the
4 specification choice of Section 4.1. Under wave fixed effects the joint pre-treatment
5 test rejects in every estimator family we run: Callaway–Sant’Anna ($\chi^2(15) = 42.9$,
6 $p < 0.001$ composite; $\chi^2(15) = 29.4$, $p = 0.014$ *warga*), the imputation estimator
7 of Borusyak, Jaravel, and Spiess (2024) (a lead two waves before exposure of
8 $+0.023$, $p = 0.04$), a Sun–Abraham lead two waves before exposure significant
9 at 5 percent, and a nearest placebo of 0.010 under the episodic estimator of de
10 Chaisemartin and D’Haultfoeuille (2026). Under province-by-wave effects it passes
11 in every family that can absorb them: the dynamic specification ($p = 0.68$ com-
12 posite, $p = 0.66$ *warga*), the saturated Sun–Abraham event study of Figure 1
13 ($\chi^2(5) = 4.69$, $p = 0.46$), and the imputation estimator, whose three leads are
14 individually insignificant there. That the verdict flips with the fixed effects and not
15 with the estimator is what the economic argument of Section 4.1 predicts, and it
16 is a stronger statement than the diagnostic alone could support: the specification
17 is not merely the one that passes, it is the one that passes across estimators that
18 make different assumptions about treatment-effect heterogeneity. What this does
19 not buy is precision. The imputation estimator that clears the pre-test under
20 province-by-wave effects cannot estimate the effect there ($+0.013$, standard error
21 0.031), because village and province-by-wave effects together absorb nearly all
22 the variation its imputation step uses. The point estimates converge and the
23 diagnostics agree on the specification; the cost is that the design has less power
24 the more of the regional counterfactual we absorb, and we report the bounds of
25 Section 5.4.1 rather than read any single pre-test as settling the matter.

26 Exposure timing and the wave-boundary artifact

27 Two timing points that Sections 1 and 4 state in one sentence each are set out
28 here.

The calendar-year artifact. We document a pitfall for work that merges the timing of physical shocks onto periodic surveys. In panels with unequally spaced waves, calendar-year exposure lags misaligned with the enumeration month both conceal the true response and manufacture a spurious one: on an identical sample, dating exposure by calendar year attenuates the two-wave effect from -0.022 to -0.003 and renders it insignificant, while producing a significant coefficient one lag later that the aligned specification shows to be zero (Table S31). We also provide the first village-level quasi-experimental estimate of the earthquake–conflict relationship for Indonesia.

Resolving a pre-period signal. An earlier calendar-year coding of first exposure produced a marginally significant pre-period coefficient; diagnostics traced it to wave-boundary timing that misplaced onset by one wave for about 21 percent of ever-treated villages, and re-timing exposure to the first wave enumerated after the event removes it (Appendix H). The joint Wald rejection under wave fixed effects remains and we carry it as an open caveat; the HonestDiD bounds (Section 5.4.1), not the pre-trend tests, are the binding constraint.

I Additional Robustness Detail

The robustness battery that does not bear on the headline contrast is reported in Supplementary Material Section S7, which carries the alternative clustering, the catalogue-window and threshold sensitivity, the reporting placebos, the displacement rings, and the estimator bridge, with their tables.

Cohort dependence, placebos and randomisation inference

Section 5.4.1 summarises these checks; the full statement follows. No single cohort drives the result: dropping each in turn leaves every estimate negative and significant, from -0.0127 to -0.0250 , and giving every cohort equal weight rather than its share of treated units leaves the effect larger, not smaller (-0.0318 ,

1 standard error 0.0122), so the pooled estimate is not an artifact of cohort-share
 2 weighting. Because two of the largest events fall in the same cohort, this is a
 3 leave-one-cohort-out and *not* a leave-one-earthquake-out check, and we do not
 4 claim the stronger version (Supplementary Material Section S7). A forward-lag
 5 placebo is a clean null (-0.0029 , $p = 0.60$), and a spatial randomisation that
 6 relocates each earthquake’s footprint to a random never-treated village produces
 7 exactly one placebo draw at least as negative as the actual estimate of -0.0199 over
 8 500 relocations, a one-sided $p = 0.004$ (Supplementary Material Table S15). The
 9 placebo distribution is centred near zero at $+0.0007$ but its dispersion is 2.1 times
 10 the clustered standard error, which is itself a warning that village-clustered inference
 11 understates uncertainty for a treatment assigned by earthquakes; recentring the
 12 draws on that mean gives a two-sided $p = 0.020$. The same exercise applied to the
 13 compositional divergence rather than to the level is weaker and we report it here
 14 rather than leave the asymmetry: relocating footprints against the village-level
 15 theft-minus-*warga* divergence puts the actual divergence of $+0.058$ at 1.7 standard
 16 deviations of the placebo distribution, with 21 of 500 draws at least as large, a
 17 one-sided $p = 0.044$. Both the level and the divergence clear 5 percent on this test,
 18 the divergence by a narrower margin.

19 **J Composition of Disorder: Identifying Assump-** 20 **tion and Benchmarks**

21 This appendix states the identifying assumption behind the composition contrast
 22 of Section 5.2.1, names the threats that survive the differencing, and reports the
 23 benchmark that estimates each outcome separately.

24 The contrast carries its own identifying assumption, and we state it rather than
 25 present the design as assumption-free. Estimating the outcomes on a common
 26 sample with outcome-interacted fixed effects is equivalent to estimating the same
 27 difference-in-differences on a single differenced outcome, theft minus communal

1 conflict. What it requires is parallel trends in that difference: absent the earthquake,
 2 the gap between reported theft and reported communal conflict in treated villages
 3 would have evolved as it did in villages not yet exposed, and exposure changes the
 4 reporting of the two items, if at all, only by a common amount. This is weaker
 5 than the level effect's assumption in one specific respect. Any village-wave shock
 6 that moves both items equally differences out, which covers a uniform disruption of
 7 the village head's reporting, post-disaster enumeration conditions, and mechanical
 8 depopulation. Three threats survive the differencing and we name them. A shock
 9 correlated with exposure that moves the two items differentially does not cancel:
 10 an incentive to document property losses for aid claims would inflate reported
 11 theft without inflating reported communal conflict, and what weighs against that
 12 account is that robbery, also a property loss on the same form, does not rise with
 13 theft. That comparison assumes the documentation incentive loads similarly on the
 14 two items, which is not automatic: an unwitnessed loss is easier to claim than one
 15 requiring an account of a violent encounter, so we treat this as weighing against
 16 the account rather than closing it. Because theft's pre-exposure rate is roughly
 17 ten times the *warga* rate, a reporting shock proportional to each item's level does
 18 not cancel in an absolute difference; the earthquake-level equivalence bound on
 19 *warga* excludes a proportional surge large enough to produce theft's rise. And the
 20 differenced outcome has a pre-trend of its own. We test it directly and report the
 21 result specification by specification, because it does not point the same way in
 22 both. Under the preferred province-by-wave effects the joint test over the interior
 23 pre-exposure leads does not reject ($p = 0.31$ against the composite, $p = 0.18$ against
 24 the definition-stable series), and the two leads nearest exposure are precise nulls
 25 of -0.001 and -0.009 . Under wave effects the same test rejects ($p = 0.018$ and
 26 $p = 0.040$). In both, the binned term collecting periods five waves and more before
 27 exposure is positive and significant, at $+0.045$ (standard error 0.020) under the
 28 preferred effects, so a joint test that includes it sits at $p = 0.06$. We do not claim
 29 the contrast dispenses with pre-trend concerns. We claim something narrower:

1 under the specification our selection rule prefers, the near pre-exposure period is
2 flat, and what remains is a level difference at horizons so distant that a single
3 cohort supplies most of the cell.

4 An earlier draft of this paper reported a rejection at $p < 0.001$ for this object
5 and rested the composition result on the earthquake-level divergence instead. Two
6 things changed. The endpoints are now binned rather than dropped, which is the
7 standard treatment and which we should have applied from the start. And the
8 crime module turns out to be fielded in all seven waves rather than five: the two
9 missing waves were absent from our extraction, not from the survey (Section 3.2).
10 Recovering them removes the ten-year hole between 2008 and 2018 that made
11 event time on this sample non-comparable in calendar time, and raises the sample
12 from 211,082 to 300,018 village-waves. The earlier rejection was in part an artifact
13 of that hole. We report the corrected tests here and in Table 20, and we continue
14 to report the earthquake-level divergence alongside them rather than in place of
15 them.

16 The definition-stable anchor

17 Because the composite communal measure changes definition in 2008, we anchor
18 the contrast on the fifth outcome, *warga*, the civilian-versus-civilian communal
19 subtype whose survey wording is near-stable across waves. The *warga* coefficient
20 on its own is -0.0085 (standard error 0.0027 , $p = 0.002$), a decline of just over half
21 a percentage point on a 1.6 percent base, and it survives the Benjamini–Hochberg
22 correction across the five outcomes ($q = 0.002$). A test of the equality of the *warga*
23 and theft effects is rejected ($p < 0.001$), and the rejection holds under kabupaten
24 clustering ($F = 5.04$, $p = 0.025$). The composition result therefore does not depend
25 on the composite outcome or on its 2008 definition change: the definition-stable
26 communal series falls while theft does not, and the equality of their effects is
27 formally rejected.

1 The contrast in event time

2 The first-time margin fails a falsification run inside the pre-exposure period, and
3 the same test has to be put to the contrast this paper rests on. It does better, but
4 not cleanly, and we report both halves. Table 20 traces the theft-minus-communal
5 difference in event time with the endpoint cells binned rather than dropped, so
6 the reference category is the wave before exposure alone and the coefficient at t_0
7 is itself the jump at exposure. Under the province-by-wave effects our selection
8 rule prefers, the three interior pre-exposure coefficients are +0.027, -0.009 and
9 -0.001 and their joint test does not reject ($p = 0.31$); the difference then rises to
10 +0.025 in the wave of exposure and +0.040 in the wave after. The definition-stable
11 version behaves the same way in the pre-period ($p = 0.18$). Under wave effects
12 the same interior test rejects ($p = 0.018$) while the jump at exposure is larger and
13 sharper (+0.034, $p = 0.007$).

14 What does not settle down in either specification is the binned term collecting
15 periods five waves and more before exposure, which is +0.045 (standard error
16 0.020) under the preferred effects. A joint test that includes it sits at $p = 0.06$.
17 That term is a single cohort observed in a single calendar wave nineteen years
18 before its own exposure, which is the configuration endpoint binning exists to
19 handle, and we regard it as weak evidence of a pre-exposure level difference rather
20 than of a trend. But it is evidence, and we report the joint test both ways rather
21 than only the version that passes.

22 We also correct the record on how this table came to read as it does. Earlier
23 drafts of this paper reported a rejection at $p < 0.001$ for this object and explained
24 it as contamination of the reference category. That explanation was wrong: the
25 earlier specification gave every event time its own dummy and pooled nothing.
26 Two things actually changed. The endpoints are now binned rather than dropped,
27 which is the standard treatment; and the crime module turns out to be fielded in
28 all seven waves rather than five, so the ten-year hole between 2008 and 2018 that
29 made event time non-comparable in calendar time is gone (Section 3.2). Before

- 1 that hole was closed, the cells being differenced drew on largely non-overlapping
2 cohorts. They no longer do.

Table 20: The Composition Contrast Before and After Exposure

The near pre-exposure period is flat under the preferred specification; the distant binned endpoint is not.

	Theft – communal		Theft – warga	
	Wave FE	Prov. × wave	Wave FE	Prov. × wave
$t_{\leq -5}$ (binned)	0.0545 (0.0172)	0.0454 (0.0202)	0.0416 (0.0169)	0.0419 (0.0199)
t_{-4}	0.0053 (0.0171)	0.0266 (0.0192)	0.0041 (0.0166)	0.0333 (0.0187)
t_{-3}	-0.0347 (0.0167)	-0.0091 (0.0187)	-0.0335 (0.0159)	-0.0053 (0.0179)
t_{-2}	-0.0301 (0.0132)	-0.0008 (0.0154)	-0.0232 (0.0126)	0.0009 (0.0148)
joint pre-trend p , interior leads	0.018	0.308	0.040	0.182
joint pre-trend p , incl. binned bin	0.000	0.061	0.000	0.066
t_0	0.0335 (0.0125)	0.0246 (0.0146)	0.0194 (0.0121)	0.0126 (0.0141)
t_{+1}	0.0685 (0.0128)	0.0402 (0.0155)	0.0543 (0.0125)	0.0259 (0.0150)
p on t_0 (the jump from the t_{-1} reference)	0.007	0.091	0.109	0.371
Memo: change t_{-2} to t_0	0.0636	0.0254	0.0426	0.0117
p	0.000	0.108	0.001	0.447

Event-study coefficients on the differenced outcome, the object the composition result rests on. The reference period is t_{-1} , and the endpoints are binned rather than dropped, so the coefficient on t_0 is itself the jump at exposure. We report the joint pre-trend test twice: over the interior leads, and again including the binned far-lead term, because excluding that term is what made an earlier draft of this table read as a clean pass. The memo row gives the change from t_{-2} , which is the most negative lead, and is therefore an upper bound on the movement at exposure rather than the movement itself. Standard errors, clustered at the village, in parentheses. $N = 229,871$ village-waves on the crime-module common sample, all seven waves.

3 The divergence at the earthquake level, worked through

- 4 The panel evidence above and the earthquake-level divergence of Section 5.2.1
5 answer different questions, and a reader is entitled to ask them together. The
6 earthquake-level estimate fixes the unit of inference, since it treats the earthquake
7 rather than the village as the assignment, but it does not by itself impose the
8 province-specific counterfactual this paper prefers elsewhere. Correct inference
9 and a credible counterfactual are not the same requirement, and we test them

1 jointly rather than leave the gap open. Table 21 reports two ways of doing so.
2 The first re-estimates D_e event by event with province-by-wave rather than wave
3 fixed effects. The second compares, within the same earthquake, villages shaken
4 at $\text{MMI} \geq \text{VI}$ against villages that felt that same shock at MMI in $[4, 6)$, taking
5 theft minus *warga* as the dependent variable directly, so that province-specific
6 trajectories difference out by construction rather than by assumption.

7 The divergence stays positive under both. It does not stay significant under
8 either. Before reading that as a failed test rather than an imprecise one, note what
9 the same within-earthquake design says about its own pre-period: running the
10 differenced outcome on pre-exposure waves only, the placebo is +0.031 (standard
11 error 0.020, $p = 0.13$). It does not reject, but it is larger than the estimate it
12 would be validating, so we do not read it as a pass. By the standard this paper
13 applies to the first-time margin in Section 5.3, a placebo of that size leaves the
14 within-earthquake contrast uninformative rather than confirmed, and we say so
15 here rather than claim the architecture validated. The treated-village-weighted
16 mean falls from the +0.076 of Section 5.2.1 to +0.0155 with province-by-wave
17 effects, and its bootstrap interval contains zero. The within-earthquake design
18 in Panel B gives +0.0221 (bootstrap standard error 0.0332), positive in 13 of 18
19 earthquakes, with an equal-event mean of +0.0476 ($p = 0.05$). Pooling over all
20 stacks gives +0.0217 (standard error 0.0100, $p = 0.03$) under village clustering
21 and the same coefficient with $p = 0.47$ once the earthquake, which is the unit of
22 assignment, is the cluster. We read the last of those as the governing number.

23 How much power this design has is a question we can answer rather than
24 assert, and we answer it with the calculation we apply to the onset margin in
25 Supplementary Material Section S11. At 80 percent power against a two-sided
26 test at 5 percent, the smallest effect a specification can reject is 2.802 times its
27 standard error. Taking the earthquake as the cluster, that minimum detectable
28 effect is 0.0832 for the pooled within-earthquake estimate, 0.0930 for its treated-
29 village-weighted counterpart and 0.0723 under province-by-wave effects, against

1 point estimates of 0.0217, 0.0221 and 0.0155. Every one of these specifications is
2 therefore between 3.3 and 4.7 times too coarse to detect an effect the size of the
3 one it produces, and even the tightest of them would need an effect nearly twice
4 its own estimate before it would reject. Their failure to reject consequently carries
5 little information about whether the divergence is present within earthquakes.
6 Part of this is mechanical, since the median earthquake reaches only one or two
7 provinces and the fixed effects absorb most of the within-event variation, and
8 the within-earthquake design retains fifteen of 24 earthquakes once both sides
9 need twenty villages. But we do not read the loss of significance as an artifact
10 of power alone, and we do not claim the exercise rescues the result. What it
11 establishes is narrower and worth stating plainly: the composition contrast is
12 identified from comparisons across regions, it is positive in sign wherever we look,
13 and it is not demonstrated within earthquake at conventional levels. The evidence
14 for recomposition rests on the consistency of that sign across weightings, designs
15 and inference units, not on a within-earthquake estimate that clears a threshold.

16 The *warga* arm of the divergence deserves its own statement. The *warga*
17 response is not merely indistinguishable from zero: it is negative and separated
18 from zero at the shock level. Estimated on the full *warga* sample rather than on the
19 narrower crime-module common sample that the divergence uses, its treated-village-
20 weighted effect is -0.0114 with an earthquake block-bootstrap 95 percent interval
21 of $[-0.0260, -0.0007]$, which excludes zero; the corresponding arm inside Table 22,
22 computed on the common sample, is -0.008 . Civilian communal violence therefore
23 falls by about a third of its 3.7 percent pre-exposure rate under earthquake-level
24 inference. The finding that it does not rise is therefore an informative equivalence
25 bound, not a mere failure to reject.

26 We do not attribute this to over-absorption; the over-absorption arithmetic
27 below sets out why.

28 What survives the paper's own specification rule is therefore the narrower claim:
29 theft and communal conflict move in opposite directions and differ significantly

Table 21: Composition Divergence at the Earthquake Level, under the Preferred Counterfactual

	Panel A Province $\times$ wave	Panel B Within-earthquake
Earthquakes with support	17	18
Earthquakes dropped for lack of support	2	6
Treated-village-weighted mean D_e	+0.0155	+0.0221
bootstrap standard error	(0.0258)	(0.0332)
bootstrap 95% CI	$[-0.0267, +0.0755]$	$[-0.0290, +0.1014]$
leave-one-earthquake-out range	$[-0.0009, +0.0322]$	$[+0.0000, +0.0445]$
Equal-event mean D_e	+0.0202	+0.0476
two-sided p	0.38	0.05
events with $D_e > 0$	8 of 17	13 of 18
Pooled over all stacks, village-clustered	—	+0.0217 (0.0100), $p = 0.03$
earthquake-clustered	—	+0.0217 (0.0297), $p = 0.47$

The paper’s headline composition result estimates the divergence $D_e = ATT_e^{\text{theft}} - ATT_e^{\text{warga}}$ earthquake by earthquake under village and wave fixed effects, and obtains +0.076 with a bootstrap standard error of 0.033. This table imposes the counterfactual the paper prefers elsewhere. Panel A repeats the event-by-event estimation with province-by-wave rather than wave fixed effects; because the median earthquake reaches only one or two provinces, the fixed effects absorb most of the within-event variation and two earthquakes lose support entirely. Panel B avoids that problem by comparing, within the same earthquake, villages shaken at $\text{MMI} \geq \text{VI}$ against villages that felt the same shock at MMI in [4, 6), with the differenced outcome (theft minus *warga*) as the dependent variable, so province-specific trajectories difference out by construction; earthquakes with fewer than twenty treated or twenty felt-control villages are dropped. The divergence stays positive in sign in every column and under both weightings. The treated-village-weighted intervals contain zero in both panels. Within earthquake the equal-event mean and the village-clustered pooled estimate do reach conventional significance, but the earthquake is the unit of assignment, and clustering there returns $p = 0.47$. We therefore report this as a limit on the composition result, not as support for it.

- 1 under both fixed-effect structures, while the sharper contrast with the definition-
- 2 stable *warga* series holds under wave effects and not under province-by-wave. The
- 3 main text states the claim at that width.

Table 22: Composition at the Shock Level: Theft versus *Warga* per Earthquake
The theft-versus-warga divergence, aggregated to the earthquake, clears earthquake-level inference under both weightings, where the communal decline alone does not.

	Theft	Warga	Divergence
Treated-village-weighted mean	0.068	-0.008	0.076
Equal-event mean	.	.	0.063
Equal-event two-sided p-value	.	.	0.023
Bootstrap standard error	.	.	0.033
95% CI lower bound	.	.	0.010
95% CI upper bound	.	.	0.142
Two-sided bootstrap p-value	.	.	0.025
Leave-one-out min	.	.	0.057
Leave-one-out max	.	.	0.095
Events with D_e greater than 0	.	.	12.000
Usable earthquakes (total)	.	.	19.000

For each of the 19 usable earthquakes (onset PODES wave 2008-2024, at least 20 in-sample first-treated villages) we estimate the absorbing treated-cohort effect on theft and on the definition-stable warga outcome against never-treated controls, with village and wave fixed effects, and form the per-event divergence: the theft effect for that earthquake minus its warga effect. The treated- village-weighted mean weights each earthquake by its number of unique first-treated villages; the equal-event mean weights every earthquake alike (blank cells are not defined for that row). The earthquake bootstrap resamples whole earthquakes with replacement (9,999 draws, seed 42) and recomputes cohort weights each draw; the leave-one-out rows give the range of the treated-weighted divergence when each earthquake in turn is dropped. Divergence is positive (theft rises faster than warga) in 12 of 19 events. Missing cells (Theft/Warga columns below row 1) are statistics defined only for the divergence.

1 The proportional scale, in full

2 Section 5.2.1 summarises the proportional evidence; this appendix reports it in full,
3 including the robbery and assault coefficients and the clustered and multiple-testing
4 variants.

5 One objection to a contrast stated in percentage points is that the outcomes
6 have very different base rates, so a difference in levels need not be a difference
7 in proportional response. We therefore re-estimate the same stacked, outcome-
8 interacted specification as a fixed-effects Poisson model, in which the coefficients
9 are proportional rather than absolute effects. The divergence survives the change
10 of scale, and on this scale it is the communal decline that is the larger movement:
11 the composite communal measure falls by 31.4 percent (-0.378 , standard error
12 0.087) and *warga* by 17.1 percent (-0.188 , standard error 0.108), while theft rises
13 by 14.0 percent (0.131 , standard error 0.017). Equality of the theft and communal
14 proportional effects is rejected at $p < 0.001$, and equality of theft and *warga* at

¹ $p = 0.003$. The comparison the framework signs survives the same change of
² scale: theft and robbery differ proportionally at $p = 0.017$. This is the demanding
³ version of the test, because robbery's base rate is an order of magnitude below
⁴ theft's, so an account in which the shock raises predation proportionally across
⁵ property crime would show up here and does not. Robbery's own proportional
⁶ coefficient is -0.0590 (standard error 0.0800), a 5.7 percent decline that is not
⁷ distinguishable from zero, which is why we rest this margin on the contrast rather
⁸ than on robbery's level. Assault's proportional coefficient is $+0.1397$ (0.0616), a
⁹ 15.0 percent rise, so the absolute null for assault should not be read as a tight
¹⁰ zero on every scale. Read proportionally the communal decline is in fact the
¹¹ larger of the two movements, which is the opposite of the impression the absolute
¹² coefficients give, and the composition result does not rest on the scale in which it is
¹³ expressed. The *warga* decline survives kabupaten clustering, at -0.0085 (standard
¹⁴ error 0.0036), so unlike in earlier drafts we do not have to set that coefficient aside.
¹⁵ Both contrasts survive spatial dependence: re-clustering at the kabupaten level,
¹⁶ the equality of theft and each communal measure is still rejected (Table 4). The
¹⁷ composite communal coefficient remains significant under kabupaten clustering
¹⁸ (-0.0218 , SE 0.0061) while the *warga* coefficient does not (SE 0.0037). Clustering
¹⁹ is not earthquake-level inference, so we still do not read this table as fresh shock-
²⁰ robust evidence that communal conflict falls; that claim rests on the staggered
²¹ estimators of Section 5. What the table does establish is the divergence: theft rises
²² where communal violence, however measured, does not. Robbery, the property-
²³ *violent* act, does not rise (-0.0141^{***}), and assault is null. The composition is
²⁴ therefore not an artifact of comparing outcomes measured on different samples,
²⁵ waves, or control sets: individual economic predation rises exactly where collective
²⁶ and property-violent acts do not. Nor is it an artifact of testing several outcomes
²⁷ at once. Correcting the five per-outcome p -values for multiple testing ([Benjamini](#)
²⁸ [and Hochberg, 1995](#)) leaves four of them significant at the 5 percent level: theft's
²⁹ rise and the communal, robbery and *warga* declines, with q -values of 0.000, 0.000,

1 0.000 and 0.002. Only assault, at $q = 0.864$, adjusts to a clear null (Table 4). We
2 none the less continue to rest the argument on the composition contrast rather
3 than on the *warga* coefficient, for the reasons given in Section 5.2.1: surviving a
4 multiple-testing correction is not the constraint that binds it.

5 The theft-versus-robbery margin

6 The contrast that carries the argument against a raised-motive account is theft
7 against robbery, and it deserves the same treatment we gave theft against communal
8 conflict rather than an appeal to the design. Because both items are recorded by
9 the same official on the same form for the same village and wave, the comparison
10 is a difference-in-differences in the relative outcome, and what it requires is parallel
11 trends in that relative outcome, not in each level separately; the biases of the two
12 component comparisons need to be equal, not zero (Olden and Møen, 2022). That
13 is why we test the difference.

14 The result is mixed and we report both halves. Over the full event window
15 the joint test of the pre-exposure leads of theft minus robbery rejects under
16 the preferred effects ($F = 4.09$, $p = 0.001$) and under wave effects ($F = 5.22$,
17 $p < 0.001$). The rejection is not spread across the pre-period: it is carried entirely
18 by the most distant lead, whose coefficient is $+0.129$ (standard error 0.037) under
19 the preferred effects, while the remaining four are individually insignificant and
20 small (-0.035 , $+0.013$, -0.013 , $+0.007$, reading from the most distant). The
21 reason is structural rather than substantive. That most distant lead is a single
22 cohort observed in a single calendar wave, nineteen years before its own exposure,
23 which is the configuration endpoint binning exists to handle; binned rather than
24 dropped, it remains positive and significant, and we report the joint test both ways
25 in Appendix J rather than only the version that passes.

26 An earlier draft defended this margin by restricting attention to a contiguous
27 block of waves, on the grounds that the crime module skipped 2011 and 2014 and
28 left event time irregularly spaced. That defence is no longer needed and we have

1 withdrawn it: the module is fielded in every wave, so spacing across the panel is
2 regular by construction and the full window is the right one to test. We nonetheless
3 treat the theft-minus-robbery margin as not cleanly pre-tested, because the distant
4 lead rejects under both specifications and we decline to read a rejection away.

5 Two further results belong here. Robbery’s own level pre-trend rejects under
6 every window we examined, which is why we do not read its level causally, exactly as
7 with theft. Assault’s does not reject ($p = 0.27$ under the preferred fixed effects, and
8 in every narrower window), so the assault comparison rests on a tested assumption
9 rather than an assumed one. Finally, our specification rule selects the simplest
10 specification that passes a conditional parallel-trends test, which is itself a pre-test
11 selection rule, and we read the resulting intervals in that light rather than as exact
12 (Roth, 2022).

13 The benchmark estimated outcome by outcome

14 The same design estimated outcome by outcome under the CS(2021) benchmark,
15 on four acts that differ in whether they are collective or individual and economic
16 or violent, is reported in Appendix Table 23. The benchmark confirms the same
17 gradient. Theft *rises* sharply, by 7.6 percentage points under CS(2021) ($p < 0.001$,
18 about 17 percent of its 45 percent base) and by 4.3 points on the common sample,
19 though its own pre-trend does not pass (-0.042 , $p = 0.04$), which is one of the two
20 reasons we do not rest the argument on theft’s level. Because theft can also rise
21 from weaker guardianship, abandoned dwellings, aid inflows or changed reporting,
22 we do not read its magnitude as a standalone causal estimate; what we read from
23 the comparison is the *contrast* with communal conflict, which does not require
24 it. Robbery does not rise, at an insignificant -0.0042 under CS ($p = 0.38$) and
25 negative on the common sample (-0.0141^{***}), so what climbs is specifically the
26 non-violent, economic act. Assault is null in both. Communal conflict *falls*.

27 Table 23 estimates the same staggered design on four outcomes that differ in two
28 dimensions: whether the act is *collective* or *individual*, and whether it is *economic*

1 or *violent*. Communal conflict (column 1) is collective violence; theft (column 2) is
2 individual and economically motivated; robbery (column 3) is individual property
3 crime with violence; assault (column 4) is individual interpersonal violence.

Table 23: Mechanism Discrimination: CS(2021) by Outcome
Window-aligned first exposure, 2005-boundary panel

	Communal conflict (collective, violent) (1)	Theft (individual, economic) (2)	Robbery (property, violent) (3)	Assault (individual, violent) (4)
CS(2021) overall ATT	−0.0177*** (0.0047)	0.0762*** (0.0113)	−0.0042 (0.0047)	−0.0062 (0.0068)
Pre-trend Pre_avg (<i>p</i>)	0.016 (0.00)	−0.042 (0.04)	−0.012 (0.27)	−0.030 (0.07)
Baseline mean	0.028	0.451	0.033	0.059
Waves	7	5	5	5
Observations	225,372	160,100	160,100	160,100
Controls	Yes	No	No	No

Notes: Callaway–Sant’Anna (doubly-robust, not-yet-treated control), window-aligned first exposure to $\text{MMI} \geq \text{VI}$, nb2005 KRB Medium/High sample. Column (1) is the main specification (baseline-by-wave controls; conditional parallel trends). Crime outcomes (PODES: any incident in the past year) are available in waves 2005, 2008, 2018, 2021, 2024 and are estimated without controls because lagged/baseline controls shorten the pre-period available for the pre-trend test in this narrower 5-wave subsample. SE clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4 Retaining the always-treated cohort

5 One sample choice differs from the level analysis and we state it rather than leave
6 it to be discovered. The level specifications exclude the cohort first exposed in
7 2005, which is always treated inside the panel; the composition sample retains it.
8 Retaining it cannot bias the contrast directly, since a village whose exposure never
9 changes contributes no within-village variation to the treatment coefficient, but
10 it does help identify the wave and province-by-wave effects. Dropping the cohort
11 moves the estimates in the direction that favours our reading rather than against
12 it: under wave effects theft rises from +0.043 to +0.062, the communal decline
13 is essentially unchanged at −0.021, and the theft-versus-communal F rises from
14 57.7 to 92.9, while under province-by-wave effects the theft-versus-robbery test is
15 unchanged ($p = 0.031$ against $p = 0.036$). We report the fuller sample because it

1 is the one the crime module defines.

2 **Over-absorption arithmetic**

3 Section 5.2.1 reports that the contrast narrows under province-by-wave effects. We
4 do not attribute that narrowing to over-absorption, because the arithmetic does not
5 support that reading. Province-by-wave effects absorb 33.5 percent of the treatment
6 variance that survives village and wave effects, which predicts a standard-error
7 inflation of 1.23; the observed inflation for theft is 1.17. The loss of precision
8 is therefore fully accounted for, and the movement in the point estimate is not.
9 Comparing each coefficient against its own wave-effects confidence interval, theft
10 and communal conflict both move outside theirs, while robbery, assault and *warga*
11 remain inside. A substantial part of both effects is accordingly between-province
12 rather than within, which is what one would expect of a shock whose footprint
13 is regional, but which we cannot separate here from province-specific trends in
14 reporting or in disorder.

15 **Dropping the never-treated: the contrast on switchers alone**

16 A referee will ask what the contrast looks like when the comparison group contains
17 no never-treated villages at all, so that identification comes only from the staggering
18 of exposure. We report it, and it is the least favourable cell in the paper. Restricting
19 to the villages that switch inside the panel, dropping both the never-treated and the
20 always-treated, the contrast is sharp under wave effects: theft against communal
21 conflict rejects at $F = 26.4$ ($p < 0.001$) and against the definition-stable series at
22 $F = 12.0$ ($p < 0.001$). Under province-by-wave effects on that same subsample it
23 is a precise null: $F = 0.78$ ($p = 0.38$) and $F = 0.03$ ($p = 0.87$).

24 Here, unlike in the full sample, the over-absorption reading is the right one
25 and we show the arithmetic rather than assert it. On the switcher-only subsample,
26 village and wave effects already remove 70.5 percent of the raw variance in exposure,
27 and province-by-wave effects remove a further 72.5 percent of what survives, leaving

1 8.1 percent of the original treatment variance to identify five outcome-specific
2 coefficients and three cross-outcome contrasts. That is a different situation from
3 the full sample, where the same specification leaves enough variation that the
4 standard-error inflation is fully accounted for by the absorption and the point
5 estimates still move. We therefore read the switcher-only province-by-wave cell as
6 uninformative rather than as a rejection, and we say so here instead of omitting it.

7 **K The First-Time Margin: Design Detail**

8 This appendix carries the design detail behind the first-time margin that Section 5.3
9 states in brief and sets aside: the matched same-earthquake design and its composite
10 estimate, the choice of counterfactual within that design, the split by prior conflict
11 history, and the three narrower boundaries that Section 5.4.2 summarises. The
12 full battery, the longer look-back, the local-geography calipers and their power
13 calculation, and the independent soil measurements are in Supplementary Material
14 Section S11.

15 **The matched design and the split by prior conflict**

16 A matched counterfactual holds the shock itself fixed: for each earthquake, treated
17 villages (mean $\text{MMI} \geq \text{VI}$) are compared with villages in the same footprint that
18 felt sub-damaging shaking (mean MMI in $[4, 6)$). Because the controls also felt the
19 earthquake, this identifies the *incremental* effect of crossing the damage threshold,
20 smaller than the total effect by construction. The composite is -0.0174 , significant
21 at 1 percent under village and kabupaten clustering but not under 24-cluster
22 earthquake-level inference ($p = 0.09$); the equal-event mean over the same 24
23 events is -0.0143 ($p = 0.11$), with 14 of 24 event estimates negative. Its pre-period
24 is not clean either: the lead two waves before exposure is $+0.0136$ ($p = 0.06$), so
25 part of a same-window comparison can reflect mean reversion, and exact-matching
26 on the full pre-period conflict path and coarsened geography sharpens rather than

1 removes that caveat.

2 One design choice has to be stated before any result is read, because it decides
3 which comparison the estimates describe. The identifying variation here is *within*
4 an earthquake, so the residual threat is that villages above and below the damage
5 threshold sit on different regional trajectories. Two instruments address it, and
6 they are not interchangeable. Restricting the matched cells so that treated and
7 comparison villages share a province removes the between-province component
8 while leaving the estimator doing what the design was built to do, comparing
9 villages that felt the same shock in the same place. Absorbing province-by-wave
10 effects *on top of* the stack-by-village and stack-by-wave effects the design already
11 carries is different: an earthquake’s footprint falls almost entirely inside one or two
12 provinces, so province-by-wave effects are close to collinear with the earthquake-
13 by-wave effects already absorbed, and what they remove is largely the identifying
14 variation rather than a confound. We therefore take same-earthquake-and-same-
15 province as the primary counterfactual for the matched design on that reasoning,
16 not on the results, and report the province-by-wave version alongside it wherever
17 it is estimated, including where it weakens the estimate.

18 Splitting the matched sample by pre-period conflict is the informative step, and
19 it cuts against the average. Since the claim is about *onset*, the classification has to
20 earn that word: a village counts as having no prior conflict when it reports none
21 in *any* PODES wave observed before its earthquake, which for the later cohorts
22 is a look-back of up to nineteen years, though the median unit has two observed
23 pre-exposure waves and the mean 2.8, and the matching cells are built under that
24 same flag so the split and the comparison set use one variable (Table S33, column
25 2). Among the roughly nine in ten treated villages with *no* such history, exposure
26 *raises* the onset of communal conflict: the matched effect is +0.0076 (standard error
27 0.0023, $p = 0.001$) against a counterfactual onset rate of 2.2 percent among matched
28 controls, a relative increase of about a third. Estimated earthquake by earthquake
29 and aggregated with a bootstrap over earthquakes, the unit at which treatment

is assigned, it is +0.0082 ($p = 0.017$), positive in 16 of 23 events. Requiring treated and comparison villages to share a province as well as an earthquake, the counterfactual set out above, leaves +0.0088 ($p = 0.008$ with the earthquake as the cluster). The aggregate local decline instead loads on the minority with a prior conflict history (-0.0620 , standard error 0.0180). Cells match on the full observed pre-period conflict path, so reversion conditional on that path is differenced out; what cannot be separated there is *differential* reversion, treated villages' pre-period elevation being more transitory than their matched controls'.

Boundaries of the onset margin

The first and most serious boundary is the pre-exposure falsification reported in Section 5.3. The remaining three are narrower. The first is the wild cluster bootstrap, $p = 0.18$ on the risk-set sample under the full-history classification, against $p = 0.017$ under the earthquake-level aggregation that matches the assignment mechanism and a within-earthquake randomisation test that places the estimate outside all 500 draws. The second is spatial: when the comparison set is made local, by distance caliper or by absorbing epicentral-distance bins, the estimate stays positive but is no longer significant, and the specifications that absorb the most geography retain as little as 4.4 percent of the treatment variance (Table S34). The confidence interval of every one of those specifications contains the baseline, and the minimum effect each could detect at conventional power exceeds the effect we estimate (Table S35), so what they establish is imprecision rather than absence. Topography is still unobserved, but shear-wave velocity is not, and both it and the site-response proxy leave the margin intact in sign (Tables S36 and S37), so the specific reading in which site conditions rather than damaging shaking drive the estimate is not supported. The third is the specification: absorbing province-by-wave effects on top of the matched design leaves +0.0042 ($p = 0.10$ under village clustering, $p = 0.27$ with the earthquake as the cluster), so the onset level is not established under the rule Section 4.1 applies to the level outcome.

1 We take the shared-province matching restriction, which leaves the estimate at
2 $+0.0088$ ($p = 0.008$ with the earthquake as the cluster), as the more informative
3 test of the same concern inside a design that is already within earthquake, and
4 we report both. Across all three boundaries the contrast between margins is the
5 firmer object than either level: under the full observed history it clears the wild
6 cluster bootstrap at $p = 0.026$.

7 **L HonestDiD Sensitivity Bounds**

8 Table 24 reports the [Rambachan and Roth \(2023\)](#) HonestDiD sensitivity analysis
9 applied to the Sun–Abraham interaction-weighted event study. We report it
10 because its leads are tighter than the pooled estimators’, not because they are
11 clean: the composite t_{-2} lead is $+0.0138$ and significant at 5 percent, which is
12 exactly the violation the sensitivity analysis is meant to price. The target is
13 the first post-exposure effect t_0 (-0.0223^{***} for the composite outcome, -0.0096^*
14 for *warga*). The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ allows post-treatment
15 violations of parallel trends up to $\bar{M}$ times the largest pre-treatment violation; the
16 breakdown value is the largest $\bar{M}$ at which the robust 95 percent identified set still
17 excludes zero.

18 On a grid fine enough to locate it, the composite breakdown value lies be-
19 tween $\bar{M} = 0.40$, where the set is $[-0.040, -0.000]$, and $\bar{M} = 0.45$, where it is
20 $[-0.042, +0.001]$: the sign survives post-treatment violations up to about two fifths
21 of the largest observed pre-treatment deviation, not up to half. The definition-
22 stable *warga* target is far more fragile, breaking down between $\bar{M} = 0.05$ and
23 $\bar{M} = 0.10$, consistent with its weaker precision; reporting only that it admits zero
24 by $\bar{M} = 0.5$ would understate how early it fails. These are moderate bounds, not
25 the near-unbounded robustness a perfectly clean pre-period would deliver. We
26 report them as the quantitative statement of how much residual violation the
27 result can absorb. The doubly-robust property of the estimator is a separate

1 safeguard, against misspecification of the nuisance models rather than against
2 failure of conditional parallel trends itself.

3 A second safeguard addresses a different worry. Because the never-treated
4 pool is large and disproportionately Javanese, one might suspect the level decline
5 reflects that pool’s own trajectory rather than the effect of exposure. Dropping the
6 never-treated villages entirely, so that identification comes only from differences in
7 exposure timing among the 2,995 villages that are eventually treated, strengthens
8 rather than weakens the estimate: the composite ATT is -0.0390 (standard error
9 0.0096) with an average pre-treatment coefficient of $+0.0025$ ($p = 0.61$), and the
10 definition-stable *warga* outcome, which does not attain conventional significance
11 on the level anywhere else in the paper, reaches -0.0196 (standard error 0.0082 ,
12 $p = 0.017$) with an average pre-treatment coefficient of -0.0033 ($p = 0.45$). We do
13 not promote this to the lead result, because the not-yet-treated estimand answers
14 a narrower question than the one the paper asks, and because the composite’s
15 individual lead one wave before exposure is -0.0236 ($p = 0.05$). But the direction
16 of the check matters: the comparison group most exposed to the objection is the
17 one whose removal makes the result larger.

18 The same analysis applied to the Callaway–Sant’Anna estimator on the wave-
19 fixed-effects panel is less forgiving. Under exact parallel trends the identified
20 set for the aggregated composite ATT is $[-0.025, -0.008]$, essentially the level
21 confidence interval, and it excludes zero; once a relative-magnitudes violation
22 of $\bar{M} = 0.5$ is allowed it widens to $[-0.057, +0.024]$ and admits zero, so the
23 breakdown value for that target lies between $\bar{M} = 0$ and $\bar{M} = 0.5$ rather than
24 at zero. The target that isolates the wave-of-exposure effect is more durable
25 on the same panel, still excluding zero at $\bar{M} = 0.5$ ($[-0.042, -0.004]$). What
26 binds on this panel is the smoothness restriction Δ^{SD} , which bounds how far the
27 post-treatment trend may depart from a linear extrapolation of the pre-trend: it
28 reaches zero at its mildest deviation ($[-0.041, 0.000]$). Reporting both restriction
29 families follows the current difference-in-differences synthesis (Roth et al., 2023).

1 Under the preferred specification the same target instead breaks down between
 2 $\bar{M} = 0.25$ and $\bar{M} = 0.50$, for the anchoring reason set out in Supplementary
 3 Material Section S7. The gap from the Sun–Abraham bounds is mechanical,
 4 not evidential, and in two distinct senses. First, the two bound different target
 5 parameters on different comparison groups: the Sun–Abraham figure is a wave-of-
 6 exposure effect estimated without covariates on never-treated controls, while the
 7 CS(2021) benchmark conditions on covariates and uses not-yet-treated controls.
 8 Second, $\bar{M}$ is scaled by each estimator’s own largest pre-treatment deviation, so
 9 an estimator with smoother leads mechanically reports a larger $\bar{M}$ on identical
 10 data. Converting both to absolute permitted deviation removes the illusion: the
 11 CS(2021) benchmark tolerates up to about 0.009 and the Sun–Abraham estimate
 12 about 0.009 to 0.017, so in outcome units the two are close to tied despite reported
 13 breakdown values that differ by a factor of two. $\bar{M}$ is thus not a common yardstick
 14 across estimators, and we do not treat the Sun–Abraham figure as evidence that
 15 its identifying assumption is stronger.

16 A further bound is worth reporting because it is the one place the sensitivity
 17 analysis does real work rather than restating an insignificant target. The effect
 18 one wave after exposure, -0.0173 ($p < 0.001$), is the event-study coefficient that
 19 survives a simultaneous band across the whole path. Applying the same analysis
 20 to it gives a breakdown between $\bar{M} = 0$ and $\bar{M} = 0.25$ under relative magnitudes,
 21 and no margin at all under smoothness. The reason is arithmetic rather than
 22 estimator choice: the largest deviation between consecutive pre-treatment leads is
 23 0.037, about 2.2 times the effect being defended, so even a small multiple of that
 24 yardstick swamps the estimate.

Table 24: HonestDiD Relative-Magnitudes Identified Sets (Sun–Abraham t_0)

$\bar{M}$	Composite ($t_0 = -0.0223$)		<i>Warga</i> ($t_0 = -0.0096$)	
	lower	upper	lower	upper
0.0 (original)	−0.032	−0.012	−0.018	−0.002
0.5	−0.047	0.001	−0.024	0.005
1.0	−0.064	0.018	−0.034	0.014
1.5	−0.083	0.037	−0.045	0.025
2.0	−0.103	0.056	−0.056	0.036

Notes: 95 percent robust identified sets under $\Delta^{RM}(\bar{M})$ (C-LF method, $\alpha = 0.05$), computed on the window-aligned Sun–Abraham event study (2 pre-periods, 4 post-periods) for the wave-of-exposure effect t_0 , under wave fixed effects. Both outcomes admit zero at $\bar{M} = 0.5$. The corresponding analysis on the preferred province-by-wave specification, where the pre-treatment leads are smaller and the breakdown point correspondingly higher, is reported in Supplementary Material Section S7.

1 M Covariate Trajectories: CS(2021) on Observ- 2 ables

Table 25: Covariate Event-Study Aggregates: Pre- and Post-Treatment Means
Baseline: 2005-boundary panel, CS(2021), always-treated cohort dropped

Covariate	Pre-treatment aggregate	p	Post-treatment aggregate	p
Any household electricity	−0.0155***	0.001	0.0044	0.246
Paved road access	0.0284***	0.007	−0.0599***	0.000
Internet available	−0.0665***	0.000	0.0346***	0.000
Police post present	−0.0043	0.599	−0.0075	0.234
Number of health centres	−0.0057	0.696	−0.0422**	0.020
Health centre present	0.0069	0.531	−0.0079	0.238
Number of primary schools	−0.2243***	0.000	−0.0940***	0.003
Distance to district capital	0.3841	0.460	1.4717***	0.000

Each row runs the benchmark Callaway–Sant’Anna specification with the covariate as the outcome and no controls, on the analysis sample with the always-treated cohort dropped. A significant pre-treatment aggregate indicates that the covariate itself trends differentially before exposure, which is why the design conditions on covariates rather than assuming unconditional parallel trends. Population and household counts cannot be tested here: BPS stopped disseminating them after the 2011 wave, so they are missing for four of the seven waves and the event-study aggregate is not estimable. Their pre-treatment growth is examined separately over 2005–2008 in Supplementary Material Section S7. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.